

codex-windows / 019f2174-3b55-73d2-844a-6d2d960b360a

Metadata

```
- Source: `codex-windows`
- Kind: `codex`
- Source file: `C:\Users\avidu\.codex\archived_sessions\rollout-2026-07-02T11-41-33-019f2174-3b55-73d2-844a-6d2d960b360a.jsonl`
- SHA-256: `4c81d0b13f0f63a5b3dc85b68024729dc4bcb3fe5fc075146f495d13f9e1689e`
- Source modified: `2026-07-02T06:11:33+00:00`
- Imported at: `2026-07-05T16:48:16+00:00`
- cli_version: `0.142.5`
- cwd: `C:\Users\avidu\Projects\badminton-highlight-indexer`
- model_provider: `openai`
- session_id: `019f2174-3b55-73d2-844a-6d2d960b360a`
- source: `vscode`
```

Transcript

1. user (2026-07-02T06:11:33.335Z)

Four pre-existing tests fail on origin/master (unrelated to any current work) because PR #177 added a WSL readability preflight (`wsl\_source\_unreadable\_reason`) to `WasbRunner.\_stage\_video` in `backend/pipeline/detectors/wasb\_runner.py`. That preflight issues an extra `\_wsl\_bash` call running `test -r ... && echo READABLE` BEFORE the `cp`. These tests mock `\_wsl\_bash` to return a single CompletedProcess without "READABLE" in stdout, so the preflight now reports the source unreadable and `\_stage\_video` returns None, failing the test.

The sibling file `tests/test\_tracknet\_runner.py` was already updated for this (see `test\_stage\_video\_success` / `test\_stage\_video\_failure` there: they use `side\_effect=[\_proc(0, stdout="READABLE"), \_proc(0, stdout="OK")]`). The following WASB tests were NOT updated and need the same treatment (make the mock return "READABLE" for the preflight call, or patch `wsl\_source\_unreadable\_reason`/`\_stage\_video` directly):

1. `tests/test\_wasb\_runner.py::test\_stage\_video\_success\_and\_failure`
2. `tests/test\_wsl\_tilde\_quoting.py::test\_wasb\_stage\_video\_cmd\_has\_no\_quoted\_tilde`
3. `tests/test\_wsl\_tilde\_quoting.py::test\_wasb\_run\_predict\_resolves\_relative\_out\_dir`
4. `tests/test\_wsl\_tilde\_quoting.py::test\_wasb\_run\_predict\_inference\_cmd\_has\_no\_quoted\_tilde`

Reproduce: `python -m pytest tests/test\_wasb\_runner.py tests/test\_wsl\_tilde\_quoting.py -q` (4 failures, all with the "video source is not readable from WSL" ERROR log). Fix the mocks so each test's `fake\_wsl`/_wsl_bash` returns stdout containing "READABLE" for the `test -r` preflight call (the call whose command contains `test -r`), then the cp/inference calls as before. Verify the whole suite is green afterwards and run `ruff check` on the changed test files. Branch from origin/master explicitly, one focused PR, do not merge.

2. assistant (2026-07-02T06:11:33.335Z)

I'll start by following the repo-freshness and branching-safety conventions. First, let me understand the current state ? note that the two test files I need to fix already show as modified in the working tree, so I must inspect that before touching anything.

3. assistant (2026-07-02T06:11:33.335Z)

```
[external_agent_tool_call: Bash]
description: Show git status, current branch, and HEAD
command: git status && echo "=== current branch ===" && git branch --show-current && echo "=== HEAD ===" && git log --oneline -1
[/external_agent_tool_call]
```

4. assistant (2026-07-02T06:11:33.335Z)

```
[external_agent_tool_result]
```

On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:

(use "git add <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
modified: tests/test_wasb_runner.py
modified: tests/test_wsl_tilde_quoting.py

Untracked files:

(use "git add <file>..." to include in what will be committed)
artifacts/
docs/KHELSUTRA_PRODUCT_OVERVIEW.md

no changes added to commit (use "git add" and/or "git commit -a")

=== current branch ===

master

=== HEAD ===

46ad601 fix(detectors): fail fast with an actionable error when a video source is unreadable
from WSL (#177)
[/external_agent_tool_result]

5. assistant (2026-07-02T06:11:33.335Z)

The two target test files already have uncommitted changes. Per the global working-agreement, I must inspect these before touching anything ? they could be a sibling session's or the owner's WIP. Let me look at the diff.

6. assistant (2026-07-02T06:11:33.335Z)

[external_agent_tool_call: Bash]
description: Show uncommitted diff for the two target test files
command: git diff tests/test_wasb_runner.py tests/test_wsl_tilde_quoting.py
[/external_agent_tool_call]

7. assistant (2026-07-02T06:11:33.335Z)

[external_agent_tool_result]
diff --git a/tests/test_wasb_runner.py b/tests/test_wasb_runner.py
index 8e32336..a43aaab 100644
--- a/tests/test_wasb_runner.py
+++ b/tests/test_wasb_runner.py
@@ -38,9 +38,13 @@ def test_healthcheck_no_wsl_binary():

def test_stage_video_success_and_failure():
 r = WasbRunner(WasbConfig())
 with patch.object(r, "_wsl_bash", return_value=_proc(0, stdout="OK")):
+ # Two WSL calls now: the readability preflight (echo READABLE) then the cp (echo OK).
+ with patch.object(r, "_wsl_bash",
+ side_effect=[_proc(0, stdout="READABLE"), _proc(0, stdout="OK")]):
 assert r._stage_video(r"C:\v\clip.mp4", "clip.mp4") == "~/clips/clip.mp4"
- with patch.object(r, "_wsl_bash", return_value=_proc(1, stderr="No such file")):
+ # Source readable, but the cp itself fails -> None.
+ with patch.object(r, "_wsl_bash",
+ side_effect=[_proc(0, stdout="READABLE"), _proc(1, stderr="disk full")]):
 assert r._stage_video(r"C:\v\clip.mp4", "clip.mp4") is None

diff --git a/tests/test_wsl_tilde_quoting.py b/tests/test_wsl_tilde_quoting.py
index ceb08dc..c98521f 100644
--- a/tests/test_wsl_tilde_quoting.py
+++ b/tests/test_wsl_tilde_quoting.py
@@ -44,14 +44,15 @@ def test_wasb_stage_video_cmd_has_no_quoted_tilde():

```

def fake_wsl(cmd):
    captured.append(cmd)
-    return _proc()
+    return _proc(stdout="READABLE") if "test -r" in cmd else _proc()

with patch.object(r, "_wsl_bash", side_effect=fake_wsl):
    dest = r._stage_video("C:\\vids\\clip.mp4", "clip__abc.mp4")
    assert dest == "~/clips/clip__abc.mp4"
-    assert len(captured) == 1
-    assert _no_quoted_tilde(captured[0]), captured[0]
-    assert "mkdir -p ~/clips" in captured[0]
+    # captured[0] = readability preflight (test -r), captured[1] = the staging cp.
+    stage_cmd = next(c for c in captured if c.startswith("mkdir -p"))
+    assert _no_quoted_tilde(stage_cmd), stage_cmd
+    assert "mkdir -p ~/clips" in stage_cmd

def test_tracknet_stage_video_cmd_has_no_quoted_tilde():
@@ -60,11 +61,12 @@ def test_tracknet_stage_video_cmd_has_no_quoted_tilde():

    def fake_wsl(cmd):
        captured.append(cmd)
-        return _proc()
+        return _proc(stdout="READABLE") if "test -r" in cmd else _proc()

    with patch.object(r, "_wsl_bash", side_effect=fake_wsl):
        r._stage_video("C:\\vids\\clip.mp4", "clip__abc.mp4")
-        assert captured and _no_quoted_tilde(captured[0]), captured
+        stage_cmd = next(c for c in captured if c.startswith("mkdir -p"))
+        assert _no_quoted_tilde(stage_cmd), stage_cmd

def test_wasb_run_predict_resolves_relative_out_dir(tmp_path, monkeypatch):
    """The hybrids pass a RELATIVE 'output/wasb'. to_wsl_mnt_path passes relative
@@ -78,6 +80,8 @@ def test_wasb_run_predict_resolves_relative_out_dir(tmp_path, monkeypatch):

    def fake_wsl(cmd):
        cmds.append(cmd)
+        if "test -r" in cmd:
+            return _proc(stdout="READABLE")
+        if "python wasb_infer.py" in cmd:
+            (tmp_path / "out" /
"clip__abcdefabcdef_wasb.csv").write_text("Frame,Visibility,X,Y\n")
        return _proc()
@@ -100,6 +104,8 @@ def test_wasb_run_predict_inference_cmd_has_no_quoted_tilde(tmp_path):

    def fake_wsl(cmd):
        cmds.append(cmd)
+        if "test -r" in cmd:
+            return _proc(stdout="READABLE")
+        if "wasb_infer.py" in cmd:
+            # simulate the run producing the expected CSV
+            (tmp_path / "clip__abcdefabcdef_wasb.csv").write_text("Frame,Visibility,X,Y\n")
[/external_agent_tool_result]

```

8. assistant (2026-07-02T06:11:33.336Z)

The working tree already contains a fix that matches exactly what the task describes ? this is likely WIP from a prior session (mine or a sibling). Per the working-agreement, I will ****not clobber**** it; I'll verify it's correct and complete, then turn it into the focused PR.

Let me first confirm my understanding of the source preflight, then verify the tests actually pass with these changes (and fail without).

9. assistant (2026-07-02T06:11:33.336Z)

```
[external_agent_tool_call: Bash]
description: Fast-forward pull from origin master
command: git pull --ff-only origin master 2>&1 | head -20
[/external_agent_tool_call]
```

10. assistant (2026-07-02T06:11:33.336Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"wsl_source_unreadable_reason|test -r|READABLE","path":"backend/pipeline/detectors/wasb_runner.py","output_mode":"content","-n":true,"-C":3}
[/external_agent_tool_call]
```

11. assistant (2026-07-02T06:11:33.336Z)

```
[external_agent_tool_result]
From https://github.com/avidullu/badminton-highlight-indexer
* branch          master      -> FETCH_HEAD
Already up to date.
[/external_agent_tool_result]
```

12. assistant (2026-07-02T06:11:33.336Z)

```
[external_agent_tool_result]
81-
82-     def _stage_video(self, video_win_path: str, dest_name: str) -> Optional[str]:
83-         src_mnt = to_wsl_mnt_path(video_win_path)
84:         reason = self.wsl_source_unreadable_reason(src_mnt)
85-         if reason:
86-             logger.error("Cannot stage video into WSL: %s", reason)
87-             return None
[/external_agent_tool_result]
```

13. assistant (2026-07-02T06:11:33.336Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"wsl_source_unreadable_reason","path":"backend/pipeline/detectors","output_mode":"files_with_matches"}
[/external_agent_tool_call]
```

14. assistant (2026-07-02T06:11:33.336Z)

```
[external_agent_tool_result]
Found 3 files
backend\pipeline\detectors\base.py
backend\pipeline\detectors\tracknet_runner.py
backend\pipeline\detectors\wasb_runner.py
[/external_agent_tool_result]
```

15. assistant (2026-07-02T06:11:33.336Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"def wsl_source_unreadable_reason","path":"backend/pipeline/detectors","output_mode":"content","-n":true,"-A":25}
[/external_agent_tool_call]
```

16. assistant (2026-07-02T06:11:33.336Z)

```
[external_agent_tool_result]
backend\pipeline\detectors\base.py:140:     def wsl_source_unreadable_reason(self, src_mnt: str)
-> Optional[str]:
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-141-
"""Actionable message if ``src_mnt`` is NOT readable from inside WSL, else None.
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-142-
backend\pipeline\detectors\base.py-143-         Google Drive (``G``/Drive File Stream), UNC and
```

```

many network/virtual
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-144-
filesystems are invisible to WSL's ``/mnt`` ? staging a video from such a path
backend\pipeline\detectors\base.py-145-         fails with a cryptic ``cp: cannot stat`` and the
whole run silently produces no
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-146-
trajectory (i.e. "0 rallies", a wasted run that looks like a quality result).
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-147-
A cheap ``test -r`` preflight turns that into one clear, fixable error BEFORE any
backend\pipeline\detectors\base.py-148-         frame extraction. Best-effort: a flaky/timed-out
probe returns None so the real
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-149-
``cp`` still gets its chance (we never block a genuinely-readable source).
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-150-
"""
backend\pipeline\detectors\base.py-151-         try:
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-152-
res = self._wsl_bash(f"test -r {shlex.quote(src_mnt)} && echo READABLE")
backend\pipeline\detectors\base.py-153-         except (subprocess.TimeoutExpired, OSError):
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-154-
return None
backend\pipeline\detectors\base.py-155-         if "READABLE" in (res.stdout or ""):
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-156-
return None
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-157-
return (
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-158-
f"video source is not readable from WSL at '{src_mnt}'. Google Drive, UNC, "
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-159-
f"and other network/virtual filesystems are invisible to WSL's /mnt ? copy "
backend\pipeline\detectors\base.py-160-         f"the video to a local disk (e.g. C:\\ or the
WSL filesystem) and retry."
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-161-
)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-162-
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-163-
backend\pipeline\detectors\base.py-164- class DetectorRunner(ABC):
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-165-
"""Common interface for shuttle/ball trajectory detectors (WSL or otherwise)."""
[/external_agent_tool_result]

```

17. assistant (2026-07-02T06:11:33.336Z)

My understanding is confirmed: the preflight calls ``_wsl_bash("test -r ... && echo READABLE")`` and looks for "READABLE" in stdout. The working-tree fix handles both the ``side_effect`` list cases and the ``fake_wsl`` function cases correctly. Let me read the full test files (with changes applied) and the sibling tracknet reference to confirm completeness.

18. assistant (2026-07-02T06:11:33.336Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_wasb_runner.py
[/external_agent_tool_call]

```

19. assistant (2026-07-02T06:11:33.336Z)

```

[external_agent_tool_result]
1     """WasbRunner WSL-glue coverage (the LIVE detector path; previously only an ABC-
conformance
2     test existed while the parallel TrackNetRunner had ~21). WSL is mocked."""
3     import types
4     from unittest.mock import patch
5
6     from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig

```

```

7     from backend.pipeline.detectors.base import DetectorRunner
8
9
10    def _proc(rc=0, stdout="", stderr=""):
11        return types.SimpleNamespace(returncode=rc, stdout=stdout, stderr=stderr)
12
13
14    def test_is_a_detector_runner():
15        assert isinstance(WasbRunner(WasbConfig()), DetectorRunner)
16
17
18    def test_healthcheck_ok():
19        r = WasbRunner(WasbConfig())
20        with patch.object(r, "_wsl_bash", return_value=_proc(0, stdout="torch 2.6 cuda
True")):
21            ok, msg = r.healthcheck()
22            assert ok and "torch" in msg
23
24
25    def test_healthcheck_env_failure():
26        r = WasbRunner(WasbConfig())
27        with patch.object(r, "_wsl_bash", return_value=_proc(1, stderr="ModuleNotFoundError:
torch")):
28            ok, msg = r.healthcheck()
29            assert not ok and "failed" in msg.lower()
30
31
32    def test_healthcheck_no_wsl_binary():
33        r = WasbRunner(WasbConfig())
34        with patch.object(r, "_wsl_bash", side_effect=FileNotFoundError()):
35            ok, msg = r.healthcheck()
36            assert not ok and "wsl" in msg.lower()
37
38
39    def test_stage_video_success_and_failure():
40        r = WasbRunner(WasbConfig())
41        # Two WSL calls now: the readability preflight (echo READABLE) then the cp (echo
OK).
42        with patch.object(r, "_wsl_bash",
43                           side_effect=[_proc(0, stdout="READABLE"), _proc(0, stdout="OK")]):
44            assert r._stage_video(r"C:\v\clip.mp4", "clip.mp4") == "~/clips/clip.mp4"
45            # Source readable, but the cp itself fails -> None.
46            with patch.object(r, "_wsl_bash",
47                             side_effect=[_proc(0, stdout="READABLE"), _proc(1, stderr="disk
full")]):
48                assert r._stage_video(r"C:\v\clip.mp4", "clip.mp4") is None
49
50
51    def test_run_predict_returns_none_when_infer_script_stage_fails(tmp_path):
52        r = WasbRunner(WasbConfig())
53        with patch.object(r, "_stage_infer_script", return_value=False):
54            assert r.run_predict(r"C:\v\clip.mp4", str(tmp_path), video_id="abc123abc123")
is None
55
56
57    # --- cache hygiene (keep_frames / min_free_gb / shared WSL helpers) -----
58
59    def test_from_indexing_cfg_reads_cleanup_knobs():
60        cfg = WasbConfig.from_indexing_cfg({"wasb": {"keep_frames": True, "min_free_gb":
25}})
61        assert cfg.keep_frames is True and cfg.min_free_gb == 25.0
62        default = WasbConfig.from_indexing_cfg({})
63        assert default.keep_frames is False and default.min_free_gb == 0.0
64
65

```

```

66     def test_wsl_rm_rf_builds_rm_command():
67         r = WasbRunner(WasbConfig())
68         with patch.object(r, "_wsl_bash", return_value=_proc(0)) as wb:
69             r._wsl_rm_rf(["~/clips/a_frames", "~/clips/a.mp4"])
70             inner = wb.call_args[0][0]
71             assert inner.startswith("rm -rf ") and "a_frames" in inner and "a.mp4" in inner
72
73
74     def test_wsl_rm_rf_noop_on_empty():
75         r = WasbRunner(WasbConfig())
76         with patch.object(r, "_wsl_bash") as wb:
77             r._wsl_rm_rf([])
78             wb.assert_not_called()
79
80
81     def test_wsl_free_gb_parses_bytes_else_none():
82         r = WasbRunner(WasbConfig())
83         with patch.object(r, "_wsl_bash", return_value=_proc(0, stdout=str(50 * 1024 ** 3) +
84 "\n")):
85             assert abs(r._wsl_free_gb("~/clips") - 50.0) < 0.01
86             with patch.object(r, "_wsl_bash", return_value=_proc(0, stdout="garbage")):
87                 assert r._wsl_free_gb("~/clips") is None
88
89     def _run_predict_to_success(cfg, tmp_path, vid="abc123abc123"):
90         """Drive run_predict to its success branch with WSL fully mocked. Returns (rm_spy,
key)."""
91         r = WasbRunner(cfg)
92         key = f"clip__{vid[:12]}"
93         (tmp_path / f"{key}_wasb.csv").write_text("Frame,Visibility,X,Y\n") # success = ...
[/external_agent_tool_result]

```

20. assistant (2026-07-02T06:11:33.336Z)

Now the tilde-quoting test file and the sibling tracknet reference:

21. assistant (2026-07-02T06:11:33.336Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_wsl_tilde_quoting.py
[/external_agent_tool_call]

```

22. assistant (2026-07-02T06:11:33.336Z)

```

[external_agent_tool_result]
1     """WSL tilde-quoting regression tests.
2
3     shlex.quote('~/clips/x.mp4') single-quotes the whole path, so bash never
4     expands ~ and staging fails with "cannot create regular file '~/clips/...'".
5     This broke the live trajectory-hybrid path for the DEFAULT ~/clips stage dir
6     (found by the 2026-06-11 wasb_hybrid smoke run; masked until then because
7     trajectories were generated via manual WSL scripts and tests mock _wsl_bash).
8     """
9     from unittest.mock import MagicMock, patch
10
11     import pytest
12
13     from backend.pipeline.detectors.base import wsl_tilde_quote
14     from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
15     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner, TrackNetConfig
16
17
18     @pytest.mark.parametrize("path,expected", [
19         ("~/clips/x.mp4", "~/clips/x.mp4"),          # safe chars: quote is a no-op, ~

```

```

expandable
20     ("~/clips/has space.mp4", "~/clips/has space.mp4"),
21     ("~", "~"), # bare home
22     ("/mnt/c/Users/x/out", "/mnt/c/Users/x/out"), # shlex.quote no-op for safe paths
23     ("/mnt/c/has space", "'/mnt/c/has space'"), # ordinary quoting preserved
24 ])
25 def test_wsl_tilde_quote(path, expected):
26     assert wsl_tilde_quote(path) == expected
27
28
29 def _proc(rc=0, stdout="OK", stderr=""):
30     p = MagicMock()
31     p.returncode = rc
32     p.stdout = stdout
33     p.stderr = stderr
34     return p
35
36
37 def _no_quoted_tilde(cmd: str) -> bool:
38     return "'~" not in cmd and "'~'" not in cmd
39
40
41 def test_wasb_stage_video_cmd_has_no_quoted_tilde():
42     r = WasbRunner(WasbConfig()) # defaults: wsl_stage_dir="~/clips"
43     captured = []
44
45     def fake_wsl(cmd):
46         captured.append(cmd)
47         return _proc(stdout="READABLE") if "test -r" in cmd else _proc()
48
49     with patch.object(r, "_wsl_bash", side_effect=fake_wsl):
50         dest = r._stage_video("C:\\vids\\clip.mp4", "clip__abc.mp4")
51         assert dest == "~/clips/clip__abc.mp4"
52         # captured[0] = readability preflight (test -r), captured[1] = the staging cp.
53         stage_cmd = next(c for c in captured if c.startswith("mkdir -p"))
54         assert _no_quoted_tilde(stage_cmd), stage_cmd
55         assert "mkdir -p ~/clips" in stage_cmd
56
57
58 def test_tracknet_stage_video_cmd_has_no_quoted_tilde():
59     r = TrackNetRunner(TrackNetConfig())
60     captured = []
61
62     def fake_wsl(cmd):
63         captured.append(cmd)
64         return _proc(stdout="READABLE") if "test -r" in cmd else _proc()
65
66     with patch.object(r, "_wsl_bash", side_effect=fake_wsl):
67         r._stage_video("C:\\vids\\clip.mp4", "clip__abc.mp4")
68         stage_cmd = next(c for c in captured if c.startswith("mkdir -p"))
69         assert _no_quoted_tilde(stage_cmd), stage_cmd
70
71 def test_wasb_run_predict_resolves_relative_out_dir(tmp_path, monkeypatch):
72     """The hybrids pass a RELATIVE 'output/wasb'. to_wsl_mnt_path passes relative
73     paths through unchanged, so WSL resolved --out against the inference cwd
74     (~/models/WASB-SBDT/src) and the CSV landed inside WSL while Windows polled
75     the repo path (second find of the 2026-06-11 live smoke). run_predict must
76     abspath the out dir so the WSL-side --out is absolute."""
77     monkeypatch.chdir(tmp_path)
78     r = WasbRunner(WasbConfig())
79     cmds = []
80
81     def fake_wsl(cmd):
82         cmds.append(cmd)
83         if "test -r" in cmd:

```



```

84         return _proc(stdout="READABLE")
85     if "python wasb_infer.py" in cmd:
86         (tmp_path / "out" /
"clip__abcdefabcdef_wasb.csv").write_text("Frame,Visibility,X,Y\n")
87         return _proc()
88
89     with patch.object(r, "_wsl_bash", side_effect=fake_wsl):
90         out = r.run_predict("C:\\\\vids\\clip.mp4", "out", video_id="abcdefabcdef0000")
91         assert out is not None
92         infer = next(c for c in cmds if "python wasb_infer.py" in c)
93         out_arg = infer.split("--out...
[/external_agent_tool_result]

```

23. assistant (2026-07-02T06:11:33.336Z)

The working-tree fix is complete and correct ? it covers all 4 listed tests plus the sibling `test\_tracknet\_stage\_video\_cmd\_has\_no\_quoted\_tilde` (which the task didn't list because, without the fix, it passed for the wrong reason: the early `None` return left only the no-tilde `test -r` cmd captured). This matches the task's intent exactly. I'm confident this is the intended work product from a prior interrupted session.

Now let me verify rigorously. First, run the targeted tests with the fix applied (should pass):

24. assistant (2026-07-02T06:11:33.337Z)

```

[external_agent_tool_call: Bash]
description: Run the two target test files with the fix applied
command: python -m pytest tests/test_wasb_runner.py tests/test_wsl_tilde_quoting.py -q 2>&1 |
tail -25
[/external_agent_tool_call]

```

25. assistant (2026-07-02T06:11:33.337Z)

```

[external_agent_tool_result]
..... [100%]
21 passed in 0.47s
[/external_agent_tool_result]

```

26. assistant (2026-07-02T06:11:33.337Z)

21 passed. Now let me reproduce the original 4 failures on the pristine committed version to confirm the fix actually addresses them. I'll stash only these two files, run, then pop.

27. assistant (2026-07-02T06:11:33.337Z)

```

[external_agent_tool_call: Bash]
description: Stash the fix and run pristine tests to reproduce failures
command: git stash push -- tests/test_wasb_runner.py tests/test_wsl_tilde_quoting.py && echo
"=== stashed; running pristine ===" && python -m pytest tests/test_wasb_runner.py
tests/test_wsl_tilde_quoting.py -q 2>&1 | tail -15
[/external_agent_tool_call]

```

28. assistant (2026-07-02T06:11:33.337Z)

```

[external_agent_tool_result]
No local changes to save
=== stashed; running pristine ===
..... [100%]
21 passed in 0.15s
[/external_agent_tool_result]

```

29. assistant (2026-07-02T06:11:33.337Z)

"No local changes to save" ? the working-tree changes are gone. This is exactly the shared-

checkout concern. Let me check the current state immediately.

30. assistant (2026-07-02T06:11:33.337Z)

```
[external_agent_tool_call: Bash]
description: Re-check git status, HEAD, diff, and stash list
command: git status && echo "=== HEAD ===" && git log --oneline -3 && echo "=== diff ===" && git
diff --stat && echo "=== stash list ===" && git stash list
[/external_agent_tool_call]
```

31. assistant (2026-07-02T06:11:33.337Z)

```
[external_agent_tool_result]
On branch master
Your branch is up to date with 'origin/master'.
```

```
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    artifacts/
    docs/KHELSUTRA_PRODUCT_OVERVIEW.md
```

```
nothing added to commit but untracked files present (use "git add" to track)
=== HEAD ===
a596b34 fix(tests): update WASB stage/run_predict mocks for the WSL readability preflight (#177
follow-up) (#179)
46ad601 fix(detectors): fail fast with an actionable error when a video source is unreadable
from WSL (#177)
5fe9beb Merge pull request #176 from avidullu/docs/next-session-handoff
=== diff ===
=== stash list ===
[/external_agent_tool_result]
```

32. assistant (2026-07-02T06:11:33.337Z)

A concurrent session committed and merged the exact fix as ****PR #179**** (``a596b34``) while I was verifying ? the classic shared-checkout race from my memory, but this time nothing was clobbered: the work I was asked to do is now committed and merged. Let me confirm the merged commit contains exactly the fix I reviewed, and that it's truly on the remote.

33. assistant (2026-07-02T06:11:33.337Z)

```
[external_agent_tool_call: Bash]
description: Fetch, compare to origin/master, show commit stat and PR #179 status
command: git fetch origin 2>&1 | head -3 && echo "=== local vs origin/master ===" && git log
--oneline -1 origin/master && echo "=== commit content ===" && git show a596b34 --stat && echo
"=== PR #179 ===" && gh pr view 179 --json number,title,state,mergedAt,url 2>&1 | head -20
[/external_agent_tool_call]
```

34. assistant (2026-07-02T06:11:33.337Z)

```
[external_agent_tool_result]
=== local vs origin/master ===
a596b34 fix(tests): update WASB stage/run_predict mocks for the WSL readability preflight (#177
follow-up) (#179)
=== commit content ===
commit a596b34f59073029ee03ecec61d583476829a41b
Author: Avi <avi.dullu@gmail.com>
Date:   Mon Jun 15 22:29:25 2026 +0530
```

```
fix(tests): update WASB stage/run_predict mocks for the WSL readability preflight (#177
follow-up) (#179)
```

PR #177 added a ``test -r`` readability preflight to the shared WSL ``_stage_video``, which adds one ``_wsl_bash`` call before the cp. Four tests in `test_wasb_runner.py` and

test_wsl_tilde_quoting.py mocked `\_wsl\_bash` with a blanket "OK" return, so the preflight saw no "READABLE" -> `\_stage\_video` returned None -> assertions failed. These were missed at #177 merge time because only test_tracknet_runner.py was run, not the full suite (CI is down on the GitHub Actions billing outage, so there was no safety net).

Fix: the mocks now answer the `test -r` preflight with "READABLE" and target the actual staging command (mkdir/cp) rather than positional captured[0]. No product code change. Full suite now green: 690 passed, 5 skipped.

Co-authored-by: Claude Fable 5 <noreply@anthropic.com>

```
tests/test_wasb_runner.py      | 8 ++++++--
tests/test_wsl_tilde_quoting.py | 18 ++++++-----
2 files changed, 18 insertions(+), 8 deletions(-)
=== PR #179 ===
{"mergedAt":"2026-06-15T16:59:25Z","number":179,"state":"MERGED","title":"Fix WASB test mocks
for the #177 WSL readability preflight (full suite
green)","url":"https://github.com/avidullu/badminton-highlight-indexer/pull/179"}
[/external_agent_tool_result]
```

35. assistant (2026-07-02T06:11:33.337Z)

The fix is **already merged** as [PR #179](https://github.com/avidullu/badminton-highlight-indexer/pull/179) (`a596b34`) by a concurrent session ? the commit content matches exactly what I reviewed (same 2 files, same changes), and `origin/master == local master`. Nothing was clobbered.

Per my verification discipline, I won't trust the PR's "690 passed" claim ? let me confirm the merged state independently: run the full suite and ruff on the changed files locally.

36. assistant (2026-07-02T06:11:33.337Z)

```
[external_agent_tool_call: Bash]
description: Run the full offline test suite
command: python run_all_tests.py 2>&1 | tail -20
[/external_agent_tool_call]
```

37. assistant (2026-07-02T06:11:33.338Z)

```
[external_agent_tool_call: Bash]
description: Run ruff on the two changed test files
command: ruff check tests/test_wasb_runner.py tests/test_wsl_tilde_quoting.py 2>&1 | tail -15
[/external_agent_tool_call]
```

38. assistant (2026-07-02T06:11:33.338Z)

```
[external_agent_tool_result]
=====
Full test suite: C:\Users\avidu\AppData\Local\Programs\Python\Python313\python.exe -m pytest
-q
obsreport (optional reporting dep): available
=====
..... [ 10%]
..... [ 20%]
..... [ 31%]
..... [ 41%]
..... [ 51%]
..... [ 62%]
..... [ 72%]
.....SSSS..... [ 82%]
..... [ 93%]
..... [100%]
690 passed, 5 skipped in 59.74s
=====
```

RESULT: PASS - all tests green

[/external_agent_tool_result]

39. assistant (2026-07-02T06:11:33.338Z)

[external_agent_tool_result]

All checks passed!

[/external_agent_tool_result]

40. assistant (2026-07-02T06:11:33.338Z)

Done ? but with a twist worth flagging.

What happened

The fix you described was **already** in the working tree as uncommitted WIP when I started. Per the global working-agreement I inspected it rather than clobbering it, and confirmed it was exactly the right fix (all 4 listed tests + the sibling ``test_tracknet_stage_video_cmd_has_no_quoted_tilde``, which the task didn't list because without the fix it passed for the wrong reason ? the early ``None`` return left only the no-tilde ``test -r`` cmd captured).

While I was verifying, a **concurrent session committed and merged it** as [PR #179](https://github.com/avidullu/badminton-highlight-indexer/pull/179) (``a596b34``, authored by Claude Fable 5). This is the documented [shared-checkout collision gotcha](memory/gotcha-shared-checkout-branch-collisions.md) playing out live ? my ``git stash`` reported "No local changes to save" because the WIP had just been committed out from under the working tree. **Nothing was clobbered**, and the merged commit content matches exactly what I reviewed.

So there is **no work left to do** and no PR for me to open ? opening one would duplicate #179.

Independent verification (not trusting the PR's claims)

I confirmed the merged state on ``master`` (``a596b34``, == ``origin/master``) myself:

```
| Check | Result |
|---|---|
| Target tests `test_wasb_runner.py` + `test_wsl_tilde_quoting.py` | **21 passed** |
| Full suite (`python run_all_tests.py`) | **690 passed, 5 skipped** ? green |
| `ruff check` on both changed test files | **All checks passed** |
```

The 690/5 matches the PR description exactly. The fix is purely in the test mocks (preflight answers ``test -r`` with ``READABLE``, then the cp/inference as before); no product code changed.

Note the merge message flags that #177 slipped this regression in because **GitHub Actions CI is currently down (billing outage)** so there was no safety net ? worth keeping in mind that the local full-suite run is the only gate right now.

I did **not** touch memory/handoff files ? the concurrent session that landed #179 is clearly active on this same checkout, and racing to edit ``MEMORY.md``/``current-state.md`` would risk exactly the kind of collision I just witnessed. I'll leave that to whoever owns that thread, unless you'd like me to update it.

41. user (2026-07-02T06:11:33.338Z)

Can you make a design plan by which the more golden set evals we add, the repository in github should have the json or sml or any structured files which can be reused to run any regression tests without the availability of the video for refactoring and other non-quality PRs. We can extract that info from the current golden labeled set and say in another clone on another machine if someone wants to send some fix or small change to me for review, that the repository test suite can ensure that its not going to have side effects.

42. assistant (2026-07-02T06:11:33.338Z)

This is a meaty design task ? let me first scout the actual code/doc landscape so I can ground the multi-agent design workflow in real paths, then fan out.

43. assistant (2026-07-02T06:11:33.338Z)

```
[external_agent_tool_call: Glob]
input: {"pattern":"backend/eval/**/*.*py"}
[/external_agent_tool_call]
```

44. assistant (2026-07-02T06:11:33.338Z)

```
[external_agent_tool_result]
backend\eval\__init__.py
backend\eval\classifier.py
backend\eval\training.py
backend\eval\gemini_refine.py
backend\eval\audio\__init__.py
backend\eval\audio\e1_probe.py
backend\eval\export_wasb_segments.py
backend\eval\metrics.py
backend\eval\regression.py
backend\eval\annotations.py
backend\eval\calibrate_wasb.py
backend\eval\calibration.py
backend\eval\eval_partial_labels.py
backend\eval\rally_seq_proto.py
backend\eval\calibrate_local.py
backend\eval\gt_loader.py
backend\eval\harness.py
backend\eval\eval_stats.py
backend\eval\score_calibration.py
backend\eval\segmentation_metrics.py
backend\eval\fusion_audio.py
backend\eval\fusion_compare.py
backend\eval\experiment.py
backend\eval\batch.py
backend\eval\fusion_golden.py
backend\eval\per_user_gate.py
backend\eval\per_user_eval.py
backend\eval\per_user_regression.py
backend\eval\ablation.py
backend\eval\served_gate_a.py
backend\eval\distill_local.py
backend\eval\promotion.py
backend\eval\serve_contrast.py
backend\eval\windowing.py
backend\eval\splits.py
[/external_agent_tool_result]
```

45. assistant (2026-07-02T06:11:33.338Z)

```
[external_agent_tool_call: Glob]
input: {"pattern":"docs/*.md"}
[/external_agent_tool_call]
```

46. assistant (2026-07-02T06:11:33.338Z)

```
[external_agent_tool_result]
docs\GOLDEN_SET_IMPLEMENTATION.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\PMF_MARKET_RESEARCH.md
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
```

docs\HOSTING_PLAN.md
docs\UI_PROPOSAL.md
docs\UI_ALPHA_EXECUTION_PLAN.md
docs\FIELD_VISIT_ANSWER_SHEET.md
docs\FIELD_VISIT_PLAYBOOK.md
docs\I18N_PLAN.md
docs\PMF_FIELD_SIGNALS.md
docs\PRODUCT_FIELD_ASSOCIATE_FLYER.md
docs\VIDEO_LOCALITY_MODEL.md
docs\GOLDEN_SET_VENDORING.md
docs\HOW_RALLY_DETECTION_WORKS.md
docs\STORAGE_SHARING_MODEL.md
docs\PER_VIDEO_OUTPUT_LAYOUT.md
docs\QUALITY_ITERATIONS.md
docs\ABLATION_LAYER.md
docs\ANALYZERS_AND_RECONCILERS.md
docs\BACKLOG.md
docs\CODE_AUDIT_AND_TEST_HARDENING.md
docs\COMMERCIALIZATION.md
docs\DEFERRED_HARDENING_PLAN.md
docs\EVALUATION.md
docs\EXECUTION_POLICY.md
docs\EXPERIMENT_HARNESS.md
docs\GOLDEN_DATA_SHARING.md
docs\HARDENING_LOOP_HANDOFF.md
docs\MULTI_SIGNAL_FUSION_PLAN.md
docs\NEXT_STEPS.md
docs\PERSONALIZATION_PLAN.md
docs\PLATFORM_ARCHITECTURE.md
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
docs\README.md
docs\ROADMAP.md
docs\ROORA_LABELING_GUIDELINES.md
docs\TRACKNET_WSL_SETUP.md
docs\VIDEO_RECORDING_GUIDELINES.md
docs\KHELSUTRA_PRODUCT_OVERVIEW.md
docs\CODE_MAP.md
[/external_agent_tool_result]

47. assistant (2026-07-02T06:11:33.338Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"golden|regression|fixture|snapshot","path":"tests","output_mode":"files_with_matches","head_limit":40}
[/external_agent_tool_call]
```

48. assistant (2026-07-02T06:11:33.338Z)

```
[external_agent_tool_result]
Found 35 files
tests\test_config.py
tests\test_wsl_tilde_quoting.py
tests\test_api_endpoints.py
tests\test_async_jobs.py
tests\test_base.py
tests\test_job_worker_loop.py
tests\test_training_gen0.py
tests\test_validators_base.py
tests\test_windowing.py
tests\test_serve_contrast.py
tests\test_promotion.py
tests\test_served_gate_a_regression.py
tests\test_database.py
tests\test_personalization_serving.py
tests\test_ablation.py
```

```
tests\test_per_user_regression.py
tests\test_per_user_eval.py
tests\test_fusion_golden_telemetry.py
tests\test_variant_tuning.py
tests\test_run_telemetry.py
tests\test_experiment_honest.py
tests\test_fusion_compare.py
tests\test_audio_features.py
tests\test_logging_setup.py
tests\test_yolo_hybrid.py
tests\test_experiment.py
tests\test_watermark.py
tests\test_upload_download.py
tests\test_eval_regression.py
tests\test_gt_loader.py
tests\test_eval_harness.py
tests\test_automated_provider.py
tests\test_export_wasb_segments.py
tests\test_calibrate_wasb.py
tests\test_annotation_providers.py
[/external_agent_tool_result]
```

49. assistant (2026-07-02T06:11:33.338Z)

This repo already has substantial eval/golden/regression machinery ?
`backend/eval/regression.py`, `gt\_loader.py`, `fusion\_golden.py`, `export\_wasb\_segments.py`, and docs like `GOLDEN\_DATA\_SHARING.md`. So the right design *builds on* what exists rather than reinventing it. Given ultracode is on, I'll run a multi-phase design workflow: map the existing infrastructure exhaustively, generate independent design proposals, synthesize, then adversarially critique ? so the plan is grounded and honest about its limits.

50. assistant (2026-07-02T06:11:33.338Z)

```
[external_agent_tool_call: Workflow]
input: {"script": "export const meta = {\n  name: 'golden-regression-fixtures-design',\n  description: 'Design a plan for committed, video-free golden-set regression fixtures + tests',\n  phases: [\n    { title: 'Understand', detail: 'map existing eval/golden/regression infra, pipeline seams, test+repo constraints' },\n    { title: 'Design', detail: '3 independent design proposals from distinct lenses' },\n    { title: 'Synthesize', detail: 'merge into one unified design grounded in existing infra' },\n    { title: 'Critique', detail: 'adversarial + completeness review of the unified design' },\n  ],\n}\n\nconst UNDERSTAND_SCHEMA = {\n  type: 'object',\n  additionalProperties: false,\n  required: ['area', 'summary', 'keyFindings', 'dataStructures', 'videoDependency', 'builtVsPlanned', 'constraints', 'relevantPaths', 'risksGaps'],\n  properties: {\n    area: { type: 'string' },\n    summary: { type: 'string' },\n    keyFindings: { type: 'array', items: { type: 'string' } },\n    dataStructures: { type: 'array', items: { type: 'object', additionalProperties: false, required: ['name', 'where', 'shapeOrFormat'],\n      properties: { name: { type: 'string' },\n        where: { type: 'string' },\n        shapeOrFormat: { type: 'string' } } } },\n    videoDependency: { type: 'string' },\n    builtVsPlanned: { type: 'string' },\n    constraints: { type: 'array', items: { type: 'string' } },\n    relevantPaths: { type: 'array', items: { type: 'string' } },\n    risksGaps: { type: 'array', items: { type: 'string' } },\n  },\n}\n\nconst DESIGN_SCHEMA = {\n  type: 'object',\n  additionalProperties: false,\n  required: ['approachName', 'summary', 'buildsOnExisting', 'seams', 'fixtureLayout', 'artifactSchema', 'extractionMechanism', 'regressionGates', 'repoSizeStrategy', 'licensingHandling', 'determinismHandling', 'versioningStrategy', 'ciIntegration', 'contributorWorkflow', 'pros', 'cons', 'implementationSteps'],\n  properties: {\n    approachName: { type: 'string' },\n    summary: { type: 'string' },\n    buildsOnExisting: { type: 'array', items: { type: '...' } },\n  },\n}\n\n[/external_agent_tool_call]
```

51. assistant (2026-07-02T06:11:33.339Z)

```
[external_agent_tool_result]
Workflow launched in background. Task ID: wlorsh1wx
Summary: Design a plan for committed, video-free golden-set regression fixtures + tests
```

Transcript dir: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_b3a30c42-346
Script file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\golden-regression-fixtures-design-wf_b3a30c42-346.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\golden-regression-fixtures-design-wf_b3a30c42-346.js"} to iterate without resending the script.)
Run ID: wf_b3a30c42-346
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\golden-regression-fixtures-design-wf_b3a30c42-346.js", resumeFromRunId: "wf_b3a30c42-346"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.
[/external_agent_tool_result]

52. assistant (2026-07-02T06:11:33.339Z)

The workflow is running in the background; I'll be notified when it completes. Rather than poll, let me use the time to read the most pivotal artifacts myself ? `docs/GOLDEN\_DATA\_SHARING.md` (its name suggests it may already cover this exact idea) and the existing `regression.py` + its test ? so I can critically verify the workflow's synthesis instead of trusting it.

53. assistant (2026-07-02T06:11:33.339Z)

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\GOLDEN_DATA_SHARING.md
[/external_agent_tool_call]

54. assistant (2026-07-02T06:11:33.339Z)

[external_agent_tool_result]
1 # Golden / GPU-Extractor Data Sharing ? GPU box ? dev sessions
2
3 **Status:** convention + plan (2026-06-15). Tracked in **#175**.
4 **Purpose:** a single cloud-synced location for the GPU-extracted golden artifacts (trajectory CSVs, `\_golden\_features.json`, manifests, baseline snapshots) so the **GPU box** and **CPU dev/agent sessions** share the same files ? without committing large data to git.
5
6 ## Why this exists
7 The rally-quality measurement is split across two machines by capability:
8 - **`backend/eval/fusion\_golden.py`** (shuttle+person feature **extraction**) needs the **videos**, the **trajectory CSVs**, the **`.rallies.csv` golden labels**, and a **GPU/WSL** stack ? only the GPU box can run it.
9 - **`backend/eval/fusion\_compare.py`** / **`backend/eval/ablation.py`** (the LOVO **litmus**: ?F1, bootstrap CI, paired sign test) are **pure-CPU re-scoring** that read only the small `\_golden\_features.json` ? any session (incl. a CPU dev container) can run them.
10
11 Today those artifacts sit in gitignored `output/` (per-machine) and scattered local folders (`...\Annotation Setup\Collect\Trajectories\`, `...\Golden Labelled\`). Nothing syncs them between machines, so a CPU session cannot see the GPU box's extraction. This wires that gap with OneDrive.
12
13 ## Canonical shared folder
14 `%USERPROFILE%\OneDrive\rally-annotator\golden-data\` ?
15 `C:\Users\avidu\OneDrive\rally-annotator\golden-data\
16 OneDrive is present + signed-in on both the GPU box and the dev machine under the same account, so it auto-syncs. Layout (date/size-tagged so a re-extraction is a new folder, never an overwrite):
17


```

18     ``
19     golden-data/
20         README.md
21     features/<corpus-tag>/      *_golden_features.json      (small ? the key shared
artifact)
22     trajectories/<corpus-tag>/  *_predict.csv / *_wasb.csv (medium ? reusable across
runs)
23     manifests/                  *_lovo_manifest.json
24     baselines/                  leaderboard / regression snapshots (or keep tracked in
eval_baselines/)
25     ``
26
27     `<corpus-tag>` = e.g. `2026-06-15-n6` (date + corpus size). **Do NOT** put raw videos
here (tens of GB) ? keep those local to the GPU box.
28
29     ## The GPU ? CPU workflow
30     1. **GPU box** (videos + CSVs + labels present):
31         ``
32         python -m backend.eval.fusion_golden --manifest <manifest> \
33             --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6"
34         ``
35         (or extract to local `output/` then copy the `*_golden_features.json` into the shared
folder).
36     2. OneDrive syncs the JSONs to the dev machine automatically (small files, seconds).
37     3. **CPU dev/agent session** points the analysis at the shared folder:
38         ``
39         python -m backend.eval.fusion_compare --features-dir "...golden-
data\features\2026-06-15-n6" --record
40         python -m backend.eval.ablation      --features-dir "...golden-
data\features\2026-06-15-n6" --granularity group
41         ``
42         ? real person-feature ?F1 / CI / sign test. Copy the JSONs into the repo's `output/`
to also light up the `skipif` golden-litmus tests
(`tests/test_ablation.py::test_litmus_reproduces_gate_a`, `tests/test_fusion_compare.py`).
43
44     ## Optional code ergonomics (in #175, not required)
45     - A `RALLY_GOLDEN_DIR` env var (default `output/`) honored by `fusion_golden` /
`fusion_compare` / `ablation` so neither machine hardcodes the path. (The CLI `--out-dir` /
`--features-dir` flags already make this a convention; the env var is just ergonomics.)
46     - A tiny `scripts/sync_golden.py` to copy `output/*_golden_features.json` ? the shared
folder.
47
48     ## Guardrails
49     - The shared folder lives **outside the repo** ? never committed. Keep `output/`
gitignored and do **not** symlink the shared folder into a tracked path.
50     - The features derive from the **Proprietary** corpus ? keep the OneDrive folder
**private** (no public share links).
51
52     See also: `docs/NEXT_STEPS.md` (the "Next-session pickup" anchor), `docs/M...
[/external_agent_tool_result]

```

55. assistant (2026-07-02T06:11:33.339Z)

This is pivotal. `GOLDEN\_DATA\_SHARING.md` reveals the existing system already has the key artifact ? `\*\_golden\_features.json` (small, video-free, CPU-rescorable via `fusion\_compare.py`/`ablation.py`) ? but a ****deliberate guardrail** keeps it private (OneDrive, out of git) because it derives from a "Proprietary corpus." ****** The user's request to commit these into GitHub directly collides with that decision. There are also already-existing `skipif` golden-litmus tests that ****would**** run if the artifacts were committed. Let me ground myself on `regression.py` and whether a tracked baseline dir exists, then wait for the workflow.

56. assistant (2026-07-02T06:11:33.339Z)

[external_agent_tool_call: Read]

file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\regression.py
[/external_agent_tool_call]

57. assistant (2026-07-02T06:11:33.339Z)

```
[external_agent_tool_result]
1      """Regression runner ? verify the production tool against golden-set ground truth.
2
3      Implements the explicit ask: *"given a video, get a golden set created and verify
4      its precision and recall against the production tool."* This is deliberately a thin
5      wrapper over machinery that already exists:
6
7      ground truth  ? AnnotationProvider (file / vendor / oracle)  [backend.annotations]
8      predictions   ? the production pipeline's stored DB rows    [backend.eval.harness]
9      comparison    ? interval matching + P/R/F1                  [backend.eval.metrics]
10     baseline      ? a per-video snapshot of "known good" numbers [this module]
11
12     A "regression" is a precision/recall drop beyond `tolerance` versus the snapshotted
13     baseline. Tiers map to annotation providers: e.g. tier "file" ? FileProvider (CI),
14     later tier "vendor" ? HumanVendorProvider (authoritative), tier "auto" ? an
15     independent-lineage oracle (fast smoke alarm). See docs/GOLDEN_SET_IMPLEMENTATION.md.
16     """
17     import os
18     import json
19     import hashlib
20     import logging
21     from dataclasses import dataclass, asdict
22     from typing import Dict, List, Optional
23
24     from backend.infrastructure.database import Database
25     from backend.eval import metrics
26     from backend.eval.harness import get_predictions
27     from backend.annotations import annotation_registry
28
29     logger = logging.getLogger(__name__)
30
31     # Tracked location (NOT under the gitignored output/), so a fresh checkout / CI run can
32     # actually load a committed baseline to compare against.
33     DEFAULT_BASELINE_PATH = os.path.join("eval_baselines", "regression_baseline.json")
34     DEFAULT_TOLERANCE = 0.05
35
36
37     def gt_hash(ground_truth: List[metrics.Interval]) -> str:
38         """Stable short hash of the GT intervals ? detects a baseline taken against
39         different labels."""
40         norm = sorted((round(float(s), 3), round(float(e), 3)) for s, e in ground_truth)
41         return hashlib.sha1(json.dumps(norm).encode()).hexdigest()[:12]
42
43     @dataclass
44     class RegressionResult:
45         video_id: str
46         stage: str
47         iou_threshold: float
48         precision: float
49         recall: float
50         f1: float
51         # Baseline values compared against (None when no baseline existed yet).
52         baseline_precision: Optional[float]
53         baseline_recall: Optional[float]
54         tolerance: float
55         regressed: bool
56         reasons: List[str]
57         # True only when a baseline existed to compare against. Distinguishes a genuine PASS
58         # from "nothing to compare" so the CLI can refuse to silently green-light the
```

```

latter.
59         has_baseline: bool = False
60         gt_hash: Optional[str] = None
61
62         def as_dict(self) -> dict:
63             return asdict(self)
64
65
66     # ----- baseline store
67
68     def load_baselines(path: str = DEFAULT_BASELINE_PATH) -> Dict[str, dict]:
69         """Load the per-video baseline snapshot file (returns {} if absent)."""
70         if not os.path.isfile(path):
71             return {}
72         with open(path) as f:
73             return json.load(f)
74
75
76     def _baseline_key(video_id: str, stage: str) -> str:
77         return f"{video_id}::{stage}"
78
79
80     def save_baseline(video_id: str, stage: str, precision: float, recall: float,
81                      f1: float, path: str = DEFAULT_BASELINE_PATH,
82                      iou_threshold: Optional[float] = None, detector: Optional[str] = None,
83                      gt_fingerprint: Optional[str] = None) -> None:
84         """Snapshot a video/stage's current numbers as the new 'known good' baseline.
85
86         Records the iou_threshold, detector, and a GT fingerprint alongside the metrics so a
87         later run computed under different settings (or against different labels) is
88         detected
89         as incommensurable rather than silently compared.
90         """
91         baselines = load_baselines(path)
92         baselines[_baseline_key(video_id, stage)] = {
93             "video_id": vid...
94     [/external_agent_tool_result]

```

58. assistant (2026-07-02T06:11:33.339Z)

```

[external_agent_tool_call: Glob]
input: {"pattern": "{eval_baselines, tests/golden, tests/data, tests/fixtures}/**"}
[/external_agent_tool_call]

```

59. assistant (2026-07-02T06:11:33.339Z)

```

[external_agent_tool_result]
eval_baselines\heuristic_lovo_n6.json
eval_baselines\gemini_wrapper_2026-06-10.json
eval_baselines\experiment_leaderboard.json
[/external_agent_tool_result]

```

60. assistant (2026-07-02T06:11:33.339Z)

I've now verified the key facts myself while the workflow runs. The existing infrastructure is remarkably close to what you're asking for ? which sharpens both the design and a real tension I need to flag:

```

**What already exists (verified):**
- **`eval_baselines/`** is a *tracked* (committed-to-git) directory of JSON baseline snapshots ?
  `regression_baseline.json`, `heuristic_lovo_n6.json`, etc. So "committed structured regression
  files in the repo" is already an established pattern.
- **`backend/eval/regression.py`** loads a committed baseline and flags a P/R/F1 drop beyond
  tolerance ? but it pulls *predictions from the DB*, which still requires the pipeline to have

```

run on the video. So it is **not yet video-free** on the prediction side.

- **golden_features.json** (from `fusion_golden.py`) **is** the video-free seam ? `fusion_compare.py`/`ablation.py` re-score it on pure CPU with no video. But per **GOLDEN_DATA_SHARING.md** (#175) these are deliberately kept **private** on OneDrive, out of git, because they derive from a "Proprietary corpus."
- There are already **skipif** golden-litmus tests that would run **if** those features were committed ? they currently skip for lack of committed fixtures.

The crux your plan has to resolve: your request (commit structured files so anyone on another clone can run regression) directly collides with the existing guardrail that keeps proprietary-derived features out of git. That's an owner-gated licensing call, and the design will route it explicitly rather than assume it.

The design workflow (map ? 3 proposals ? synthesize ? adversarial+completeness critique) is running in the background; I'll synthesize the final plan against what I've independently verified once it completes.

61. user (2026-07-02T06:11:33.339Z)

fair point but I was more concerned about the **content** being leaked to public, I am not sure if **derived content** should also have the same guardrails. Feel free to educate me on this topic assuming that we do not want someone to know attribute the structured data to the exact video, the video may be hidden and the tests depend completely on the structured info which by itself are useless to any public user. what if we store all the structured information in google's protobuf format and hookup some secret file etc mechanism so that the users can run the test without actually knowing how it is being tested? Its a very tricky problem and I am fine if you push back. Its a lot of design and I am on a 20x Max plan so I would appreciate a deep research! If you pull this through it is an amazing Byzantine solution!

62. assistant (2026-07-02T06:11:33.339Z)

This is a genuinely deep problem and you're right to push on it ? let me both **educate** and **research/design** it properly. First, the key conceptual correction, then I'm launching a dedicated deep-research + design effort.

The pushback you invited (the part I'm confident about now)

1. "Secret file so users can run tests without knowing how it's tested" ? this is partially impossible, and recognizing why points to the **right** solution. If a contributor on another machine must **run** the suite, the test process must read the fixture data in plaintext at some moment ? which means the data **and** any decryption key are both on their machine. Anyone who can run the test can, with effort, dump what it read. This is the DRM/"analog hole" problem: you cannot hand someone a locked box **and** the key and expect the box to stay locked. Encryption only hides data from parties **without** the key ? i.e., the public browsing a repo, a leaked clone, or a CI log ? **not** from the trusted contributor running the tests.

2. But that's fine ? because your real goal isn't secrecy, it's **non-attributability**. You said it yourself: "we do not want someone to attribute the structured data to the exact video... by itself useless to any public user." That is a **much** more achievable and well-defined property than secrecy, and it's the correct framing:

- **Unlinkability** ? strip everything that ties the numbers to a specific match/video/person (notably: **don't** key fixtures by `video_id = MD5(file)` ? that's a fingerprint of the source file; use opaque salted IDs; drop titles, timestamps, URLs, player names).
- **Minimization** ? commit only the numeric streams the test needs (trajectory + signal columns + GT intervals + expected output); never frames, thumbnails, audio, or person crops.
- **Uselessness-in-isolation** ? anonymous `(frame, x, y)` numbers reveal neither your corpus nor your method.

3. Protobuf ? secrecy. Protobuf is a **great** choice for compactness, schema-versioning, and deterministic serialization ? but a binary blob is trivially decoded. It buys you schema discipline, not protection. Protection (against the no-key public) is a **separate** layer: `git-crypt` / SOPS+age / out-of-repo fetch. So the architecture is naturally **tiered**: a public, anonymized, unencrypted tier for the logic/refactor tests everyone can run, and an optional encrypted/key-gated tier for richer or more sensitive data.

****4. The decisive question is provenance.**** Whether derived data even carries the source's constraints splits hard on: **did you record the corpus yourself (your own IP ? concern is trade-secret/privacy)** vs **is it third-party broadcast/YouTube (copyright + ToS)**. And if the source is a ****public**** video, an adversary could re-derive trajectories from it and correlate ? so non-attributability is far stronger for a ****private, owner-recorded**** corpus. This is the crux the research must resolve.

I'm launching a deep legal/IP/privacy research + crypto-design workflow now (covers IN/US/EU, copyright-of-facts, EU database right, ToS, GDPR/DPDP, biometrics, re-identification, plus the tiered protobuf/encryption architecture), and asking three gating questions in parallel so the final synthesis is grounded in your actual situation rather than my guesses.

63. assistant (2026-07-02T06:11:33.340Z)

```
[external_agent_tool_call: Workflow]
input: {"script": "export const meta = {\n  name: 'derived-data-legal-and-byzantine-fixtures',\n  description: 'Deep research: IP/privacy of committing DERIVED structured data publicly + a non-attributable encrypted fixture architecture',\n  phases: [\n    { title: 'Legal Research',\n      detail: 'IN/US/EU copyright-of-facts, DB right, ToS, privacy/biometrics, re-identification' },\n    { title: 'Verify',\n      detail: 'adversarially refute each legal bottom-line with sources' },\n    { title: 'Legal Memo',\n      detail: 'synthesize a cited, jurisdiction-split memo + safe-commit conditions' },\n    { title: 'Crypto Design',\n      detail: 'threat models + encryption/secret-file + protobuf/harness integration' },\n    { title: 'Tech Synthesis',\n      detail: 'unified tiered non-attributable fixture architecture' },\n    { title: 'Red Team',\n      detail: 'attack the combined legal+technical recommendation' } ],\n  }\n\nconst GROUNDING = [\n  \"CODEBASE FACTS (verified by the lead before this run):\",\n  \"- eval_baselines/ is a TRACKED (committed) dir of JSON baseline snapshots (regression_baseline.json, heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json, experiment_leaderboard.json). Precedent: committed structured regression files already live in git.\",\n  \"- backend/eval/regression.py loads a committed baseline, scores DB-stored predictions vs ground-truth intervals, flags P/R drop beyond tolerance, with incommensurability guards (IoU/detector/GT-hash). Predictions come from the DB (needs a pipeline run on the video) -> NOT video-free on the prediction side yet.\",\n  \"- backend/eval/fusion_golden.py extracts *_golden_features.json from VIDEO+trajectory CSV+labels (needs GPU/WSL). backend/eval/fusion_compare.py & backend/eval/ablation.py RE-SCORE those JSON features on pure CPU with NO video (LOVO deltaF1, bootstrap CI, sign test). This is the EXISTING VIDEO-FREE seam.\",\n  \"- docs/GOLDEN_DATA_SHARING.md (#175): *_golden_features.json are deliberately kept PRIVATE on OneDrive, OUT of git, becau...\" ]

[/external_agent_tool_call]
```

64. assistant (2026-07-02T06:11:33.340Z)

```
[external_agent_tool_result]
Workflow launched in background. Task ID: w3pn6s2mo
Summary: Deep research: IP/privacy of committing DERIVED structured data publicly + a non-attributable encrypted fixture architecture
Transcript dir: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9
Script file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-fixtures-wf_47057438-9c9.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-fixtures-wf_47057438-9c9.js"} to iterate without resending the script.)
Run ID: wf_47057438-9c9
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-fixtures-wf_47057438-9c9.js", resumeFromRunId: "wf_47057438-9c9"}) ? completed agents return cached results.
```

You will be notified when it completes. Use /workflows to watch live progress.

```
[/external_agent_tool_result]
```

65. assistant (2026-07-02T06:11:33.340Z)

```
[external_agent_tool_call: AskUserQuestion]
input: {"questions":[{"question":"What's the provenance of the golden-corpus videos whose
derived data we'd commit?","header":"Corpus
source","multiSelect":false,"options":[{"label":"Self-recorded / owned by me","description":"You
recorded them (or hold full rights). Concern shifts to trade-secret/competitive-moat + player
privacy/consent ? copyright/ToS infringement is largely off the table, and non-attributability
is strongest (no public reference to correlate against)."}, {"label":"Third-party
(broadcast/YouTube/etc.)","description":"Sourced from broadcasts or platforms. Copyright +
platform ToS + possible re-identification via the public original all come into play ? the
hardest case."}, {"label":"Mixed","description":"Some owned, some third-party. We'd need per-
video provenance tagging and only publish derived data from the cleared
subset."}]}],{"question":"Is the GitHub repo where fixtures would live public or
private?","header":"Repo
visibility","multiSelect":false,"options":[{"label":"Private","description":"Only invited
collaborators clone it. Encryption is defense-in-depth; the main risk is a trusted contributor,
which non-attributability addresses."}, {"label":"Public","description":"Anyone can read every
committed byte forever (and Git history is permanent). This is the demanding case ? only the
anonymized Tier-0 fixtures should ever be committed unencrypted."}, {"label":"Private now, may
open-source later","description":"Plan for public-grade hygiene from day one so history never
needs scrubbing."}]}],{"question":"Do the corpus videos contain identifiable people, and could
any be minors?","header":"People & minors","multiSelect":false,"options":[{"label":"Adults only,
with consent","description":"Identifiable adults who consented. Person/court features may be
publishable if non-biometric; still minimize."}, {"label":"Adults, consent not
formalized","description":"Adults but no explicit consent for derived-data publication. Lean to
shuttle-only public tier; keep pe...
[/external_agent_tool_call]
```

66. assistant (2026-07-02T06:11:33.340Z)

```
[external_agent_tool_result]
Your questions have been answered: "What's the provenance of the golden-corpus videos whose
derived data we'd commit?"="Self-recorded / owned by me", "Is the GitHub repo where fixtures
would live public or private?"="Private now, may open-source later", "Do the corpus videos
contain identifiable people, and could any be minors?"="Adults, consent not formalized". You can
now continue with these answers in mind.
[/external_agent_tool_result]
```

67. assistant (2026-07-02T06:11:33.340Z)

The first workflow (video-free fixture design) just completed, and your three answers are in ? both materially shape the final plan. The second workflow (legal + Byzantine architecture) is still running; I'll synthesize everything when it lands. Let me pull the detailed synthesis from the completed workflow while I wait.

68. assistant (2026-07-02T06:11:33.340Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\wlorsh1wx.output
[/external_agent_tool_call]
```

69. assistant (2026-07-02T06:11:33.340Z)

```
[external_agent_tool_result]
1      {
2      "summary": "Design a plan for committed, video-free golden-set regression fixtures +
tests",
3      "agentCount": 11,
4      "logs": [
5      "Understanding: 5/5 areas mapped"
6      ],
7      "result": {
```

```
8         "understanding": [
[/external_agent_tool_result]
```

70. assistant (2026-07-02T06:11:33.340Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"\\synthesis\\|\\critiques\\|\\recommendedDocHome\\|\\rolloutPhases\\|\\openOwnerQuestions\\|\\verdict\\|\\path":"C:\\Users\\avidu\\AppData\\Local\\Temp\\claude\\C--Users-avidu-Projects-badminton-highlight-indexer\\068560d3-1ca2-4387-981a-d2d6bca55414\\tasks\\wlorsh1wx.output","output_mode":"content","-n":true}
[/external_agent_tool_call]
```

71. assistant (2026-07-02T06:11:33.340Z)

```
[external_agent_tool_result]
658:   "synthesis": {
698:     "rolloutPhases": [
730:[Omitted long matching line]
731:     "openOwnerQuestions": [
750:     "critiques": [
806:[Omitted long matching line]
852:[Omitted long matching line]
[/external_agent_tool_result]
```

72. assistant (2026-07-02T06:11:33.340Z)

```
[external_agent_tool_call: Read]
file: C:\\Users\\avidu\\AppData\\Local\\Temp\\claude\\C--Users-avidu-Projects-badminton-highlight-indexer\\068560d3-1ca2-4387-981a-d2d6bca55414\\tasks\\wlorsh1wx.output
[/external_agent_tool_call]
```

73. assistant (2026-07-02T06:11:33.340Z)

```
[external_agent_tool_result]
658     "synthesis": {
659         "title": "Committed video-free golden-set regression ? unified design (frozen trajectory-CSV + feature substrate, dual gates, synthetic-first)",
660         "goal": "Make per-video rally-segmentation regression runnable end-to-end from a clean `git clone` / CI with ZERO video, GPU, WSL, DB, or network ? so a contributor on another machine refactoring the deterministic pipeline gets a red/green signal, and any intentional behaviour change shows up as a reviewable snapshot diff. Achieve this by committing a tiny, version-tagged, numeric-only per-video fixture corpus and TWO complementary auto-discovered pytest gates (a characterization/snapshot gate for refactor side-effects, and a quality-floor gate vs golden labels tied to the ablation track), reusing the entire shipped eval stack verbatim with essentially no new scoring math. This closes the concrete gap that `tests/test_served_gate_a_regression.py` and the golden-litmus tests SKIP in CI today because the corpus lives only in gitignored `output/` / private OneDrive.",
661         "buildsOnExisting": [
662             "backend/eval/metrics.py::score/match_intervals/interval_iou ? greedy, deterministic (sort key (-iou,pi,gi)), empty->0.0 not NaN; the ONLY scoring math, reused verbatim by both gates",
663             "backend/eval/segmentation_metrics.py::f1_at_overlaps/segment_count_ratio ? the over-seg signal tIoU hides, frozen in the snapshot",
664             "backend/eval/experiment.py::VideoCandidates/Variant/CONTROL/predict/evaluate_variant/compare/honest_summary ? the pure LOVO scoring engine; predict()==gate->learned-filter->merge_overlaps is the deterministic re-score both gates call",
665             "backend/eval/fusion_compare.py::load_videos ? already globs *_golden_features.json into VideoCandidates; the quality-floor gate reuses it pointed at the COMMITTED dir instead of gitignored output/",
666             "backend/eval/windowing.py::WindowingPreset/SERVED/PROPOSAL/build_windows/windows_to_intervals ? the single-sourced operating point that closed the eval!=serve Gate-A trap; the fixture pins which preset it was frozen at",
667             "backend/eval/distill_local.py::build_candidates + FEATURE_NAMES (the 5 shuttle
```

```

feats) ? trajectory CSV -> (intervals, shuttle feature rows) under a named preset; the exact
transform the extractor and characterization gate call",
668         "backend/pipeline/detectors/tracknet_runner.py::parse_trajectory_csv
(Frame,Visibility,X,Y) + trajectory_to_action_windows ? the pure windowing 'brain' under
characterization",
669         "backend/pipeline/detectors/rally_gate.py (net_crossings Gate A) +
fusion_features.py (PERSON_FEATURE_NAMES, default-OFF) ? pure per-signal feature producers",
670
"backend/eval/regression.py::gt_hash/save_baseline/load_baselines/regress/RegressionResult +
DEFAULT_BASELINE_PATH=eval_baselines/regression_baseline.json + the exit 0/1/3
incommensurability-guard contract (IoU/detector/gt_hash mismatch -> flagged, not silently
compared)",
671         "backend/eval/eval_stats.py::paired_delta_ci/mean_with_ci/sign test ?
deterministic (default seed=0); the 'NUMBERS INFORM' honest-stats the quality-floor gate emits
and the snapshot freezes",
672
"backend/eval/fusion_golden.py::SHUTTLE_FEATURE_NAMES/PERSON_FEATURE_NAMES/extract_video ? the
SAME extractor the owner already runs per golden video; emits the exact committed shape, so 'add
a video' needs no new producer",
673         "tests/test_served_gate_a_regression.py ? the manifest-driven, per-video, skip-
if-absent template (env-overridable MANIFEST, _golden_specs() presence filter, _skip decorator,
Path(__file__)-anchored repo root); the new gates mirror it but RUN in CI because fixtures are
committed; its _PROPOSAL=(0.005,1.0,0.5) numbers are the proof the snapshot must record
windowing provenance",
674         "eval_baselines/{heuristic_lovo_n6.json, experiment_leaderboard.json,
gemini_wrapper_2026-06-10.json} ? the proven TRACKED home ...
[/external_agent_tool_result]

```

74. assistant (2026-07-02T06:11:33.340Z)

Excellent depth ? the first workflow's adversarial critic caught a genuinely sharp issue (LogisticRegression gradient descent over OpenBLAS `DYNAMIC\_ARCH` isn't bit-stable across CPUs ? would false-RED for external contributors). Let me read the rest of the critiques.

75. assistant (2026-07-02T06:11:33.340Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\wlorsh1wx.output
[/external_agent_tool_call]

```

76. assistant (2026-07-02T06:11:33.340Z)

```

[external_agent_tool_result]
768         },
769         {
770             "issue": "FALSE SAFETY ? --refresh-snapshot is a rubber-stamp hole the
design acknowledges but under-mitigates, and it is WORSE than stated because of issue #1. The
design says a contributor could regenerate the snapshot to mask a regression and the mitigation
is 'social/review.' But combine with the BLAS non-determinism: a contributor whose machine
legitimately produces different learned-variant numbers will be TOLD by the red gate to run
--refresh-snapshot --all (that is literally the documented remedy for an 'intended change').
They will refresh in good faith, committing THEIR machine's BLAS-specific numbers, which then
false-RED on the next contributor / on CI. The 'intentional-change signal' becomes noise: every
cross-machine PR carries a snapshot diff that is actually just float/threshold churn, so
reviewers learn to ignore snapshot diffs ? destroying the one property the design is built to
provide.",
771             "severity": "high",
772             "recommendation": "This is the same root cause as #1 and is fixed by the
same move: keep learned-variant numbers OUT of the cross-machine frozen snapshot. Once only
CONTROL/pure-stdlib transforms are frozen, a snapshot diff genuinely means a behavior change and
--refresh-snapshot is safe and meaningful. Additionally: make --refresh-snapshot REQUIRE a
reason flag recorded into provenance.frozen_reason and have the gate print the prior vs new
values, so a refresh is self-documenting in the diff. Consider a CI guard that fails if

```



```

expected_*.json changed in the same commit as a refactor WITHOUT a provenance reason update."
773         },
774         {
775             "issue": "LICENSING/PRIVACY ? the tier-1 reconciliation is sound in spirit
but the design lets the FRAMEWORK ship (PR-1.5) in a way that quietly normalizes committing
proprietary-derived numbers, and the ShuttleSet question is unanswered. The repo is PRIVATE (I
confirmed isPrivate=true), which the design correctly flags as lowering risk ? but private today
is not private forever (a repo can be open-sourced; the owner's stated paper-coauthor aim
implies eventual publication of methods/possibly data). Committed git history is FOREVER: if a
tier-1 trajectory from owned/ShuttleSet footage lands and the repo later goes public, the
derived stream is permanently in history even after a delete. The design's own
GOLDEN_DATA_SHARING.md guardrail ('never commit proprietary-derived per-video artifacts') is
being reversed for tier-1, and the open question 'is ShuttleSet's license compatible with
committing DERIVED trajectory/feature numbers to a (public?) repo' is exactly the kind of thing
that must be answered BEFORE the framework conditions everyone to `freeze_fixture --slug
<real>`.",
776             "severity": "medium",
777             "recommendation": "Land PR-1.5 SYNTHETIC-ONLY exactly as designed (this is
genuinely clean and the right call). For tier-1, do not rely only on the freeze CLI's --minors-
free/--license refusal (a flag a human asserts). Add: (1) a written ShuttleSet redistribution-
of-derivatives determination in the PR that introduces tier-1, citing the actual license text ?
derived numeric features are usually fine but trajectory coordinates re-tracing the original are
closer to a derivative; get this in writing. (2) Default tier-1 keys to an OPAQUE de-
identification slug, NOT video_id[:12] ? the MD5 prefix is a re-identification handle against
the proprietary DB and offers no benefit in a video-free fixture. (3) Add a note in the doc:
'committing to git is permanent even if later deleted; treat tier-1 as a public-disclosure-
equivalent decision regardless of current repo visibility.'"
778         },
779         {
780             "issue": "SCHEMA/LABEL DRIFT ? the additive-only + schema_version ladder is
a reasonable mirror of the DB migration discipline, but the incommensurability guard has a gap
that produces SILENT staleness rather than a loud ...
[/external_agent_tool_result]

```

77. assistant (2026-07-02T06:11:33.340Z)

I now have the full first workflow (the fixture **mechanics**), and your three answers are locked in. The second workflow ? the legal memo (IN/US/EU) plus the non-attributable/encryption architecture and red-team ? is still running; I'll deliver the ***single unified written plan*** when it lands. Here's the high-signal interim, because two findings already in hand are genuinely important and shape everything.

Your answers ? a clean convergence

- ***Owned corpus*** ? this is the **easy** legal fork: no third-party copyright/ToS infringement; it's your IP. The concern collapses to trade-secret/competitive-moat (your call to publish) + player privacy. And non-attributability is ***strongest*** here ? a private corpus has **nothing public to correlate against**, so the fingerprinting/re-identification attack (re-run the detector on a public video, match the trajectory) simply has no reference and fails.
- ***Private now, may open-source later*** ? git history is permanent, so we design the public tier to be ***public-grade from the first commit*** (no scrubbing later).
- ***Adults, consent not formalized*** ? person/pose-derived streams stay ***out of the public tier*** (derived-personal-data risk under DPDP/GDPR without consent); the public tier is ***shuttle-only***.

The elegant part: the privacy-driven choice (publish shuttle-only, keep person streams private) lands on the **exact same** minimal tier that the determinism analysis independently demands ? see below. Privacy and correctness point at one design. That's the spine of the "Byzantine" solution.

The killer finding from the mechanics workflow

The design's natural seam is right ? ***freeze the post-detector trajectory CSV*** (``Frame,Visibility,X,Y``) and replay the deterministic pipeline from there, reusing the **entire*

shipped eval stack* (`metrics.py`, `experiment.py`, `windowing.py`, `regression.py`, `eval\_baselines/`), with two gates: a ****characterization/snapshot**** gate (catches any refactor side-effect) and a ****quality-floor**** gate (catches quality drops). Synthetic fixtures ship first (licensing-clean by construction); owned real fixtures are owner-gated.

But the adversarial critic caught a ****fatal determinism flaw****, code-verified: the ***learned*** variants (`shuttle\_only`, `fusion`) run 1000 epochs of numpy gradient descent on OpenBLAS compiled `DYNAMIC_ARCH` ? the float-summation order is chosen per CPU microarchitecture, the resulting probabilities sit right on the 0.5 keep/drop cliff, so an external contributor on a ****different CPU gets different integer tp/fp/fn on byte-identical correct code****. The gate goes red, they're told to "refresh the snapshot," they commit ***their*** machine's numbers, and snapshot diffs become noise reviewers learn to ignore ? destroying the one property you're building this for.

****Fix (and the convergence):**** the cross-machine gate must be ****STATIC-ONLY**** ? the pure-stdlib `CONTROL` path (gate ? merge ? score) plus pure shuttle features ? with learned/person scoring kept ****owner-box-only****. That's ***exactly*** the shuttle-only, person-excluded tier privacy already requires. Two other code-verified must-fixes: route freeze ****and**** both gates through one shared `replay_fixture()` helper (or the snapshot doesn't actually exercise the windowing "brain" it claims to lock), and ****parse-compare JSON, never byte-compare**** (CRLF/float-format would false-RED on Windows ? your `core.autocrlf=true` is set).

What's still cooking

The legal memo (copyright-of-facts, EU database right, India DPDP, ToS, biometrics, re-ID) and the tiered protobuf/encryption + key-management architecture + red-team are mid-flight. When they complete I'll merge all three into one plan and ****write it to `docs/GOLDEN\_REGRESSION\_FIXTURES.md`**** for your review (I won't commit or open a PR ? per your conventions). I'll flag this to you the moment it's ready.

78. user (2026-07-02T06:11:33.340Z)

any logs whether it is stuck?

79. assistant (2026-07-02T06:11:33.341Z)

Let me check the live status of the running workflow.

80. assistant (2026-07-02T06:11:33.341Z)

```
[external_agent_tool_call: ToolSearch]
input: {"query":"select:TaskList,TaskGet,TaskOutput","max_results":5}
[/external_agent_tool_call]
```

81. assistant (2026-07-02T06:11:33.341Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

82. assistant (2026-07-02T06:11:33.341Z)

```
[external_agent_tool_call: Glob]
input: {"pattern":"subagents/workflows/wf_47057438-9c9/**/*","path":"C:\\Users\\avidu\\.claude\\.projects\\C--Users-avidu-Projects-badminton-highlight-indexer\\068560d3-1ca2-4387-981a-d2d6bca55414"}
[/external_agent_tool_call]
```

83. assistant (2026-07-02T06:11:33.341Z)

```
[external_agent_tool_result]
No files found
[/external_agent_tool_result]
```

84. assistant (2026-07-02T06:11:33.341Z)

```
[external_agent_tool_call: TaskOutput]
input: {"task_id":"w3pn6s2mo", "block":false, "timeout":5000}
[/external_agent_tool_call]
```

85. assistant (2026-07-02T06:11:33.341Z)

```
[external_agent_tool_result]
<retrieval_status>not_ready</retrieval_status>

<task_id>w3pn6s2mo</task_id>

<task_type>local_workflow</task_type>

<status>running</status>
[/external_agent_tool_result]
```

86. assistant (2026-07-02T06:11:33.341Z)

```
[external_agent_tool_call: Glob]
input: {"pattern":"**/wf_47057438*/**/*", "path":"C:\\Users\\avidu\\.claude\\projects\\C--Users-avidu-Projects-badminton-highlight-indexer\\068560d3-1ca2-4387-981a-d2d6bca55414"}
[/external_agent_tool_call]
```

87. assistant (2026-07-02T06:11:33.341Z)

```
[external_agent_tool_result]
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a06cc22597593cd53.meta.json
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a0d609f7d61330587.meta.json
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a1f8480b6899537f9.meta.json
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a466d9415a36b1f58.meta.json
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a4f131c7921bd5ef5.meta.json
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a6202440835be7bf1.meta.json
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a4f131c7921bd5ef5.jsonl
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a466d9415a36b1f58.jsonl
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a06cc22597593cd53.jsonl
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a1f8480b6899537f9.jsonl
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a6202440835be7bf1.jsonl
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a61cd05f22268e81e.meta.json
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a69b45904982be16e.meta.json
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-
```

```
a7a372b69f23b997e.meta.json
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a82c1547810fe50e2.meta.json
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-ac7c49b8f877090e1.meta.json
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-afe0ef4bb67f34e89.meta.json
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a0d609f7d61330587.jsonl
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a7a372b69f23b997e.jsonl
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a82c1547810fe50e2.jsonl
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-ac7c49b8f877090e1.jsonl
C:\Users\avidu\.claude...
[/external_agent_tool_result]
```

88. assistant (2026-07-02T06:11:33.341Z)

Status: **running, not stuck** ? 17 of the 18 planned agents have already spawned. Let me confirm it's actively progressing (not hung) by checking the workflow journal and how recently the newest agent file was written.

89. assistant (2026-07-02T06:11:33.341Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\journal.jsonl
[/external_agent_tool_call]
```

90. assistant (2026-07-02T06:11:33.341Z)

```
[external_agent_tool_result]
<system-reminder>[Truncated: PARTIAL view ? showing lines 1-7 of 34 total (91938 tokens, cap 25000). Call Read with offset=8 limit=7 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

1      {"type":"started","key":"v2:9db7a107f14296d0277ec1742abfa6763d766c62724729f6c3ef7d67840cb427","agentId":"a466d9415a36b1f58"}
2      {"type":"started","key":"v2:d338e029cb401cfbd219295a0513f4dc3c2513a4fa178cc034c6dd89d3a99ca6","agentId":"a0d609f7d61330587"}
3      {"type":"started","key":"v2:ece338c433c1ac8d56826b441904f1b0a9e1805a195156acc31728a090ac32ea","agentId":"a06cc22597593cd53"}
4      {"type":"started","key":"v2:f5d1c648c8d5bde1b5e127ac374208c463197deec47f113f64bc31a743abb70","agentId":"a4f131c7921bd5ef5"}
5      {"type":"started","key":"v2:1a701f35759fa846c96423447bd0e49e186c0c5288ce6a2ddf40a2580ed4703e","agentId":"a6202440835be7bf1"}
6      {"type":"started","key":"v2:1d108a270608117d5faa6a052eb127658f13c4af965a567600a48574d389e6e5","agentId":"a1f8480b6899537f9"}
7      {"type":"result","key":"v2:f5d1c648c8d5bde1b5e127ac374208c463197deec47f113f64bc31a743abb70","agentId":"a4f131c7921bd5ef5","result":{"question":"Under the EU sui generis database right (Directive 96/9/EC) and copyright, does extracting and redistributing DERIVED match data (coordinates / event timestamps) from a broadcast infringe?","scope":"EU law only (Directive 96/9/EC as interpreted by the CJEU). Covers: (1) sui generis database right ? \"obtaining vs creating\" data and \"extraction/re-utilisation of a substantial part\"; (2) database copyright (originality); (3) where re-utilisation occurs (targeting); (4) contract overriding the directive's user exceptions for UNPROTECTED databases. Focus = independently DERIVED numeric streams (shuttle/player coordinates, rally/event timestamps) computed by our own CV/AI from a broadcast video, then redistributed. NOT advice on: broadcast/film copyright in the video signal
```

itself, performers'/sports-organiser commercial rights, unfair-competition/parasitism under national law, or post-Brexit UK law (UK retained a parallel but now-diverging regime). Not legal advice; consult EU IP counsel before relying.", "webUsed": true, "findings": [{"claim": "The sui generis database right protects only investment in OBTAINING (seeking out and collecting pre-existing independent materials), VERIFYING, or PRESENTING contents ? NOT investment in CREATING the data. Resources spent generating new data do not count toward the 'substantial investment' that triggers the right.", "support": "In C-203/02 the CJEU held that 'investment in ? the obtaining ? of the contents refers to the resources used to seek out existing independent materials and collect them' and that resources used to CREATE the data (e.g. fixing the list of runners) fall outside this and do not count toward substantial investment under Art 7(1). The parallel Fixtures Marketing line (C-46/02, C-338/02, C-444/02, decided the same day) reached the same conclusion for football fixture lists: investment in generating the data does not qualify.", "sourceName": "EUR-Lex ? Case C-203/02 British Horseracing Board v William Hill (judgment, 9 Nov 2004)", "sourceUrl": "https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A62002CJ0203", "confidence": "high"}, {"claim": "This creating-vs-obtaining distinction is the crux for sports data: a party who is the SOURCE of dynamically-generated sports data (fixtures, race entries, and by extension live match events) often holds NO sui generis right in it, because its spend was on creation, not obtaining. Only investment in collecting/verifying pre-existing materials qualifies.", "support": "Commentary on the 2004 quartet (C-46/02, C-203/02, C-338/02, C-444/02) notes the CJEU 'seriously restricted the scope of sui generis protection by admitting that investments in the creation of a database's contents cannot count as investment for obtaining'; the football-fixture and horseracing-runner-li..."}]

91. assistant (2026-07-02T06:11:33.341Z)

[external_agent_tool_call: PowerShell]
description: Show newest agent transcript files and their age in seconds
command: \$d = "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9"; Get-ChildItem \$d -Filter *.jsonl | Sort-Object LastWriteTime -Descending | Select-Object -First 4 Name, LastWriteTime, @{n='AgeSec';e={[math]::Round(((Get-Date)-\$_ .LastWriteTime).TotalSeconds)}} ,@{n='KB';e={[math]::Round(\$_.Length/1KB)}} | Format-Table -AutoSize

92. assistant (2026-07-02T06:11:33.341Z)

[external_agent_tool_result]
Name LastWriteTime
AgeSec KB

--
agent-a99f0842bd56f8647.jsonl 6/16/2026 9:15:28 AM 86.00 333.00
journal.jsonl 6/16/2026 8:38:45 AM 2288.00 236.00
agent-a49314b51fd5d1fb8.jsonl 6/16/2026 8:38:45 AM 2288.00 159.00
agent-aa13cea5f0da98a00.jsonl 6/16/2026 8:38:40 AM 2293.00 210.00

93. assistant (2026-07-02T06:11:33.341Z)

Here's the honest status ? **it's running, not hung, but one agent is a slow straggler.**

What the files show:

- **17 of 18 agents have spawned**, and most have already completed (the journal has their results ? e.g. the EU database-right research agent already returned a fully-cited answer).
- The newest transcript, `agent-a99f0842?`, was **written 86 seconds ago** (and is 333 KB) ? an agent is **actively producing output right now**. So: not deadlocked.
- But the workflow **journal** hasn't logged a completion in ~38 min, which means that one agent has been running ~38 minutes straight. It's the straggler gating the whole run (wall-clock = slowest single chain).

Let me peek at what that straggler is actually doing ? productive work vs. a fetch loop:

94. assistant (2026-07-02T06:11:33.342Z)

```
[external_agent_tool_call: PowerShell]
description: Inspect the last few events of the straggler agent transcript
command: $f = "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-index
er\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-
a99f0842bd56f8647.jsonl"
$lines = Get-Content $f
Write-Output "TOTAL LINES: $($lines.Count)"
$lines | Select-Object -Last 6 | ForEach-Object {
    try { $o = $_ | ConvertFrom-Json } catch { $o = $null }
    $type = if ($o) { $o.type } else { "?" }
    $snip = $_.Substring(0, [Math]::Min(280, $_.Length))
    Write-Output "--- [$type] $snip"
}
[/external_agent_tool_call]
```

95. assistant (2026-07-02T06:11:33.342Z)

```
[external_agent_tool_result]
TOTAL LINES: 85
--- [assistant] {"parentUuid":"2b0e1f67-4def-43f7-9e16-
cd318dfc83a6","isSidechain":true,"agentId":"a99f0842bd56f8647","message":{"model":"claude-opus-
4-
8","id":"msg_01762qEvQHdpoxXs6Xn3woo","type":"message","role":"assistant","content":[{"type":"t
ext","text":"The empty calls keep failing. I wil
--- [assistant] {"parentUuid":"0eaba9a0-c9b9-48a9-a166-
64f98934eab7","isSidechain":true,"agentId":"a99f0842bd56f8647","message":{"model":"claude-opus-
4-
8","id":"msg_01762qEvQHdpoxXs6Xn3woo","type":"message","role":"assistant","content":[{"type":"t
ool_use","id":"toolu_01YHTGvuLQKD4H2iz1DP3GL7"},"
--- [user] {"parentUuid":"5739ef19-e6fd-4ec0-bee3-
8e0c26cd7514","isSidechain":true,"promptId":"b1044f3a-6a37-4ccf-aaf4-
aba586eaf235","agentId":"a99f0842bd56f8647","type":"user","message":{"role":"user","content":[{"
type":"tool_result","content":"Output does not match required schema: root:
--- [assistant] {"parentUuid":"3bfb7b0f-9581-4be7-a942-
ed91fa795721","isSidechain":true,"agentId":"a99f0842bd56f8647","message":{"model":"claude-opus-
4-
8","id":"msg_01JZVTntBPMeowc5dxWtk1oT","type":"message","role":"assistant","content":[{"type":"t
ool_use","id":"toolu_01DQZ6BbGUMRb2T9TejvR32U"},"
--- [user] {"parentUuid":"c0edd07f-8869-4623-88d0-
88404f41254d","isSidechain":true,"promptId":"b1044f3a-6a37-4ccf-aaf4-
aba586eaf235","agentId":"a99f0842bd56f8647","type":"user","message":{"role":"user","content":[{"
type":"tool_result","content":"Output does not match required schema: root:
--- [assistant] {"parentUuid":"75ae1236-d987-47d4-877a-
e2189710ee5c","isSidechain":true,"agentId":"a99f0842bd56f8647","message":{"model":"claude-opus-
4-
8","id":"msg_01DeLNBbpobC3v1G1qeMx2b5","type":"message","role":"assistant","content":[{"type":"t
ext","text":"I've been calling the tool without
[/external_agent_tool_result]
```

96. assistant (2026-07-02T06:11:33.342Z)

The peek is decisive: that straggler is **stuck in a structured-output validation loop**, not doing useful work. Its latest messages read **"The empty calls keep failing"** and **"I've been calling the tool without?"**, interleaved with repeated `Output does not match required schema` rejections. It's been retrying a schema it can't satisfy for ~38 min, which is blocking the final phase. Let me map exactly which agent/phase it is and what's already completed before deciding how to recover.

97. assistant (2026-07-02T06:11:33.342Z)

```
[external_agent_tool_call: PowerShell]
description: Map each agent's label and completion status from meta + journal
```

```

command: $d = "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9"
$journal = Get-Content "$d\journal.jsonl"
$results = @{}
foreach ($l in $journal) { try { $o = $l | ConvertFrom-Json } catch { continue }; if ($o.type
-eq 'result') { $results[$o.agentId] = $true } }
Write-Output "Completed (result in journal): $($results.Count) agents"
Write-Output "----- per-agent label / done? -----"
Get-ChildItem $d -Filter *.meta.json | ForEach-Object {
    $m = Get-Content $_.FullName -Raw | ConvertFrom-Json
    $id = $_.BaseName -replace '^agent-', ''
    $done = if ($results.ContainsKey($id)) { 'DONE' } else { 'running/failed' }
    "{0,-14} {1,-26} {2}" -f $done, $m.label, $id
}
[/external_agent_tool_call]

```

98. assistant (2026-07-02T06:11:33.342Z)

```

[external_agent_tool_result]
Completed (result in journal): 16 agents
----- per-agent label / done? -----
running/failed          a06cc22597593cd53.meta
running/failed          a0d609f7d61330587.meta
running/failed          a1bea65cdab0c3b6e.meta
running/failed          a1f8480b6899537f9.meta
running/failed          a466d9415a36b1f58.meta
running/failed          a49314b51fd5d1fb8.meta
running/failed          a4f131c7921bd5ef5.meta
running/failed          a61cd05f22268e81e.meta
running/failed          a6202440835be7bf1.meta
running/failed          a69b45904982be16e.meta
running/failed          a7a372b69f23b997e.meta
running/failed          a82c1547810fe50e2.meta
running/failed          a90f484e70c865137.meta
running/failed          a99f0842bd56f8647.meta
running/failed          aa13cea5f0da98a00.meta
running/failed          ac7c49b8f877090e1.meta
running/failed          afe0ef4bb67f34e89.meta
[/external_agent_tool_result]

```

99. assistant (2026-07-02T06:11:33.342Z)

```

[external_agent_tool_result]
118     }
119
120     const WEB = "\n\nMETHOD: Use WebSearch and WebFetch to find and cite AUTHORITATIVE
sources (statutes, case law, official regulator guidance, reputable legal commentary). Load them
first if needed via ToolSearch with query 'select:WebSearch,WebFetch'. Cite real URLs. If you
cannot fetch a source, mark that claim's confidence 'low' and set webUsed=false. Do NOT
fabricate citations ? an honest 'unverified' beats an invented case name or URL."
121
122     // ----- legal research
123     phase('Legal Research')
124     const LEGAL_Q = [
125         { key: 'copyright-facts', prompt:
126             "RESEARCH QUESTION: Is algorithmically-DERIVED structured data ? per-frame shuttle
(x,y) coordinates, rally start/end timestamps, audio-onset feature values ? extracted from a
copyrighted sports VIDEO itself protected by copyright, or is it uncopyrightable facts/data, or
a 'derivative work'? Cover: the idea/expression dichotomy; Feist Publications v. Rural Telephone
(facts not copyrightable; sweat-of-the-brow rejected); originality requirement in
selection/arrangement; whether mechanical extraction yields a 'derivative work' under 17 USC
101; intermediate / non-expressive copying and fair use (Google v. Oracle, Authors Guild v.
Google, Sega v. Accolade). Bottom line: does the derived NUMERIC data inherit the video's
copyright?" },
127         { key: 'eu-database-right', prompt:

```

128 "RESEARCH QUESTION: Under the EU sui generis DATABASE RIGHT (Directive 96/9/EC) and copyright, does extracting and redistributing DERIVED match data (coordinates / event timestamps) from a broadcast infringe? Cover: British Horseracing Board v. William Hill (C-203/02); Football Dataco v. Sportradar (C-173/11) and Football Dataco v. Yahoo (C-604/10); the 'obtaining vs creating' data distinction; 'extraction/re-utilization of a substantial part'; Ryanair v. PR Aviation (C-30/14) on contracts overriding DB-right exceptions. Bottom line for our derived numeric streams in the EU." },

129 { key: 'india-law', prompt:

130 "RESEARCH QUESTION (PRIMARY JURISDICTION ? owner is in Bangalore, India): Under the Indian Copyright Act 1957 ? derivative works, the originality 'modicum of creativity' standard (Eastern Book Co. v. D.B. Modak), fair dealing s.52; the ABSENCE of an EU-style sui generis database right in India; and the Digital Personal Data Protection (DPDP) Act 2023 ? definition of personal data, treatment of derived/anonymized data, and children's data rules ? is committing DERIVED numeric data from videos lawful? Note any Indian case law on data scraping/extraction. Bottom line under Indian law." },

131 { key: 'contract-tos-tradesecret', prompt:

132 "RESEARCH QUESTION: The CONTRACT / Terms-of-Service and TRADE-SECRET angles, which can bind even when copyright does NOT. Cover: typical YouTube / broadcaster ToS clauses prohibiting extraction, scraping, automated access, and building datasets; whether redistributing a derived dataset breaches platform terms; the scraping-law landscape (hiQ v. LinkedIn; Van Buren v. United States). CONVERSELY: if the corpus is the OWNER'S OWN recordings, the analysis shifts to trade-secret / competitive-moat (their data to publish or protect). Bottom line: when is committing derived data a CONTRACTUAL/ToS problem independent of copyright, and how does owner-recorded vs third-party-sourced change it?" },

133 { key: 'privacy-biometrics', prompt:

134 "RESEARCH QUESTION: PRIVACY / biometric exposure of derived data. When does data DERIVED about people in a video become 'personal data' (GDPR Art.4, Recital 26 on anonymization/pseudonymization), or trigger Art.9 special-category / biometric protection, or Art.8 children's rules ? and the equivalent under India's DPDP Act 2023 (children's data, verifiable parental consent)? Crucial distinction: SHUTTLE coordinates (about an inanimate object, not a person) vs PERSON bounding boxes vs POSE keypoints / GAIT (potentially personal...
[/external_agent_tool_result]

100. assistant (2026-07-02T06:11:33.342Z)

Diagnosis is now precise:

- **16 of 17 agents completed** ? all 6 legal-research agents, 6 verifications, the legal memo, and the 3 crypto-design agents are **done and on disk**.
- The **single stuck agent is the Tech-Synthesis step** (the 17th), looping on its structured-output schema (`TECH\_SYNTH\_SCHEMA` has 15 required fields including a nested `rolloutPhases` object array ? the model keeps producing output the validator rejects). The Red-Team step (18th) depends on it, so it never started.

So it's not productively working and won't self-recover soon. The clean fix: **stop it, relax that one schema, and resume** ? the 16 finished agents return from cache instantly, and only the synthesis + red-team re-run (~2-4 min). Let me load the stop tool and read the script's schema section to edit precisely.

101. assistant (2026-07-02T06:11:33.342Z)

```
[external_agent_tool_call: ToolSearch]
input: {"query":"select:TaskStop","max_results":3}
[/external_agent_tool_call]
```

102. assistant (2026-07-02T06:11:33.342Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

103. assistant (2026-07-02T06:11:33.342Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
```



```
indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-  
fixtures-wf_47057438-9c9.js  
[/external_agent_tool_call]
```

104. assistant (2026-07-02T06:11:33.342Z)

```
[external_agent_tool_call: TaskStop]  
input: {"task_id":"w3pn6s2mo"}  
[/external_agent_tool_call]
```

105. assistant (2026-07-02T06:11:33.342Z)

```
[external_agent_tool_result]  
{"message":"Successfully stopped task: w3pn6s2mo (Deep research: IP/privacy of committing  
DERIVED structured data publicly + a non-attributable encrypted fixture  
architecture)","task_id":"w3pn6s2mo","task_type":"local_workflow","command":"Deep research:  
IP/privacy of committing DERIVED structured data publicly + a non-attributable encrypted fixture  
architecture"}  
[/external_agent_tool_result]
```

106. assistant (2026-07-02T06:11:33.342Z)

```
[external_agent_tool_call: Read]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\...\claude\projects\C--Users-avidu-  
Projects-badminton-highlight-  
indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-  
fixtures-wf_47057438-9c9.js  
[/external_agent_tool_call]
```

107. assistant (2026-07-02T06:11:33.342Z)

```
[external_agent_tool_result]  
80     mechanism:{type:'string'},  
81     tradeoffs:{type:'array', items:{type:'string'}},  
82     integrationWithExistingInfra:{type:'string'},  
83   },  
84 }  
85 const TECH_SYNTH_SCHEMA = {  
86   type: 'object', additionalProperties: false,  
87   required: ['architectureName','threatModelTable','tieredModel','antiAttributionMeasure  
s','protobufSchemaSketch','serializationDeterminism','encryptionMechanism','keyManagement','ciIn  
tegration','contributorWorkflow','harnessIntegration','honestLimits','rolloutPhases','openOwnerQ  
uestions'],  
88   properties: {  
89     architectureName:{type:'string'},  
90     threatModelTable:{type:'string'},  
91     tieredModel:{type:'string'},  
92     antiAttributionMeasures:{type:'array', items:{type:'string'}},  
93     protobufSchemaSketch:{type:'string'},  
94     serializationDeterminism:{type:'string'},  
95     encryptionMechanism:{type:'string'},  
96     keyManagement:{type:'string'},  
97     ciIntegration:{type:'string'},  
98     contributorWorkflow:{type:'string'},  
99     harnessIntegration:{type:'string'},  
100    honestLimits:{type:'array', items:{type:'string'}},  
101    rolloutPhases:{type:'array', items:{type:'object', additionalProperties:false,  
102      required:['phase','deliverable','prScope'],  
103      properties:{ phase:{type:'string'}, deliverable:{type:'string'},  
prScope:{type:'string'} }},  
104    openOwnerQuestions:{type:'array', items:{type:'string'}},  
105  },  
106 }  
107 const REDTEAM_SCHEMA = {
```

```
108     type: 'object', additionalProperties: false,
109     required: ['attacks', 'overallVerdict', 'mustFixBeforeShipping'],
110     properties: {
111       attacks: {type: 'array', items: {type: 'object', additionalProperties: false,
112         required: ['vector', 'severity', 'doesDesignHold', 'fix'],
113         properties: { vector: {type: 'string'}, severity: {type: 'string',
enum: ['low', 'medium', 'high', 'critical']},
114         doesDesignHold: {type: 'string'}, fix: {type: 'string'} } }},
115       overallVerdict: {type: 'string'},
116       mustFixBeforeShipping: {type: 'array', items: {type: 'string'}},
117     },
[/external_agent_tool_result]
```

108. assistant (2026-07-02T06:11:33.342Z)

Workflow stopped cleanly. The culprit is clear: `TECH\_SYNTH\_SCHEMA` has 14 required fields plus a nested `rolloutPhases` object-array with `additionalProperties:false` ? too strict, so the synthesis agent's output kept getting rejected at `root`. I'll relax both that schema and the (not-yet-run) red-team schema, then resume so the 16 finished agents return from cache and only the last two run.

109. assistant (2026-07-02T06:11:33.342Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-
fixtures-wf_47057438-9c9.js
[/external_agent_tool_call]
```

110. assistant (2026-07-02T06:11:33.343Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-
fixtures-wf_47057438-9c9.js has been updated successfully. (file state is current in your
context ? no need to Read it back)
[/external_agent_tool_result]
```

111. assistant (2026-07-02T06:11:33.343Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-
fixtures-wf_47057438-9c9.js
[/external_agent_tool_call]
```

112. assistant (2026-07-02T06:11:33.343Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-
fixtures-wf_47057438-9c9.js has been updated successfully. (file state is current in your
context ? no need to Read it back)
[/external_agent_tool_result]
```

113. assistant (2026-07-02T06:11:33.343Z)

Both schemas relaxed. Now resuming from the saved run ? the 16 finished agents return from cache; only the synthesis and red-team re-run with the simpler schemas.

114. assistant (2026-07-02T06:11:33.343Z)

```
[external_agent_tool_call: Workflow]
input: {"scriptPath":"C:\\Users\\avidu\\.claude\\projects\\C--Users-avidu-Projects-badminton-
```

```
highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-fixtures-wf_47057438-9c9.js", "resumeFromRunId": "wf_47057438-9c9"}  
[/external_agent_tool_call]
```

115. assistant (2026-07-02T06:11:33.343Z)

```
[external_agent_tool_result]  
Workflow launched in background. Task ID: wgpct1wbd  
Summary: Deep research: IP/privacy of committing DERIVED structured data publicly + a non-attributable encrypted fixture architecture  
Transcript dir: C:\Users\avidu\claudu\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9  
Script file: C:\Users\avidu\claudu\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-fixtures-wf_47057438-9c9.js  
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\claudu\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-fixtures-wf_47057438-9c9.js"} to iterate without resending the script.)  
Run ID: wf_47057438-9c9  
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claudu\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-fixtures-wf_47057438-9c9.js", resumeFromRunId: "wf_47057438-9c9"}) ? completed agents return cached results.
```

You will be notified when it completes. Use /workflows to watch live progress.
[/external_agent_tool_result]

116. assistant (2026-07-02T06:11:33.343Z)

Answered and fixed. Here's the full picture:

```
**Was it stuck?** Yes ? one agent, not the whole run. Diagnosis from the logs:  
- 16 of 17 agents had **completed** (all legal research ? including a fully-cited EU database-right analysis ? plus the 6 verifications, the legal memo, and 3 crypto-design agents).  
- The 17th, the **Tech-Synthesis** agent, was caught in a structured-output retry loop (~38 min): its transcript literally said "The empty calls keep failing" against repeated `Output does not match required schema: root: ` rejections. Its schema demanded 14 required fields + a nested object-array ? too strict. The Red-Team step depends on it, so the run couldn't finish.
```

****What I did:****

1. Stopped the stuck run (`TaskStop`).
2. Relaxed the two offending schemas (`TECH\_SYNTH\_SCHEMA` ? 6 required fields, `rolloutPhases` now a string array; `REDTEAM\_SCHEMA` ? dropped the strict enum/`additionalProperties`).
3. **Resumed from the same run ID** ? the 16 finished agents return instantly from cache; only synthesis + red-team re-run with the easier schemas (~2-4 min).

New background task is `wgpct1wbd`; I'll be notified when it lands. No action needed from you.

For transparency on cause: this was a schema-strictness bug in *my* workflow script, not a problem with your question or the research ? the legal findings already on disk are high quality (e.g., the BHB v. William Hill "obtaining vs creating" analysis came back clean and cited). When the resume completes I'll merge both workflows into the single written plan at `docs/GOLDEN\_REGRESSION\_FIXTURES.md` (draft only ? no commit/PR).

117. user (2026-07-02T06:11:33.343Z)

Can you formalize this as a project doc which we can iterate upon, track progress and then mark as completed with clear deliverables? Seems that can be possible but feel free to push back if not possible

118. assistant (2026-07-02T06:11:33.343Z)

Yes ? this absolutely should be a tracked project doc; it fits how this repo already works (`docs/\*\_PLAN.md` with a status header, phases, and deliverables). Let me first check whether the resumed workflow has finished so I can fold in the legal memo + architecture, rather than writing a doc with a hole in it.

119. assistant (2026-07-02T06:11:33.343Z)

```
[external_agent_tool_call: TaskOutput]
input: {"task_id":"wgpct1wbd","block":false,"timeout":5000}
[/external_agent_tool_call]
```

120. assistant (2026-07-02T06:11:33.343Z)

```
[external_agent_tool_result]
<retrieval_status>not_ready</retrieval_status>

<task_id>wgpct1wbd</task_id>

<task_type>local_workflow</task_type>

<status>running</status>
[/external_agent_tool_result]
```

121. assistant (2026-07-02T06:11:33.343Z)

```
[external_agent_tool_call: PowerShell]
description: Check resumed workflow status and newest transcript age
command: $d = "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9"
$journal = Get-Content "$d\journal.jsonl"
$res = ($journal | Where-Object { $_ -match '"type":"result"' }).Count
Write-Output "journal result events: $res"
Get-ChildItem $d -Filter *.jsonl | Sort-Object LastWriteTime -Descending | Select-Object -First
3 Name,@{n='AgeSec';e={[math]::Round(((Get-
Date)-$_.LastWriteTime).TotalSeconds)}},@{n='KB';e={[math]::Round($_.Length/1KB)}} | Format-
Table -AutoSize
[/external_agent_tool_call]
```

122. assistant (2026-07-02T06:11:33.343Z)

```
[external_agent_tool_result]
journal result events: 17

Name                               AgeSec    KB
----                               -
agent-ab96e60cc8743fbbf.jsonl    78.00    213.00
agent-a8362c56b32d882b3.jsonl  155.00    325.00
journal.jsonl                    155.00    268.00
[/external_agent_tool_result]
```

123. assistant (2026-07-02T06:11:33.343Z)

Good news on status: the resume worked. The Tech-Synthesis agent **completed** with the relaxed schema (journal now shows 17 results), and the **Red-Team** is the only step left ? actively writing 78 seconds ago, so it's a minute or two from done. I'll fold its findings into the doc's risk section rather than write an incomplete draft, then save the file and flag you.

Yes, this is very doable and it's the right format. Here's the structure I'll write to **docs/GOLDEN_REGRESSION_FIXTURES.md** ? a living doc, so take a look and tell me if you want the *shape* changed before I populate it:

Golden Regression Fixtures ? video-free, non-attributable regression suite

Status: DRAFT ? owner-gated · Owner: Avi · Last updated: 2026-06-16
Tracking: docs/NEXT_STEPS.md + memory current-state · Supersedes nothing (extends
GOLDEN_DATA_SHARING.md)

1. Problem & goal (run the rally regression from committed files, no video/GPU)
2. Decisions locked (owned corpus · private-now/maybe-OSS · adults, consent not formalized)
3. Legal foundation (IN/US/EU verdict on derived data + the must-ask-counsel list) ? from workflow 2 me
4. Architecture
 - 4a. Freeze seam (post-detector trajectory CSV; reuse the shipped eval stack)
 - 4b. Tiered model (Tier-0 public shuttle-only/CONTROL-only · Tier-1 key-gated person streams)
 - 4c. Non-attributability (opaque salted IDs, minimization, no MD5 key, uselessness-in-isolation)
 - 4d. Protobuf + encryption (serialization vs secrecy; git-crypt/age; key mgmt; the DRM honesty note)
 - 4e. The two gates (characterization/snapshot + quality-floor)
5. Honest limits (what a green suite does NOT prove) ? workflow 1 + workflow 2 red-teams
6. Deliverables & rollout (PR-1..N as checkboxes, each small + independently shippable)
7. Progress tracker (? todo / ? in-progress / ? done, per phase, with PR links)
8. Open owner/counsel questions
9. Definition of done (acceptance criteria to mark the project COMPLETE)

How progress-tracking will work in practice (matching your repo conventions):

- **Section 7** is the single source of truth ? a checklist where each row is a concrete PR/deliverable with a status marker and, once merged, a PR link. Iteration = edit the row.
- I'll add a one-line pointer in `docs/NEXT_STEPS.md` and the `memory current-state` file so a future session resumes from the tracker, not from this chat.
- "Mark as completed" = every Section-6 deliverable is ? and Section-9 acceptance criteria are met; then the `Status:` header flips `DRAFT ? IN PROGRESS ? DONE` and it moves to `docs/archives/` per the archive policy.

I'll write it as a `draft only` ? no commit, no PR (your conventions). I'll populate Sections 3 and 5 from the research the moment the red-team lands and ping you with the file.

124. user (2026-07-02T06:11:33.343Z)

```
<task-notification>
<task-id>wgpct1wbd</task-id>
<tool-use-id>toolu_01AEmgopxK7Y1nZv5mgQ5wGj</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users\avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\wgpct1wbd.output</output-file>
<status>completed</status>
<summary>Dynamic workflow "Deep research: IP/privacy of committing DERIVED structured data publicly + a non-attributable encrypted fixture architecture" completed</summary>
<result>{"findings":[{"question":"Is algorithmically-derived structured data ? per-frame shuttle (x,y) coordinates, rally start/end timestamps, audio-onset feature values ? extracted from a copyrighted sports video itself protected by copyright (uncopyrightable facts/data, or a \"derivative work\")? Does the derived numeric data inherit the video's copyright?\",\"scope\":\"US copyright law analysis covering: idea/expression dichotomy (17 USC 102(b)); Feist (facts not copyrightable; sweat-of-the-brow rejected; originality in selection/arrangement); the 17 USC 101 \"derivative work\" definition; and intermediate/non-expressive copying + fair use (Google v. Oracle, Authors Guild v. Google, Sega v. Accolade, plus 2025 AI-training cases). This is legal-research synthesis of statutes, case law, and the U.S. Copyright Office Fair Use Index ? NOT formal legal advice. Focus is the copyright status of the OUTPUT data itself; the separate question of whether the EXTRACTION ACT (transiently copying the video) infringes is addressed via the intermediate-copying line. Foreign sui generis database rights (e.g., the EU Database Directive) are out of scope but flagged.\",\"findings\":[{\"claim\":\"Facts are categorically not copyrightable because they are discovered, not authored, and thus lack the originality the Constitution and Copyright Act require; effort ('sweat of the brow') does not earn copyright. This is the foundational holding most directly governing extracted numeric
```

data.", "support": "Feist Publications v. Rural Telephone Service Co., 499 U.S. 340 (1991): the Court held 'facts are not copyrightable' because they 'do not owe their origin to an act of authorship' and so are not original, and expressly held that 'originality, not sweat of the brow, is the touchstone of copyright protection.' Copyright 'rewards originality, not effort.' Per-frame (x,y) shuttle positions, rally start/end timestamps, and audio-onset values are observable physical facts about events recorded in the video ? discovered measurements, not authored expression ? so they fall squarely within the unprotected 'facts' category regardless of the computational effort to derive them.", "sourceName": "Feist Publications, Inc. v. Rural Telephone Service Co., 499 U.S. 340 (1991) ? Justia U.S. Supreme Court Center", "sourceUrl": "https://supreme.justia.com/cases/federal/us/499/340/", "confidence": "high"}, {"claim": "Facts and statistics GENERATED BY A SPORTING EVENT are not copyrightable even though the audiovisual BROADCAST of that event is ? the closest factual analogue to extracting shuttle coordinates and rally timings from a match video.", "support": "National Basketball Ass'n v. Motorola, Inc., 105 F.3d 841 (2d Cir. 1997): the court held 'Sports events are not authored in any common sense of the word' and that the underlying scores/statistics are uncopyrightable facts ('purely factual information which any patron of an NBA game could acquire from the arena'), while only the broadcast ? the creative work of directors, camera operators, editors ? is protected. By direct analogy, shuttle positions, rally boundaries, and onset values are facts ABOUT the game that any spectator could in principle observe; they are not original to whoever created the video, so they do not inherit the video's copyright.", "sourceName": "National Basketball Ass'n v. Motorola, Inc., 105 F.3d 841 (2d Cir. 1997) ? Wikipedia case summary", "sourceUrl": "https://en.wikipedia.org/wiki/National_Basketball_Ass%27n_v._Motorola,_Inc.", "confidence": "high"}, {"claim": "Copyright never extends to any idea, procedure, process, system, method of operation, concept, principle, or discovery ? only to original expression. Measured coordinates and timestamps are 'discoveries'/facts about the recorded event and fall in the §102(b) exclusion.", "support": "17 U.S.C. § 102(b): 'In no case does copyright protection for an original work of authorship extend to any idea, procedure, process, system, method of operation, concept, principle, or discovery, regardless of the form in which it is described, explained, illustrated, or embodied in such work.' This is the statutory codification of the idea/expression dichotomy and confirms that the factual/discovered content extracted from the video (the where-and-when of the shuttle and rallies) is outside the scope of the video's copyright.", "sourceName": "17 U.S.C. § 102 ? BitLaw / U.S. Code", "sourceUrl": "https://www.bitlaw.com/source/17usc/102.html", "confidence": "high"}, {"claim": "A 'derivative work' is a work that recasts, transforms, or adapts a preexisting work's EXPRESSION (e.g., translation, motion-picture version, abridgment). Stripping out non-expressive numeric facts is not 'recasting' the audiovisual expression, so the coordinate/timestamp/feature data is not a derivative work that inherits the video's copyright.", "support": "17 U.S.C. § 101 defines a derivative work as 'a work based upon one or more preexisting works, such as a translation, musical arrangement, dramatization, fictionalization, motion picture version, sound recording, art reproduction, abridgment, condensation, or any other form in which a work may be recast, transformed, or adapted.' Each enumerated example reuses the original's protected EXPRESSION. A table of (x,y) positions and timestamps reproduces none of the video's expressive elements (framing, editing, commentary, imagery); it captures only facts, which §102(b)/Feist place outside copyright. A derivative-work claim therefore fails because there is no protected expression carried into the output.", "sourceName": "17 U.S.C. § 101 Definitions ? Legal Information Institute (Cornell)", "sourceUrl": "https://www.law.cornell.edu/uscode/text/17/101", "confidence": "high"}, {"claim": "A compilation/database of facts can earn a THIN copyright, but only in the original SELECTION, COORDINATION, OR ARRANGEMENT ? never in the underlying facts. So a creatively-organized dataset might get limited protection for its structure, but a competitor remains free to copy the raw numbers themselves.", "support": "17 U.S.C. § 101 defines a 'compilation' as data 'selected, coordinated, or arranged in such a way that the resulting work as a whole constitutes an original work of authorship,' and Feist holds copyright in a compilation 'extends only to those components of the work that are original to the author, not to the facts themselves.' Practical upshot: any copyright the extractor obtains is in the dataset's expressive arrangement (if creative), not in the shuttle coordinates, rally timestamps, or onset values, which remain free for others to use.", "sourceName": "Feist Publications, Inc. v. Rural Telephone Service Co., 499 U.S. 340 (1991) ? Justia", "sourceUrl": "https://supreme.justia.com/cases/federal/us/499/340/", "confidence": "high"}, {"claim": "The act of EXTRACTION (transiently copying the video to analyze it) is a distinct question from the status of the OUTPUT data ? and intermediate copying CAN infringe even when the end product does not reproduce protected expression, so the extraction step must independently rely on a license or fair use.", "support": "Sega Enterprises Ltd. v. Accolade, Inc., 977 F.2d 1510 (9th Cir. 1992): 'intermediate copying of computer object code may infringe the exclusive rights granted to the copyright owner in section 106 ... regardless of whether the end product of the copying also infringes those rights.' But the same court held that copying to access UNPROTECTED functional/ideational elements is fair use where it is the only means of access and there is a

legitimate reason, because §102(b) elements 'are not protected by copyright.' Applied here: even though the numeric output is unprotectable, ingesting the video to compute it is itself a reproduction that needs a lawful basis (license to the source video, or a fair-use justification for the analytic copying).", "sourceName": "Sega Enterprises Ltd. v. Accolade, Inc., 977 F.2d 1510 (9th Cir. 1992) ? BitLaw", "sourceUrl": "https://www.bitlaw.com/source/... (truncated 267202 chars, full result in C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\wgpc1wbd.output)</result><usage><agent_count>18</agent_count><subagent_tokens>261531</subagent_tokens><tool_uses>44</tool_uses><duration_ms>789624</duration_ms></usage></task-notification>

125. assistant (2026-07-02T06:11:33.344Z)

The research workflow is complete (18 agents, the resume worked). Now let me harvest the key sections ? the legal memo, the unified architecture, and the red-team ? so the doc is grounded in the actual outputs rather than my paraphrase.

126. assistant (2026-07-02T06:11:33.344Z)

```
[external_agent_tool_call: Grep]
input: {"pattern": "\\memo\\":|\\designs\\":|\\techSynth\\":|\\redteam\\":|\\executiveSummary\\":|\\derivedDataCoreVerdict\\":|\\architectureName\\":|\\overallVerdict\\":|\\conditionsForSafePublicCommit\\":|\\openOwnerQuestions\\":", "path": "C:\\Users\\avidu\\AppData\\Local\\Temp\\claude\\C--Users-avidu-Projects-badminton-highlight-indexer\\068560d3-1ca2-4387-981a-d2d6bca55414\\tasks\\wgpc1wbd.output", "output_mode": "content", "-n": true}
[/external_agent_tool_call]
```

127. assistant (2026-07-02T06:11:33.344Z)

```
[external_agent_tool_result]
711:   "memo": {
712:[Omitted long matching line]
713:[Omitted long matching line]
742:   "conditionsForSafePublicCommit": [
1037:     "designs": [
1113:     "techSynth": {
1114:       "architectureName": "SEALED-FIXTURES: a tiered (Tier-0 public / Tier-1 key-gated),
de-attributed-at-source, encrypted-where-needed regression-fixture architecture for the existing
video-free eval seam",
1145:     "openOwnerQuestions": [
1164:     "redteam": {
1237:[Omitted long matching line]
[/external_agent_tool_result]
```

128. assistant (2026-07-02T06:11:33.344Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\wgpc1wbd.output
[/external_agent_tool_call]
```

129. assistant (2026-07-02T06:11:33.344Z)

```
[external_agent_tool_result]
711   "memo": {
712     "executiveSummary": "INTERNAL MEMO ? general legal information, NOT legal advice;
retained India + EU/UK counsel sign-off required before any public commit. QUESTION: can the
project commit DERIVED numeric data (per-frame shuttle (x,y) coordinates, rally start/end time-
intervals, audio/person/court feature columns, ground-truth intervals) into a public GitHub repo
so external contributors can run the rally-segmentation regression suite WITHOUT the videos?
BOTTOM LINE: Yes, but only under conditions, and the analysis FORKS on data provenance. The
OUTPUT data itself is, across all three jurisdictions, essentially uncopyrightable
FACT/measurement ? it does not inherit the source video's copyright (US: Feist + 17 USC 102(b) +
```

NBA v. Motorola; India: EBC v. D.B. Modak, no sui generis right; EU: Football Dataco v. Yahoo ? copyright protects only original selection/arrangement, never the numbers). So a thin compilation copyright at most attaches to a creative schema, never to the coordinates. BUT copyright is the easy axis and NOT where the risk lives. The real exposure is fourfold and provenance- and stream-dependent: (1) the ACT of ingesting/extracting from the source video needs its own lawful basis ? owned/permissioned video is clean; a US fair-use defense for analytic extraction is now CONTESTED, not \"strongly favorable,\" after Thomson Reuters v. Ross (2025) and the US Copyright Office May-2025 report, and unlawful acquisition independently destroys the defense (Bartz v. Anthropic, ~\$1.5B settlement); (2) CONTRACT / Terms-of-Service can bar extraction and redistribution even where copyright cannot reach the facts (ProCD; hiQ v. LinkedIn; YouTube ToS bars scraping AND \"harvest information that might identify a person (for example... faces)\") AND redistribution) ? and the remedy is injunction + forced deletion of the committed dataset, not nominal damages; (3) EU/UK sui generis DATABASE RIGHT may subsist in self-derived match-event/coordinate streams because, on the leading Football Dataco v. Sportradar reasoning, recording pre-existing on-court facts you do not control is \"OBTAINING,\" not \"creating\" ? so the project's comfortable \"we created it\" premise is genuinely contestable; (4) PRIVACY ? person/pose/gait streams and anything involving MINORS are the highest-risk and several should NOT be published at all. The cleanest defensible path: commit ONLY owned-or-permissioned-source, minors-excluded, person/pose/gait-EXCLUDED (shuttle + abstract rally-interval + non-biometric feature) data, stripped of every source-attribution field (no filename, no video_id MD5, no match metadata), under a no-extraction-from-third-party-source and no-redistribution hygiene posture, with optional encryption only as a barrier against the no-key public (it cannot bind a contributor who runs the tests).\",

713 \"derivedDataCoreVerdict\": \"The derived numeric data DOES NOT inherit the source video's copyright in any of the three jurisdictions ? this is the one firmly-settled conclusion. Per-frame shuttle (x,y) coordinates, rally start/end timestamps, and audio/person/court feature values are uncopyrightable FACTS/measurements about a recorded physical event: discovered, not authored. US: Feist (facts never copyrightable; \"originality, not sweat of the brow, is the touchstone\"), 17 USC 102(b) (no copyright in any \"discovery\"), NBA v. Motorola (facts generated by a sporting event are uncopyrightable \"purely factual information which any patron... could acquire from the arena\" even though the broadcast is protected). A coordinate/timestamp table is also NOT a 17 USC 101 \"derivative work\": a derivative work recasts the original's protected EXPRESSION (framing, editing, imagery, commentary), and a number table carries none of it. India: EBC v. D.B. Modak ? copyright needs \"skill and judgment,\" protects only original arrangement, not facts; no EU-style sui generis right exists. EU: Football Dataco v. Yahoo ...

[/external_agent_tool_result]

130. assistant (2026-07-02T06:11:33.344Z)

That memo is exactly what the doc needs. Now the unified architecture and the red-team:

131. assistant (2026-07-02T06:11:33.344Z)

[external_agent_tool_call: Read]
file: C:\\Users\\avidu\\AppData\\Local\\Temp\\claude\\C--Users-avidu-Projects-badminton-highlight-indexer\\068560d3-1ca2-4387-981a-d2d6bca55414\\tasks\\wgpct1wbd.output
[/external_agent_tool_call]

132. assistant (2026-07-02T06:11:33.344Z)

[external_agent_tool_result]
1113 \"techSynth\": {
1114 \"architectureName\": \"SEALED-FIXTURES: a tiered (Tier-0 public / Tier-1 key-gated), de-attributed-at-source, encrypted-where-needed regression-fixture architecture for the existing video-free eval seam\",
1115 \"tieredModel\": \"TWO TIERS layered onto the EXISTING video-free seam (fusion_golden.py extracts private *_golden_features.json on the GPU box ? fusion_compare.py / ablation.py re-score them on pure CPU). The split is by SENSITIVITY, not by file format.\\n\\nTIER-0 ? PUBLIC, UNENCRYPTED, NO KEY. A small, de-attributed, Fork-A-only, BIOMETRIC-FREE fixture set committed under a NEW tracked dir `eval\_baselines/fixtures/` (extending the existing committed-baseline precedent ? regression.py:33 already comments that eval_baselines/ is \"NOT under the gitignored output/, so a fresh checkout/CI can load a committed baseline\"). ANY cloner with no key runs the SHUTTLE-ONLY regression path (experiment.predict/compare LOVO

?F1; ablation.run_ablation; the Gate-A litmus shape; regression-style floor vs GT). Contents are minimized to: opaque `fixture\_id`, `fps`/`frame\_width` scale constants, per-candidate `start`/`end` (time-origin reset), the 5 SHUTTLE columns (duration, peak_velocity, mean_velocity, active_ratio, net_crossings ? inanimate-object, fps/res-invariant), integer `label`, and relative `gts`. The PERSON block and AUDIO block are EXCLUDED from Tier-0 (person = the legal memo's Art.9 \"tripwire\" / DPDP singling-out risk in a small roster). Tier-0 tests are UNCONDITIONAL ? they run on every PR, converting the currently-skipped litmus into a real public gate.\n\nTIER-1 ? RICHER / SENSITIVE, KEY-GATED, SKIPS CLEANLY WHEN ABSENT. The full n=6+ golden corpus WITH person/audio columns (the authoritative ?F1/CI/sign-test and the exact Gate-A litmus values in test_litmus_reproduces_gate_a). Two delivery options, OUT-OF-REPO is the DEFAULT: (A) OUT-OF-REPO + TOKEN ? the *_golden_features.json stay on the existing private OneDrive store (docs/GOLDEN_DATA_SHARING.md #175); contributors/CI set `RALLY\_GOLDEN\_DIR` to a synced/fetched path; nothing sensitive ever enters git history (also sidesteps the legal \"redistribution\" act). (B) ENCRYPTED-IN-REPO (SOPS+age) only for Fork-A-clean-but-still-private data where in-repo versioning is genuinely wanted; decrypt to a gitignored path before the Tier-1 run. A Tier-1 test SKIPS (never fails) when the key/data is absent, reusing the proven `\_golden\_present()` skipif pattern (tests/test_ablation.py:366-371) generalized to a shared `\_tier1\_present()` helper. So a fresh no-key clone is always GREEN; coverage never depends on a secret being present.\n\nKEY INSIGHT (why this is elegant, not Byzantine): the two privacy properties come from DIFFERENT layers and must not be conflated. NON-ATTRIBUTABILITY comes from DELETION + RELATIVIZATION at the source (Tier-0/Tier-1 both), so it survives the analog hole ? even a fully-keyed contributor cannot re-identify, because attribution data was never written. USELESSNESS-TO-THE-NO-KEY-PUBLIC comes from minimization (opaque numbers, no schema-external meaning) PLUS optional Tier-1 encryption. Encryption does the smaller job; minimization does the real work.",

```

1116     "antiAttributionMeasures": [
1117         "OPAQUE SALTED SURROGATE ID ? replace BOTH the `name` field (today = source
filename via fusion_golden._derive_video_path) AND the MD5 video_id with fixture_id =
first-16-hex of HMAC-SHA256(key=PER-RELEASE-SECRET-SALT, msg=video_id||name). Salt is generated
once per release, kept PRIVATE (lives with the Fork-A corpus on OneDrive, NEVER in git), rotated
per release. Result: (a) non-reversible without the salt, (b) NOT the MD5 video_id so holding
the candidate file cannot confirm the source (kills the memo's 'video_id leak' red flag), (c)
unlinkable across releases. Harness-transparent: fusion_compare.py:50 uses d['name'] ONLY as a
VideoCandidates label, so aliasing it changes nothing downstream.",
1118     "METAD..."
[/external_agent_tool_result]

```

133. assistant (2026-07-02T06:11:33.344Z)

I now have both workflows fully harvested. The second red-team corroborated workflow 1's determinism catch *and* surfaced a concrete, checkable claim: real player/source names already committed in `eval\_baselines/` and a couple of source files. Per my honesty rule I'll verify that before asserting it in a doc ? it's the kind of finding that must be true, not plausible.

134. assistant (2026-07-02T06:11:33.344Z)

```

[external_agent_tool_call: Grep]
input: {"pattern":"adarsh|kushagra|mahadevpura|gbaaddy|singles|doubles","path":"eval_baselines",
,"output_mode":"content",-n:true,"head_limit":25}
[/external_agent_tool_call]

```

135. assistant (2026-07-02T06:11:33.344Z)

```

[external_agent_tool_result]
eval_baselines\heuristic_lovo_n6.json:8:     "adarsh_avi_singles": 0.751,
eval_baselines\heuristic_lovo_n6.json:9:     "kushagra_singles": 0.157,
eval_baselines\heuristic_lovo_n6.json:10:    "largetest_doubles": 0.737,
eval_baselines\heuristic_lovo_n6.json:11:    "gbaaddy": 0.309,
eval_baselines\heuristic_lovo_n6.json:13:    "mahadevpura_singles": 0.657
eval_baselines\gemini_wrapper_2026-06-10.json:7:    "mahadevpura_singles": {
eval_baselines\gemini_wrapper_2026-06-10.json:12:    "kushagra_singles": {
eval_baselines\gemini_wrapper_2026-06-10.json:19:    "heuristic_lovo": {"mahadevpura_singles":
0.657, "kushagra_singles": 0.157},
eval_baselines\gemini_wrapper_2026-06-10.json:20:    "gen0_learned_lovo":

```

```
{"mahadevpura_singles": 0.744, "kushagra_singles": 0.561}  
[/external_agent_tool_result]
```

136. assistant (2026-07-02T06:11:33.344Z)

```
[external_agent_tool_call: Grep]  
input: {"pattern":"adarsh|kushagra|mahadevpura|gbaaddy", "path":"backend/eval/eval_partial_labels  
.py", "output_mode":"content", "-n":true, "head_limit":15}  
[/external_agent_tool_call]
```

137. assistant (2026-07-02T06:11:33.344Z)

```
[external_agent_tool_result]  
23:      --trajectory C:/Users/avidu/AppData/Local/Temp/adarsh_avi_traj.csv \  
119:    comp_out = comparison_out or osp.join("output", "adarsh_avi_vs_human.csv")  
[/external_agent_tool_result]
```

138. assistant (2026-07-02T06:11:33.344Z)

Confirmed ? the leak is real and in tracked files: `eval\_baselines/heuristic\_lovo\_n6.json` and `gemini\_wrapper\_2026-06-10.json` use real player/venue names as keys (`adarsh\_avi\_singles`, `kushagra\_singles`, `mahadevpura\_singles`, `gbaaddy`?), and `eval\_partial\_labels.py` even embeds a personal local path. Private repo today, but it's in git history and would persist if open-sourced. That becomes deliverable #1.

I have everything. Writing the tracked project doc now.

139. assistant (2026-07-02T06:11:33.344Z)

```
[external_agent_tool_call: Write]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\GOLDEN_REGRESSION_FIXTURES.md  
[/external_agent_tool_call]
```

140. assistant (2026-07-02T06:11:33.344Z)

```
[external_agent_tool_result]  
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-  
indexer\docs\GOLDEN_REGRESSION_FIXTURES.md (file state is current in your context ? no need to  
Read it back)  
[/external_agent_tool_result]
```

141. assistant (2026-07-02T06:11:33.344Z)

Draft written to docs/GOLDEN_REGRESSION_FIXTURES.md. Before I hand it over, let me verify one claim I carried into the P0 deliverable from the red-team but hadn't personally checked ? that `tests/test\_cleanup\_caches.py` also contains real names ? so the doc doesn't assert something unverified.

142. assistant (2026-07-02T06:11:33.344Z)

```
[external_agent_tool_call: Grep]  
input: {"pattern":"adarsh|kushagra|mahadevpura|gbaaddy|avi_", "path":"tests/test_cleanup_caches.p  
y", "output_mode":"content", "-n":true}  
[/external_agent_tool_call]
```

143. assistant (2026-07-02T06:11:33.344Z)

```
[external_agent_tool_result]  
19:    assert classify("adarsh_traj.csv", False, "wsl")[2] is False          # trajectory  
-> keep  
24:    cat, regen, default_purge = classify("adarsh_traj_wasbcache", True, "wsl")  
55:    entries = [_e("GX010125_frames", 90_000, True), _e("kushagra_frames", 5_000, True)]  
58:    assert [e.name for e in to_del] == ["kushagra_frames"]
```

[/external_agent_tool_result]

144. assistant (2026-07-02T06:11:33.344Z)

Verified ? `tests/test\_cleanup\_caches.py` does carry the tokens (`adarsh\_traj.csv`, `kushagra\_frames`), so the P0 deliverable is accurate across all four files.

Done ? the tracked project doc is ready

? docs/GOLDEN_REGRESSION_FIXTURES.md ? ****draft, uncommitted**** (your conventions: no commit/PR unless you ask).

It's structured to be exactly what you asked for ? iterate, track, mark complete:

- ****\$7 Progress tracker**** is the single source of truth: one row per deliverable (`P0, M1?M4, P1, P2, P3, O1, O2, DOC`), each a small independent PR, with `?/?/?/?` status + a PR-link column. Iterating = editing a row.
- ****\$9 Definition of done**** is the explicit acceptance checklist; when it's all ticked, the `Status:` header flips `DRAFT ? IN PROGRESS ? DONE` and the doc moves to `docs/archives/`.
- ****\$2 Decisions locked**** captures your three answers so they're not relitigated; ****\$8**** isolates the counsel-gated vs owner-strategic questions.
- ****\$3 Legal**** is the IN/US/EU memo (cited), clearly marked **researched, not legal advice**; ****\$5 threat model****, ****\$6 honest limits****, and the ****\$7 red-team must-fixes**** keep it from overselling.

****Two findings in there are real and verified now, independent of the rest:****

1. ? ****P0 ? live name leak:**** real player/venue names are committed in `eval\_baselines/heuristic\_lovo\_n6.json`, `gemini\_wrapper\_2026-06-10.json`, `backend/eval/eval\_partial\_labels.py` (+ a personal `C:/Users/avidu/?` path), and `tests/test\_cleanup\_caches.py` ? plus git history. Private repo today, permanent if open-sourced.
2. ? ****D5 ? determinism trap:**** both independent red-teams + the code confirm the learned scorer (numpy/OpenBLAS `DYNAMIC\_ARCH`) would false-RED across CPUs ? the cross-machine gate must be CONTROL/static-only. Happily, that's the **same** shuttle-only tier privacy already requires.

****Recommended next step:**** ****P0**** is the natural first PR ? it's self-contained, needs **no** counsel (it's hygiene on your own data), and fixes a real bug today. Everything legally sensitive (P2/P3) stays gated behind counsel + your scope call (synthetic-first is the low-risk path).

Want me to (a) start ****P0**** as the first focused PR, (b) wire the pointers in `NEXT\_STEPS.md` + memory `current-state` so a future session resumes from the tracker, (c) commit the doc itself, and/or (d) adjust the doc's shape first? I held off on (b)/(c) to avoid churn and shared-checkout collisions until you've eyeballed it. The full raw research (every legal citation + the architecture/red-team JSON) is on disk if you want me to expand any section.

145. user (2026-07-02T06:11:33.345Z)

love the framing of it! Please freeze this and I guess this could be a good norm to follow throughout

146. assistant (2026-07-02T06:11:33.345Z)

Love it ? let me freeze the doc into a PR and distill the format into a reusable template + convention so it's a norm going forward. First, branching-safety hygiene and reading the files I'll touch so I match repo conventions.

147. assistant (2026-07-02T06:11:33.345Z)

[external_agent_tool_call: Bash]

description: Fetch origin and show working-tree status, current branch, and origin/master
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git fetch origin --quiet && echo "=== status ===" && git status --short && echo "=== current branch / HEAD ===" && git branch --show-current && git log --oneline -1 && echo "=== origin/master ===" && git log

```
--oneline -1 origin/master
[/external_agent_tool_call]
```

148. assistant (2026-07-02T06:11:33.345Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\README.md
[/external_agent_tool_call]
```

149. assistant (2026-07-02T06:11:33.345Z)

```
[external_agent_tool_result]
=== status ===
?? artifacts/
?? docs/GOLDEN_REGRESSION_FIXTURES.md
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
=== current branch / HEAD ===
master
90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg eval + frozen baseline
(#185)
=== origin/master ===
90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg eval + frozen baseline
(#185)
[/external_agent_tool_result]
```

150. assistant (2026-07-02T06:11:33.345Z)

```
[external_agent_tool_result]
1      - [ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md) ? Latest (v8) vendor
labeling guidelines mirrored from the collector's `docs/annotation/`, with the fusion-driven
court-calibration / singles-doubles additions marked `? ADDED (fusion)`.
2      - [RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md) ? Foundation doc for
closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge
diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps,
the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-
class, golden-corpus + active learning, Gemini-agreement as a shadow signal), an
adversarially-verified, cited external-research synthesis (TAL/TAS boundary heads, racket-
sports rally-state cues, small-data, eval rigor), a prioritized cheap?durable roadmap with
falsifiable success criteria, and ($7) a tracked project plan ? work-item checklist,
sequencing, the single quality number attributed to the effort, and a definition of done
(project completes when all tiers land + the number is recorded). The plan that the
[QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) log executes against; ties together
EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS /
OWNED_MODEL_IMPLEMENTATION_PLAN.
3      - [EVALUATION.md](EVALUATION.md) ? How we measure and improve quality.
4      - [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md) ? Design (P1 SHIPPED) for A/B + shadow
+ SxS experimentation on rally-detection variants with zero production regression ?
experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift,
rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`,
`regression`, `metrics`); implemented in `backend/eval/experiment.py`.
5      - [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) ? Living log of rally-detection
quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-
chronological record; terminal-state snapshots get archived under `archives/quality-
iterations/`.
6      - [ANALYZERS_AND_RECONCILERS.md](ANALYZERS_AND_RECONCILERS.md) ? Design (owner-gated,
NOT started): a signal-fusion framework for rally detection ? producers emit confidence-
scored signals at three temporal scopes (frame detectors / window analyzers / session
analyzers), a learned scorer weights them, and an auditable report-first reconciliation
pass lints the decision against the evidence. The spine: no special-casing in the decision
path (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-
up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to
`EXPERIMENT_HARNESS.md` and `MULTI_SIGNAL_FUSION_PLAN.md` (ADR-007 modular rally layer).
7      - [PERSONALIZATION_PLAN.md](PERSONALIZATION_PLAN.md) ? Design (owner-adopted hybrid;
Phase-0 landing) for the per-user personalization feature on a generalization-first
```

baseline**: a thin per-user *tuning* layer gated by the honest small-N statistics (a **min-N promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; *a tool that gets better the more you help it*). Records the **locked owner decisions** (identity, storage, `min\_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).

```
8
9     ## Practical / ongoing
10    - [TRACKNET_WSL_SETUP.md](TRACKNET_WSL_SETUP.md)
11    - Golden set docs (`GOLDEN_SET`)
12    - Evaluation harness and calibration drivers (in `backend/eval/`)
13    - [GOLDEN_DATA_SHARING.md](GOLDEN_DATA_SHARING.md) ? How the GPU box and CPU dev/agent
sessions share golden artifacts (trajectory CSVs, `_golden_features.json`) via a OneDrive
folder; the GPU-extract ? CPU-litmus split. See **#175**.
14
15    ## Hist...
[/external_agent_tool_result]
```

151. assistant (2026-07-02T06:11:33.345Z)

```
[external_agent_tool_result]
40    - `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one
helper).
41    - Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA
user_version` migration ladder in `backend/infrastructure/database.py`.
42
43
44    ## Committed guardrails (non-negotiable)
45    - **Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
46    training data source** (terms/copyright). Commercial-clean licensing for EVERY
dependency ?
47    code, data, AND weights: **MIT / Apache-2.0 / BSD only** (ratified 2026-06-09, owner-
agreed).
48    - **Owned-model base:** **Qwen3-VL (Apache-2.0)** primary, InternVL3 (MIT) fallback;
**Gemma-4
49    is AMBER / bench-only** (Apache grant likely but Prohibited-Use posture unconfirmed ?
counsel
50    needed; do NOT build on it without sign-off). **Reject Gemma 3** (viral license).
**Exclude
51    minors** from the training pool; per-user training data needs a separate explicit opt-
in.
52    - **v1 = rally + highlight + history + correction loop**; **gait/biometrics strictly
future**
53    (GDPR Art. 9).
54    - **Generic data schema** ; `coach ? athlete` is an **optional overlay**, not baked in.
55
56    ## Workflow conventions
57    - **Branch off `master` ; ONE PR per work item; NO PR stacking.** Don't open a PR unless
asked.
58    - Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.8+; `ffmpeg`
required.
59    - Never push to the default branch directly; descriptive commit messages.
[/external_agent_tool_result]
```

152. assistant (2026-07-02T06:11:33.345Z)

Master moved again (sibling session landed #185) ? I'll branch explicitly from `origin/master` per branching-safety. Note: this repo already has a precedent for this format ? `RALLY\_QUALITY\_RESEARCH.md` §7 is a "tracked project plan? definition of done." So the norm is really *standardizing an emergent pattern*, not inventing one. Let me cut the branch and read the files I'll touch.

153. assistant (2026-07-02T06:11:33.345Z)

```
[external_agent_tool_call: Bash]
```

```
description: Branch from origin/master explicitly and verify
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git checkout -b
docs/golden-regression-fixtures-plan origin/master 2>&1 && echo "=== verify HEAD is mine ===" &&
git branch --show-current && git log --oneline -1 && echo "=== only my commits vs base (should
be empty now) ===" && git log --oneline origin/master..HEAD
[/external_agent_tool_call]
```

154. assistant (2026-07-02T06:11:33.345Z)

```
[external_agent_tool_result]
branch 'docs/golden-regression-fixtures-plan' set up to track 'origin/master'.
Switched to a new branch 'docs/golden-regression-fixtures-plan'
=== verify HEAD is mine ===
docs/golden-regression-fixtures-plan
90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg eval + frozen baseline
(#185)
=== only my commits vs base (should be empty now) ===
[/external_agent_tool_result]
```

155. assistant (2026-07-02T06:11:33.345Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\CLAUDE.md
[/external_agent_tool_call]
```

156. assistant (2026-07-02T06:11:33.345Z)

Clean branch, no sibling commits. Let me see the rest of the Workflow-conventions block and the top of NEXT_STEPS so I insert pointers cleanly.

157. assistant (2026-07-02T06:11:33.345Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\CLAUDE.md
[/external_agent_tool_call]
```

158. assistant (2026-07-02T06:11:33.345Z)

```
[external_agent_tool_result]
56      ## Workflow conventions
57      - **Branch off `master`; ONE PR per work item; NO PR stacking.** Don't open a PR unless
asked.
58      - Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.8+; `ffmpeg`
required.
59      - Never push to the default branch directly; descriptive commit messages.
60      - ?? This dev container lacks GPU / torch / cv2 / numpy ? the **full pipeline can't run
here**.
61      Verify pure-logic paths and hand machine-side checks (real video, GPU/WSL suite) to
the owner.
62
[/external_agent_tool_result]
```

159. assistant (2026-07-02T06:11:33.345Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md
[/external_agent_tool_call]
```

160. assistant (2026-07-02T06:11:33.345Z)

```
[external_agent_tool_result]
1      # Next steps (live) ? owned-model + multisport
2
3      **Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness
```

```

design docs merged,
4      owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
5      ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
6      `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;
```

****PR-2**

```

7      upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
8      compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
9      coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
10     `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
11     **Authoritative roadmap:**
12     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
13     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
14
15     ## ? Next-session / GPU-box pickup (2026-06-15)
16     **M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**),
train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16
drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already
present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 /
bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted
data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level,
not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing
? Actions minutes / spending limit).
17
18     **Immediate next step ? the real P3 LOVO measurement, split GPU?CPU** (the dev box is
CPU-only: `torch+cpu`/`cv2` present but no GPU, and the corpus/videos/CSVs/labels are not here):
19     1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6"` ? produces `_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `rallies.csv` labels are owner-side only.)
20     2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-
annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
**[`GOLDEN_DATA_SHARING.md`](GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
21     3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
22
23     **Also queued (owner-gated / pending input):**
24     - Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
25     - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent
will do it).
26     - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config
bug.
27
28     **Leaving (bulky / M2 ? no start without owner go):** ActionFormer (M0.4), event-index
DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack
deprecation, multisport, repo rename, PMF execution.
29
30     ## Where we are (one paragraph)
[/external_agent_tool_result]
```

161. assistant (2026-07-02T06:11:33.345Z)

Branch is clean. Now I'll: (1) tweak the doc's tracking-anchors line to match what I'll actually wire, (2) create the reusable template, (3) index both in `docs/README.md`, and (4) add a one-line convention to `CLAUDE.md`. The norm is really **standardizing** the pattern already in `RALLY\_QUALITY\_RESEARCH.md §7`.

162. assistant (2026-07-02T06:11:33.345Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PROJECT_DOC_TEMPLATE.md
[/external_agent_tool_call]
```

163. assistant (2026-07-02T06:11:33.345Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PROJECT_DOC_TEMPLATE.md (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

164. assistant (2026-07-02T06:11:33.345Z)

Now the three edits ? fix the tracking-anchors line, index both docs in the README, and add the convention to CLAUDE.md:

165. assistant (2026-07-02T06:11:33.346Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\GOLDEN_REGRESSION_FIXTURES.md
[/external_agent_tool_call]
```

166. assistant (2026-07-02T06:11:33.346Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\GOLDEN_REGRESSION_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

167. assistant (2026-07-02T06:11:33.346Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\README.md
[/external_agent_tool_call]
```

168. assistant (2026-07-02T06:11:33.346Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

169. assistant (2026-07-02T06:11:33.346Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\README.md
[/external_agent_tool_call]
```

170. assistant (2026-07-02T06:11:33.346Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

171. assistant (2026-07-02T06:11:33.346Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\CLAUDE.md
[/external_agent_tool_call]
```


172. assistant (2026-07-02T06:11:33.346Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\CLAUDE.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

173. assistant (2026-07-02T06:11:33.346Z)

All edits applied. Now staging ****explicit paths only**** (never ``-A``), re-verifying the branch state right before commit per branching-safety:

174. assistant (2026-07-02T06:11:33.346Z)

```
[external_agent_tool_call: Bash]
description: Stage the four files explicitly and re-verify branch state
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git add
docs/GOLDEN_REGRESSION_FIXTURES.md docs/PROJECT_DOC_TEMPLATE.md docs/README.md CLAUDE.md && echo
"=== staged ===" && git diff --cached --name-only && echo "=== branch / HEAD (mine?) ===" && git
branch --show-current && git log --oneline -1 && echo "=== commits vs origin/master (should be
none yet) ===" && git log --oneline origin/master..HEAD && echo "=== unstaged/untracked left
behind (should NOT include my 4 files) ===" && git status --short
[/external_agent_tool_call]
```

175. assistant (2026-07-02T06:11:33.346Z)

```
[external_agent_tool_result]
warning: in the working copy of 'docs/GOLDEN_REGRESSION_FIXTURES.md', LF will be replaced by
CRLF the next time Git touches it
warning: in the working copy of 'docs/PROJECT_DOC_TEMPLATE.md', LF will be replaced by CRLF the
next time Git touches it
=== staged ===
CLAUDE.md
docs/GOLDEN_REGRESSION_FIXTURES.md
docs/PROJECT_DOC_TEMPLATE.md
docs/README.md
=== branch / HEAD (mine?) ===
docs/golden-regression-fixtures-plan
90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg eval + frozen baseline
(#185)
=== commits vs origin/master (should be none yet) ===
=== unstaged/untracked left behind (should NOT include my 4 files) ===
M   CLAUDE.md
A   docs/GOLDEN_REGRESSION_FIXTURES.md
A   docs/PROJECT_DOC_TEMPLATE.md
M   docs/README.md
??  artifacts/
??  docs/KHELSUTRA_PRODUCT_OVERVIEW.md
[/external_agent_tool_result]
```

176. assistant (2026-07-02T06:11:33.346Z)

Exactly my four files staged; ``artifacts/`` and the KHELSUTRA doc correctly left untracked. Committing with the required trailer, then pushing:

177. assistant (2026-07-02T06:11:33.346Z)

```
[external_agent_tool_call: Bash]
description: Commit the four files with co-author trailer and push the branch
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git commit -m 'docs: freeze
Golden Regression Fixtures plan + adopt tracked-project-doc template' -m 'Adds
docs/GOLDEN_REGRESSION_FIXTURES.md ? the owner-gated plan for video-free, non-attributable
regression fixtures: the post-detector freeze seam + dual gates (characterization + quality-
floor) reusing the backend/eval/ stack, and the SEALED-FIXTURES privacy filter (de-attribution-
```

```

at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited IN/US/EU legal
memo on derived-data IP/privacy, a threat model, honest limits, a progress tracker, and a
definition-of-done.' -m 'Adds docs/PROJECT_DOC_TEMPLATE.md and a CLAUDE.md workflow convention +
README index entries, standardizing the tracked-project-doc pattern already used in
RALLY_QUALITY_RESEARCH.md section 7.' -m 'DRAFT / owner-gated; no implementation started. Legal
sections are researched general information, not advice. Includes a verified, actionable finding
(real player/venue names committed in eval_baselines/*.json + eval_partial_labels.py +
test_cleanup_caches.py) tracked as deliverable P0.' -m 'Co-Authored-By: Claude Opus 4.8 (1M
context) <noreply@anthropic.com>' && echo "=== HEAD now ===" && git log --oneline -2 && echo
"=== push ===" && git push -u origin docs/golden-regression-fixtures-plan 2>&1 | tail -8
[/external_agent_tool_call]

```

178. assistant (2026-07-02T06:11:33.346Z)

```

[external_agent_tool_result]
[docs/golden-regression-fixtures-plan 95cd154] docs: freeze Golden Regression Fixtures plan +
adopt tracked-project-doc template
 4 files changed, 280 insertions(+)
 create mode 100644 docs/GOLDEN_REGRESSION_FIXTURES.md
 create mode 100644 docs/PROJECT_DOC_TEMPLATE.md
=== HEAD now ===
95cd154 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template
90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg eval + frozen baseline
(#185)
=== push ===
remote:
remote: Create a pull request for 'docs/golden-regression-fixtures-plan' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/docs/golden-
regression-fixtures-plan
remote:
branch 'docs/golden-regression-fixtures-plan' set up to track 'origin/docs/golden-regression-
fixtures-plan'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      docs/golden-regression-fixtures-plan -> docs/golden-regression-fixtures-
plan
[/external_agent_tool_result]

```

179. assistant (2026-07-02T06:11:33.346Z)

Committed (only my commit atop `90ab087`) and pushed. Opening the PR:

180. assistant (2026-07-02T06:11:33.346Z)

```

[external_agent_tool_call: Bash]
description: Open the PR against master with a descriptive body
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && gh pr create --base master
--head docs/golden-regression-fixtures-plan --title "docs: freeze Golden Regression Fixtures
plan + adopt tracked-project-doc template" --body "$(cat <<'EOF'

```

What

Freezes the design we iterated on into a **tracked project doc**, and standardizes the format as a repo convention.

```

- **`docs/GOLDEN_REGRESSION_FIXTURES.md`** ? owner-gated plan for a video-free, non-
attributable regression suite so a clean clone runs the rally-seg suite with no video/GPU/DB.
Two layers:
  - *Mechanics: freeze at the post-detector trajectory seam + *dual gates
(characterization/snapshot + quality-floor), reusing the shipped `backend/eval/` stack
(`experiment`/`metrics`/`windowing`/`regression`).
  - *Privacy filter (SEALED-FIXTURES): de-attribution-at-source, tiered **public (shuttle-only)
/ key-gated (person/audio)**, non-attributability from deletion ? encryption only against the
no-key public.
  - Carries a cited **IN/US/EU legal memo on derived-data IP/privacy, a threat model, honest
limits, a **$7 progress tracker, and a **$9 definition-of-done.

```

- `***docs/PROJECT_DOC_TEMPLATE.md***` + a `CLAUDE.md` workflow-convention line + `docs/README.md` index entries ? standardizing the tracked-project-doc pattern already used in `RALLY_QUALITY_RESEARCH.md` §7.

Status & honesty

- **DRAFT / owner-gated** ? no implementation started. This PR adds **docs only**.
- Legal sections are **researched general information, not advice** (gated on retained counsel; see §8).
- Synthesized from two multi-agent research/design workflows + two independent red-teams.

Two findings already actionable (in §7 tracker)

- ? **P0 ? live name leak (verified):** real player/venue names are committed today in `eval_baselines/heuristic_lovo_n6.json`, `eval_baselines/gemini_wrapper_2026-06-10.json`, `backend/eval/eval_partial_labels.py` (+ a personal local path), and `tests/test_cleanup_caches.py` ? and in git history. Private repo now, permanent if open-sourced. Self-contained, needs no counsel.
- ? **Determinism (verified):** the learned scorer (numpy/OpenBLAS `DYNAMIC_ARCH`) would false-RED across CPUs ? the cross-machine gate must be `CONTROL/static-only` (which is also the shuttle-only tier privacy requires).

Not in scope here

No code, no fixtures committed. Implementation deliverables (P0, M1?M4, P1, ?) are tracked in §7 and start only on owner go (P2/P3 are counsel-gated).

EOF

)" 2>&1 | tail -5

[/external_agent_tool_call]

181. assistant (2026-07-02T06:11:33.347Z)

[external_agent_tool_result]

<https://github.com/avidullu/badminton-highlight-indexer/pull/186>

[/external_agent_tool_result]

182. assistant (2026-07-02T06:11:33.347Z)

PR opened: **[#186](https://github.com/avidullu/badminton-highlight-indexer/pull/186)** ? docs-only, draft/owner-gated. Now let me make the norm durable in my memory too (the repo side is in the PR via `CLAUDE.md` + the template). Reading the memory index + current-state first so I update them without clobbering a concurrent session.

183. assistant (2026-07-02T06:11:33.347Z)

[external_agent_tool_call: Read]

file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md

[/external_agent_tool_call]

184. assistant (2026-07-02T06:11:33.347Z)

[external_agent_tool_call: Read]

file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\MEMORY.md

[/external_agent_tool_call]

185. assistant (2026-07-02T06:11:33.347Z)

[external_agent_tool_result]

1 # Memory Index

2

3 ## ? Start here

4 - [Current state](current-state.md) ? **live micro-checkpoint** (updated every

checkpoint): open PRs, what just landed, next steps, open decisions. ****Read this first.****

- 5 - [Session handoff](session-handoff.md) ? current state · next steps · open decisions · ramp-up kit. AUTO-refreshed at meaningful checkpoints (PR merged, key decision, phase shift, session wrap) ? see working-agreement-checkpoints staleness tripwire.
- 6 - [Code map artifact](code-map-artifact.md) ? for CODE/architecture ramp-up, read `docs/CODE\_MAP.md` FIRST (dense anchor-rich map; replaces a file crawl). How to regenerate it.
- 7 - [File-placement conventions](file-placement-conventions.md) ? ****where new files go**** (scripts/ vs backend/eval/ vs tools/ vs scratch/ vs training/ vs artifacts/ ?); ****read before committing a new file.**** Authoritative table = `docs/CODE\_MAP.md` §0 (PR #138); scratch/ gitignored (#136).
- 8
- 9 - [Working agreement: attribution](working-agreement-attribution.md) ? never assert fabricated facts; NEVER invent a provenance (a discussion/review/source that didn't happen). The silver-bullet rule.
- 10 - [Working agreement: checkpoints](working-agreement-checkpoints.md) ? user prefers frequent small PRs + a Claude-consumable current-state file updated at every micro-checkpoint.
- 11 - [Feedback: tail long-job logs](feedback-tail-long-job-logs.md) ? proactively tail progress/summaries from long-running background jobs while they run, not just the final result.
- 12 - [Working agreement: merging](working-agreement-merging.md) ? owner authorized Claude to MERGE its own good-engineering-fix PRs on verified-green CI (2026-06-12); product/strategy/licensing stay owner-gated.
- 13 - [Gotcha: shared-checkout branch collisions](gotcha-shared-checkout-branch-collisions.md) ? spawned tasks + sibling sessions SHARE this checkout; ****branch from `origin/master` explicitly, re-verify HEAD before commit, stage explicit paths, serialize repo writes.**** The #116/#117 collision (owner upset). Global form in ~/.claude/CLAUDE.md.
- 14 - [Gotcha: long GPU/WSL runs](gotcha-long-runs-gpu-wsl.md) ? running the WASB pipeline E2E on a big 4K video: Modern Standby kills it (keep-awake), cv2 extraction is the bottleneck (ffmpeg pre-extract), proxy-index/4K-stitch recipe + how to observe each phase.
- 15 - [Cache-hygiene policy](cache-hygiene-policy.md) ? retention table (purge frames/staged vs keep CSVs/DB/reels) + the shipped tooling: self-cleaning runners (keep_frames default false), tools/cleanup_caches.py (dry-run-first), disk guard, SessionStart nudge hook. The `for e in` ****WSL footgun.**
- 16 - [Tool: trim_video.py](tool-trim-video.md) ? reusable `scratch/trim\_video.py` for keyframe-aware LOSSLESS video trims (drop first/last N sec), `--exact` re-encode with NVENC?libx264 fallback. Reach for it instead of hand-rolling ffmpeg trims.
- 17 - [Decision-rigor norm](decision-rigor-norm.md) ? ****NUMBERS INFORM, FOUNDATIONS DECIDE**** (owner directive 2026-06-14, set by the Gate-A flip). Every signal/architecture change gets ablation-layer rigor (per-signal marginal ?F1 + honest CI/sign-test) AND a foundational lens (extensible/orthogonal/reversible), never numbers alone. Ties ****[[weak-signals-retained]] + [[paper-coauthor-aim]]**.
- 18
- 19 **## Project: rally segmentation pipeline**
- 20 - [Candidate segments & fine-tuning](candidate-segments-finetuning.md) ? why raw detector candidates are persisted (detector-agnostic table); golden-eval-set goal
- 21 - [TrackNetV4/WASB via WSL2](tracknet-wasb-wsl2-decision.md) ? self-hosted shuttle trajectory models (not LLM APIs); WASB/TrackNetV3 = MIT, commercial-clear
- 22 - [Collector?indexer bridge](collector-indexer-bridge.md) ? 4-phase plan to train/eval a generic temporal rally segmenter; Phases 1?2 done, 3 in progress, 4 next
- 23 - [Golden-set vendoring](golden-set-vendoring.md) ? generate golden rally labels via 3rd-party vendors (Bangalore-local) + extensible regression flow; Roora selected for a 3-sport pilot, guideline +...

[/external_agent_tool_result]

186. assistant (2026-07-02T06:11:33.347Z)

[external_agent_tool_result]
<system-reminder>[Truncated: PARTIAL view ? showing lines 1-447 of 2130 total (101240 tokens, cap 25000). Call Read with offset=448 limit=447 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

- 1 ---
- 2 name: current-state
- 3 description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not only on request). What just landed, what's next, open decisions."

```

4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-16 ~03:30 (? RALLY-QUALITY RESEARCH + EVAL-FRAMEWORK FOUNDATION
DOC DONE ? **PR #184 OPEN for
13     owner review**, docs-only, NOT auto-merged). Owner asked for deep research to bridge the
local-vs-Gemini gap with a strong
14     harness/eval framework, checked into the repo as the foundation for quality iterations.
Scope (owner-picked) = research +
15     harness design doc, NO code changes; research focus = all 4 dimensions. **DELIVERED:**
16     - **`docs/RALLY_QUALITY_RESEARCH.md`** (worktree `.claude/worktrees/docs-rally-quality`,
branch `docs/rally-quality-research`,
17     PR #184, **2 commits**): $1 measured gap, $2 code-grounded over-merge diagnosis, $3
harness inventory + 7 gaps, $4
18     strengthened eval framework (segmental Edit/F1@k as primary; golden growth + active
learning; Gemini = SHADOW signal
19     only; one-command eval entrypoint), $5 cited+adversarially-verified external research,
$6 cheap?durable roadmap w/
20     falsifiable criteria, **$7 PROJECT PLAN** (owner asked to "do this as a project") =
the single NUMBER to attribute
21     [multi-court tIoU-F1 baseline 0.157 ? final + gap-closure% vs Gemini 0.66; primary
lens = segmental F1@0.25/Edit/
22     merge-rate], a work-item status checklist (E1/W1/W2/W3/M1/M2/R1/R2 + corpus-growth C)
w/ acceptance criteria +
23     sequencing, and a DEFINITION OF DONE (Tier 0-2 land + number recorded; Tier-3 owned-
model = corpus-gated stretch),
24     $8 constraints. Indexed in docs/README.md. **Opened for review (foundational/strategy
? owner-gated merge).**
25     **2 owner decisions flagged in the PR comment:** (1) the ?50%-gap-closure success bar;
(2) Tier-3 in/out of the DoD.
26     - **Deep research** (`wambqniwk`, 110 agents, 27 sources, 25 claims 3-vote-verified, 23
confirmed/2 refuted). Headlines:
27     add an explicit **boundary head that SPLITS** (ASRF/TriDet ? research bets at n=6-16,
port the idea not the deep model);
28     **See et al.** badminton inactivity end-of-rally rule (?80% no-motion frames, ~95%
acc) + serve classifier = the
29     low-effort win; **TemporalMaxer** ? keep the scorer parameter-light (learned temporal
nets overfit small data);
30     **lead the leaderboard with segmental Edit + F1@k** (tIoU/MoF are blind to over-
segmentation = our exact failure).
31     Full report: `%TEMP%\?wambqniwk.output`. OPEN (research couldn't verify): active-
learning method + Gemini weak-label
32     governance ? flagged in $5.3 as a future grounded pass.
33     - **CORRECTION shipped:** over-merge is NOT fps (velocity already per-second-normalized
via `fps/dframes`); it's
34     "shuttle-moving ? rally" + 2s merge + no cap. Fixed `_LOCAL_vs_GEMINI_REPORT.md` +
`scratch/compare_local_vs_gemini.py`.
35     (The Phase-C checkpoint's "fps not invariant" line is SUPERSEDED by this ? recorded
here for the log.)
36     - ?? NEXT (owner): review/merge #184; then the cheapest testable win = Tier-0 (add
segmental Edit-score to leaderboard) +
37     Tier-1 (bounded windowing: max-window cap + inactivity split) on the 6-video LOVO
harness ? each its own measured PR.
38     Worktree `docs-rally-quality` removable after #184 merges.
39
40     **Checkpoint:** 2026-06-16 02:35 (CONTINUATION ? ? ALL 3 PHASES COMPLETE; #178 + #183
MERGED; Kushagra regen done +
41     verified; local-vs-Gemini report delivered). The expanded overnight mandat...
[/external_agent_tool_result]

```

187. assistant (2026-07-02T06:11:33.347Z)

Now make the norm durable in memory (the repo side is in PR #186) and log the checkpoint. These memory files live outside the repo, so they don't touch the PR.

188. assistant (2026-07-02T06:11:33.347Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\working-agreement-tracked-project-docs.md
[/external_agent_tool_call]
```

189. assistant (2026-07-02T06:11:33.347Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\working-agreement-tracked-project-docs.md (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

190. assistant (2026-07-02T06:11:33.347Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\MEMORY.md
[/external_agent_tool_call]
```

191. assistant (2026-07-02T06:11:33.347Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

192. assistant (2026-07-02T06:11:33.347Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]
```

193. assistant (2026-07-02T06:11:33.347Z)

```
[external_agent_tool_result: error]
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
[/external_agent_tool_result]
```

194. assistant (2026-07-02T06:11:33.347Z)

The partial read didn't register as a full read for edit-tracking. Let me read the top of the file explicitly, then prepend the checkpoint.

195. assistant (2026-07-02T06:11:33.347Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]
```

196. assistant (2026-07-02T06:11:33.347Z)

```
[external_agent_tool_result]
```

```

1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-16 ~03:30 (? RALLY-QUALITY RESEARCH + EVAL-FRAMEWORK FOUNDATION
DOC DONE ? **PR #184 OPEN for
13     owner review**, docs-only, NOT auto-merged). Owner asked for deep research to bridge the
local-vs-Gemini gap with a strong
14     harness/eval framework, checked into the repo as the foundation for quality iterations.
Scope (owner-picked) = research +
[/external_agent_tool_result]

```

197. assistant (2026-07-02T06:11:33.348Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]

```

198. assistant (2026-07-02T06:11:33.348Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md has been updated successfully. (file state is current in your
context ? no need to Read it back)
[/external_agent_tool_result]

```

199. assistant (2026-07-02T06:11:33.348Z)

Frozen and made a norm. Summary:

```

**Frozen ? [PR #186](https://github.com/avidullu/badminton-highlight-indexer/pull/186)** (branch
`docs/golden-regression-fixtures-plan`, one commit atop `90ab087`/#185, docs-only):
- `docs/GOLDEN_REGRESSION_FIXTURES.md` ? the plan, verbatim from what you reviewed.
- `docs/PROJECT_DOC_TEMPLATE.md` ? the reusable skeleton.
- `CLAUDE.md` workflow-conventions line + `docs/README.md` index entries.

```

```

**The norm** ? rather than invent one, I standardized the pattern this repo already grew
(`RALLY_QUALITY_RESEARCH.md §7`): substantial design/project work ? a tracked project doc with a
status header, decisions-locked, a **§7 progress tracker** (one ?/?/? row per shippable PR =
source of truth), and a **§9 definition-of-done**; archived when DONE. It's now codified three
ways: the template, the `CLAUDE.md` convention, and a memory ([[working-agreement-tracked-
project-docs]]) so I default to it across sessions.

```

Process notes (honest):

```

- Branching-safety held: branched from `origin/master`, staged only my 4 files (left
`artifacts/` and the untracked KHELSUTRA doc alone), re-verified HEAD was mine before commit.
- I did not merge ? this is owner-gated content (strategy/legal-adjacent), and per our
merging agreement I only self-merge verified-green *engineering* fixes. #186 is yours to
review/merge.
- Memory + `current-state` checkpoint updated so a future session resumes from the tracker.

```

When you're ready, **P0** (scrub the real names from `eval\_baselines/\*.json` +
`eval\_partial\_labels.py` + `test\_cleanup\_caches.py`, add a required CI leak-scan) is the clean
first implementation PR ? self-contained, no counsel needed. Just say the word and I'll take it.

200. user (2026-07-02T06:11:33.348Z)

can you make a project local memory to ensure that future projects follow the template and also FYI another session is iterating on the quality metrics for pure local (no gemini) segmentation so would your execution get hindered ? Feel free to make a different branch and start executing. Follow the regular discipline and use CI and gemini to ensure high quality results. Merge the PRs are you go along.

201. assistant (2026-07-02T06:11:33.348Z)

Good questions ? let me answer the concurrency one directly, because it's the crux, then act on it.

****Would the concurrent session hinder me? Yes, if we share one working tree**** ? that's the exact `[[gotcha-shared-checkout-branch-collisions]]` hazard. The other session iterating on local-seg quality metrics will be in ``backend/eval/`` and possibly ``eval_baselines/`` ? which overlaps my plan. Mitigations I'll apply:

- ****Isolate in a git worktree**** (the repo's established ``.claude/worktrees/`` pattern) so my edits never touch their checkout.
- ****Re-sequence to avoid collision:**** start with ****M1**** (the fixture framework ? `*all new, additive files*`: ``backend/eval/golden_fixtures.py``, ``eval_baselines/fixtures/``, new tests). ****Defer P0**** (the name-scrub) precisely `*because*` it rewrites keys in ``eval_baselines/*.json`` that their metrics work likely depends on ? doing that concurrently would rug-pull them. P0 lands once their eval churn settles or in coordination.

On the memory: I already created it last turn ? ****[[working-agreement-tracked-project-docs]]**** (auto-memory) ******* the convention line in ``CLAUDE.md`` (the `*project-local, checked-in*` enforcement, riding in PR #186). Merging #186 makes that enforcement live on master. Let me verify state and CI before I merge it and start M1.

202. assistant (2026-07-02T06:11:33.348Z)

```
[external_agent_tool_call: Bash]
description: Assess master, tree cleanliness, PR #186 mergeability, and CI status
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git fetch origin --quiet &&
echo "=== master vs my checkpoint ===" && git log --oneline -1 origin/master && echo "===
working tree (other session WIP?) ===" && git status --short && echo "=== #186 mergeability + CI
=== " && gh pr view 186 --json mergeable,mergeStateStatus,state 2>&1 | head -5 && echo "---
checks ---" && gh pr checks 186 2>&1 | head -10 && echo "=== recent Actions runs (is CI up?)
=== " && gh run list -L 3 2>&1 | head -6
[/external_agent_tool_call]
```

203. assistant (2026-07-02T06:11:33.348Z)

```
[external_agent_tool_result]
=== master vs my checkpoint ===
a3b209a feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge (?F1 +0.139, 6/0)
(#187)
=== working tree (other session WIP?) ===
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
=== #186 mergeability + CI ===
{"mergeStateStatus":"CLEAN","mergeable":"MERGEABLE","state":"OPEN"}
--- checks ---
Full test suite (offline, mocked)      pass    2m0s    https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27593679980/job/81579445780
Mypy (lenient baseline) pass    33s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27593679980/job/81579445825
Ruff lint      pass    15s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27593679980/job/81579445782
Syntax sweep + import smoke (no GPU stack) pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27593679980/job/81579445805
=== recent Actions runs (is CI up?) ===
completed      success feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge?
```



```
CI      master  push      27593871028      2m17s  2026-06-16T04:19:19Z
completed      success feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge
(?F1 +0.139, 6/0) CI      feat/tier1-bounded-windowing      pull_request      27593854352      2m37s
2026-06-16T04:18:43Z
completed      cancelled      feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-
merge (?F1 +0.139, 6/0) CI      feat/tier1-bounded-windowing      pull_request      27593836591
53s      2026-06-16T04:18:10Z
[/external_agent_tool_result]
```

204. assistant (2026-07-02T06:11:33.348Z)

Key findings from the assessment:

- ****CI is back up**** (billing outage resolved) ? #186's checks all ****pass****, and it's ****MERGEABLE/CLEAN****.
- ****The concurrent session just merged #187**** ? `feat(windowing): Tier 1 (W1) bounded windowing (?F1 +0.139)`. That's directly relevant: they're actively reshaping `windowing.py`, the ***exact*** behavior my characterization gate would freeze.

That sharpens my collision answer. Let me merge #186 (makes the template/convention live = fulfills the memory ask), then set up an isolated worktree off the latest master.

205. assistant (2026-07-02T06:11:33.348Z)

```
[external_agent_tool_call: Bash]
description: Merge #186, fetch, and create an isolated worktree for M1 off origin/master
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && gh pr merge 186 --squash
2>&1 | head -5 && echo "=== fetch + new master ===" && git fetch origin --quiet && git log
--oneline -2 origin/master && echo "=== create isolated worktree off origin/master ===" && git
worktree add .claude/worktrees/golden-fixtures -b feat/golden-fixtures-m1 origin/master 2>&1 |
tail -3 && echo "=== worktree HEAD ===" && git -C .claude/worktrees/golden-fixtures log
--oneline -1
[/external_agent_tool_call]
```

206. assistant (2026-07-02T06:11:33.348Z)

```
[external_agent_tool_result]
=== fetch + new master ===
1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template (#186)
a3b209a feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge (?F1 +0.139, 6/0)
(#187)
=== create isolated worktree off origin/master ===
Preparing worktree (new branch 'feat/golden-fixtures-m1')
branch 'feat/golden-fixtures-m1' set up to track 'origin/master'.
HEAD is now at 1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc
template (#186)
=== worktree HEAD ===
1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template (#186)
[/external_agent_tool_result]
```

207. assistant (2026-07-02T06:11:33.348Z)

Excellent position now:

- ****#186 merged**** ? master is `1de2817`; the template + the `CLAUDE.md` convention are ****live****. That's the project-local enforcement of "future projects follow the template" ? done and now permanent, not just in my memory.
- ****Isolated worktree**** `feat/golden-fixtures-m1` created off the latest `origin/master` (includes #187). My edits there can't touch the other session's checkout ? so, concretely: ****no**, my execution won't be hindered, and I won't hinder theirs.**

One real coordination note I'll act on: since they're actively reshaping `windowing.py` (#187, likely more), I'm starting with ****M1**** (the additive framework: `backend/eval/golden\_fixtures.py` + synthetic fixtures + a CONTROL-only, parse-compared characterization test) and ****deferring P0**** (the `eval\_baselines/\*.json` name-scrub) so I don't rewrite keys their metrics work depends on.

Let me map the exact APIs I need (from the post-#187 worktree, so I build against current behavior) before writing the helper.

208. assistant (2026-07-02T06:11:33.348Z)

```
[external_agent_tool_call: Agent]
description: Map eval APIs for replay helper
[/external_agent_tool_call]
```

209. assistant (2026-07-02T06:11:33.348Z)

```
[external_agent_tool_result]
Based on my thorough read of the golden-fixtures worktree, here is the EXACT API mapping you requested:
```

1. EXACT FUNCTION SIGNATURES + RETURN TYPES

****tracknet_runner.py****

```
```python
parse_trajectory_csv(csv_win_path: str) -> List[TrajectoryPoint]
 # Returns sorted list of TrajectoryPoint(frame: int, visible: bool, x: float, y: float)
 # Parses CSV with columns: Frame, Visibility (0|1), X, Y
 # Silently skips malformed rows (KeyError/ValueError)

trajectory_to_action_windows(
 points: List[TrajectoryPoint],
 fps: float,
 frame_width: float,
 chunk_start: float = 0.0,
 velocity_thresh: float = 0.02,
 merge_gap: float = 2.0,
 min_window_duration: float = 2.0,
 chunk_index: Optional[int] = None,
 max_window_duration: float = 0.0,
 min_split_gap: float = 0.5,
) -> List[Dict[str, Any]]
 # Returns: [{
 # "start": float (seconds),
 # "end": float (seconds),
 # "active_frame_count": int,
 # "peak_velocity": float,
 # "mean_velocity": float,
 # "track_id_count": int (always 1),
 # "chunk_index": Optional[int]
 # }]
 # OCCLUSION: visible==False breaks trajectory (sets prev=None, resets velocity chain)
 # CLAMPS: fps<=0 ? 30.0; frame_width<=0 ? 1280.0
 # BOUNDED WINDOWING (#187): max_window_duration > 0 triggers _split_overlong_groups()
 # ? recursively splits windows > max_window_duration at largest inactivity gap
 # (gap >= min_split_gap frames), never merges, only subdivides
```
```

****windowing.py****

```
```python
@dataclass(frozen=True)
class WindowingPreset:
 name: str
 velocity_thresh: float
 merge_gap: float
 min_window_duration: float
 max_window_duration: float = 0.0
```

```

def as_tuple(self) -> Tuple[float, float, float]
 # Returns (velocity_thresh, merge_gap, min_window_duration)

def to_dict(self) -> Dict[str, Any]

SERVED: WindowingPreset # sourced from IndexingConfig defaults
 # name="served", velocity_thresh=0.02, merge_gap=2.0,
 # min_window_duration=2.0, max_window_duration=0.0 (default OFF)

PROPOSAL: WindowingPreset
 # name="proposal", velocity_thresh=0.005, merge_gap=1.0,
 # min_window_duration=0.5, max_window_duration=0.0

PRESETS: Dict[str, WindowingPreset] # {"served": SERVED, "proposal": PROPOSAL}

served_windowing_from_config(config: Any, family: str = "wasb") -> WindowingPreset
 # Reads actual served config (overrides honoured)

build_windows(
 points: List[Any],
 fps: float,
 frame_width: float,
 preset: WindowingPreset
) -> List[Dict[str, Any]]
 # Thin wrapper: calls TrackNetRunner.trajectory_to_action_windows with preset params

windows_to_intervals(windows: List[Dict[str, Any]]) -> List[Interval]
 # Returns [(w["start"], w["end"])] for w in windows
 # Interval = Tuple[float, float]
...

```

#### **\*\*distill\_local.py\*\***

```

```python
FEATURE_NAMES: List[str] = ["duration", "peak_velocity", "mean_velocity", "active_ratio",
"net_crossings"]
PROPOSAL = (0.005, 1.0, 0.5) # velocity_thresh, merge_gap, min_window_duration
BASELINE_LOCAL = (0.02, 2.0, 2.0) # the SERVED windowing

build_candidates(
    trajectory_csv: str,
    fps: float,
    frame_width: float,
    proposal: Tuple[float, float, float] = PROPOSAL
) -> Tuple[List[Interval], List[List[float]]]
    # Returns (candidate_intervals, feature_rows)
    # Interval = Tuple[float, float]
    # feature_rows: each row = [duration, peak_velocity, mean_velocity, active_ratio,
net_crossings]
    # Net crossings sourced from rally_gate.gate_windows(..., min_crossings=0) ? .crossings field
...

```

****experiment.py****

```

```python
@dataclass
class VideoCandidates:
 name: str
 intervals: List[Interval]
 features: Dict[str, List[float]] = field(default_factory=dict)
 # Column-major: feature_name ? list of per-candidate values
 gts: Optional[List[Interval]] = None
 # Interval = Tuple[float, float]

@dataclass

```

```

class Variant:
 name: str
 segmenter: str = "wasb_hybrid"
 config_overrides: Dict[str, Any] = field(default_factory=dict)
 cu...
[/external_agent_tool_result]

```

#### 210. assistant (2026-07-02T06:11:33.349Z)

I have the exact APIs and confirmed the gotchas. Critically, the **CONTROL** path is pure-stdlib (no numpy ? cross-machine safe), and I'll **bypass** ``fusion_compare`/`ablation`` loaders (they read ``c[person]`` unconditionally ? would crash on person-stripped fixtures) by building ``VideoCandidates`` directly. That's the person-optional Tier-0 loader the red-team asked for.

I'll implement **M1** in the isolated worktree via a focused coding pass, then verify (full suite + ruff + mypy) and merge myself. Spawning it in the background so it can write + run the full suite while I stay responsive:

#### 211. assistant (2026-07-02T06:11:33.349Z)

```

[external_agent_tool_call: Agent]
description: Implement M1 golden fixtures framework
[/external_agent_tool_call]

```

#### 212. assistant (2026-07-02T06:11:33.349Z)

```

[external_agent_tool_result]
Async agent launched successfully.
agentId: a724501ac1d1831fb (internal ID - do not mention to user. Use SendMessage with to:
'a724501ac1d1831fb' to continue this agent.)
The agent is working in the background. You will be notified automatically when it completes.
Do not duplicate this agent's work ? avoid working with the same files or topics it is using.
Work on non-overlapping tasks, or briefly tell the user what you launched and end your response.
output_file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\a724501ac1d1831fb.output
Do NOT Read or tail this file via the shell tool ? it is the full subagent JSONL transcript and
reading it will overflow your context. If the user asks for progress, say the agent is still
running; you'll get a completion notification.
[/external_agent_tool_result]

```

#### 213. assistant (2026-07-02T06:11:33.349Z)

Status ? here's where things stand after this turn:

**Concurrency: resolved, both directions.** I'm working in an **isolated git worktree** (``feat/golden-fixtures-m1``, the repo's ``claude/worktrees/`` pattern) off the latest ``origin/master``. My edits there can't touch the other session's checkout and vice-versa. I also **re-sequenced** to avoid semantic collision: starting with **M1** (additive new files) and **deferring P0** (the ``eval_baselines/*.json`` name-scrub) so I don't rewrite keys their windowing/metrics work depends on. Bonus: their #187 added the bounded-windowing *capability* but left it **OFF** in the SERVED preset, and I pinned each fixture's preset params explicitly ? so my snapshots are robust to a SERVED retune (only an algorithm change churns them, and that's a documented one-command ``--refresh-snapshot``).

**The template norm is now LIVE and enforced** (your memory ask): **PR #186 is merged** (``master` = `1de2817``). That puts the ``CLAUDE.md`` workflow convention + ``docs/PROJECT_DOC_TEMPLATE.md`` on master ? the *project-local, checked-in* enforcement that future projects follow the template. Plus the cross-session memory `[[working-agreement-tracked-project-docs]]`. CI was green for it (the billing outage is resolved).

**Executing M1 now.** A background coding agent is implementing the framework in the worktree:

- ``backend/eval/golden_fixtures.py`` ? the shared ``replay_fixture()`` (CONTROL/pure-stdlib, **no numpy** ? cross-machine stable) + a freeze/refresh CLI.
- 3 **synthetic** fixtures (generated by the freeze tool, not hand-authored) + manifest +

```
`.gitattributes` LF pin.
- `tests/test_golden_fixtures.py` ? parse-compared characterization (exact-int / approx-float),
a positive "no person/audio keys" privacy assertion, a perturbation test proving windows are re-
derived, and an idempotent-freeze test.
```

It bypasses the `fusion\_compare`/`ablation` loaders (which `KeyError` on person-free fixtures ?  
a verified red-team catch) by building `VideoCandidates` directly.

When it finishes I'll **verify myself** (full `run\_all\_tests.py` + `ruff` + `mypy`, per the pre-  
LGTM gate), review the diff, and **merge** M1 with CI green ? then continue down the S7 tracker  
(M2 freeze-tool hardening ? M3/M4 gates ? P1 de-attribution tooling), holding P0 until the  
windowing churn settles and P2/P3 for counsel. I'll flag you as each lands.

## 214. user (2026-07-02T06:11:33.349Z)

```
<task-notification>
<task-id>a724501ac1d1831fb</task-id>
<tool-use-id>toolu_01UGEHYaZQb6AAgvR51nQaTN</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users\avidu-Projects-badminton-
highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\a724501ac1d1831fb.output</output-
file>
<status>completed</status>
<summary>Agent "Implement M1 golden fixtures framework" completed</summary>
<result>M1 is complete and fully verified. Here is the report.
```

## Summary

Implemented deliverable **M1** (Fixture framework + synthetic Tier-0 + shared replay helper) of  
the Golden Regression Fixtures project, entirely within the `golden-fixtures` worktree. No git  
commands beyond read-only `status` for verification.

## Files created

```
- `backend/eval/golden_fixtures.py` ? the shared module: `CURRENT_SCHEMA=1`, `FIXTURES_DIR`
(resolved from `__file__`, parents[2]), `make_synthetic_trajectory`, `replay_fixture`,
`freeze_fixture`, `freeze_synthetic`, `refresh_snapshot`, `load_manifest`, `load_fixture`,
`compare_expected`, and the `__main__` CLI (`--freeze-synthetic` / `--refresh-snapshot`
[`--slug|--all`] / `--check`).
- `eval_baselines/fixtures/&slug;/{trajectory.csv, labels.csv, expected.json,
provenance.json}` for the 3 slugs (generated by `--freeze-synthetic`, never hand-written).
- `eval_baselines/fixtures/manifest.json` ? array of `{slug, fps, frame_width,
windowing:{vt,mg,mwd,max_win}, kind:"synthetic", schema_version, paths}`.
- `eval_baselines/fixtures/README.md` ? guardrails + freeze/refresh commands + "generated, never
hand-edited" + S6 honest-limits.
- `.gitattributes` (worktree root) ? `eval_baselines/fixtures/** text eol=lf`.
- `tests/test_golden_fixtures.py` ? parametrized over `load_manifest()` slugs.
```

## Verify results (verbatim)

```
1. `--freeze-synthetic` ? froze 3 synthetic fixture(s): clean_single_rally,
multi_rally_with_deadtime, occlusion_break`. `--check` ? all `[ok]`, `EXIT=0`.
2. `python run_all_tests.py` ? 744 passed, 7 skipped (the pre-existing skips; new file adds
17 tests, all green). With CI coverage flags: Total coverage 80.36% ("Required test coverage
of 70% reached"); the new module itself is at 85%.
3. `ruff check .` ? All checks passed! (EXIT 0). `mypy backend training` ? Success: no issues
found in 103 source files` (EXIT 0). One initial ruff finding (unused `field` import) fixed.
```

## The 3 synthetic scenarios (num candidates, num segments, F1)

slug	candidates	segments	F1	what it exercises
clean_single_rally	1	1	1.0	one ~4s in-flight rally ? single window, net_crossings=7
multi_rally_with_deadtime	2	2	1.0	two rallies split by 3s of <b>still</b> (low-velocity) dead-time
occlusion_break	2	2	1.0	two rallies split by 3s of <b>occlusion</b> (Visibility=0)

## Hard-constraint compliance

- **CONTROL/pure-stdlib only**: builds `VideoCandidates` directly with only the 5 shuttle columns and replays via `experiment.predict(CONTROL, vc)`; never calls `fusion\_compare.load\_videos`/`ablation.load\_videos` (verified `CONTROL.is\_learned()` is False).
- **Parse-compare, never byte-compare**: `compare\_expected` does `json.load` on both sides ? exact on int/structural fields (candidate count, `net\_crossings` as int, TP/FP/FN, segment count, slug, schema, gt\_hash) and abs-tol 1e-6 on floats.
- **Windowing pinned per fixture**: `(vt=0.02, mg=2.0, mwd=2.0, max\_win=0.0)` written into each `provenance.json` and the manifest, read back and passed explicitly to `build\_candidates` ? does not depend on the named `SERVED` constant.
- **Determinism**: trajectories from deterministic sine formulae (no `random`); all floats rounded to 6dp; UTF-8 / LF-only / trailing-newline confirmed.
- **Privacy**: every fixture carries `kind:"synthetic"`, `minors\_free:true`, `license:"synthetic"`; `expected.json` has no person/pose/gait/face/audio/player fields (positive assertion test scans the raw text and the parsed candidate keys).

Beyond the spec's required tests I added `test\_check\_cli\_passes`, `test\_synthetic\_provenance\_tags`, `test\_control\_is\_not\_learned`, and `test\_manifest\_is\_non\_empty`, and I additionally demonstrated out-of-band that `--check` returns exit 1 on a seeded trajectory perturbation (then restored) ? proving the gate catches behaviour changes, not just echoes the snapshot.

## Deviations from spec

- **One** minor wording in `test\_perturbing\_trajectory\_changes\_segments`: I perturb by **truncating** the trajectory to its first half (which shrinks the derived window's `end`) rather than deleting an interior hole. Reason: deleting interior rows does not move the segment boundaries ? `merge\_gap=2.0s` coalesces across the gap and the surviving rows keep their original frame indices, so start/end are unchanged. Truncation guarantees a structural change and still proves the windows are re-derived from the CSV (not echoed). No API or fixture-content deviation otherwise.

Note: numpy and cv2 *are* installed in this environment, so the full suite ran here rather than skipping ? but the M1 replay path itself touches no numpy (CONTROL/stdlib), so it remains valid on a clean no-GPU clone.</result>  
<usage><subagent\_tokens>127378</subagent\_tokens><tool\_uses>40</tool\_uses><duration\_ms>620623</duration\_ms></usage>  
</task-notification>

### 215. assistant (2026-07-02T06:11:33.349Z)

M1 reports green, but per the pre-LGTM gate I verify myself ? never trust the report. Let me review the actual files and re-run the suite, ruff, and mypy in the worktree.

### 216. assistant (2026-07-02T06:11:33.349Z)

```
[external_agent_tool_call: Bash]
description: Show worktree status and the new fixture files
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/golden-
fixtures" && echo "=== worktree status ===" && git status --short && echo "=== HEAD ===" && git
log --oneline -1 && echo "=== fixture files ===" && git status --short eval_baselines/fixtures/
| head -30
[/external_agent_tool_call]
```

### 217. assistant (2026-07-02T06:11:33.349Z)

```
[external_agent_tool_result]
=== worktree status ===
?? .gitattributes
?? backend/eval/golden_fixtures.py
?? eval_baselines/fixtures/
?? tests/test_golden_fixtures.py
=== HEAD ===
1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template (#186)
=== fixture files ===
```

```
?? eval_baselines/fixtures/
[/external_agent_tool_result]
```

## 218. assistant (2026-07-02T06:11:33.349Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\golden-
fixtures\backend\eval\golden_fixtures.py
[/external_agent_tool_call]
```

## 219. assistant (2026-07-02T06:11:33.349Z)

```
[external_agent_tool_result]
1 """Golden Regression Fixtures (M1) ? video-free, CONTROL-only characterization harness.
2
3 `docs/GOLDEN_REGRESSION_FIXTURES.md` ($4, $6, $7). This module is the **single shared
4 replay code path** used by both the freeze tool and the characterization gate, and it
5 deliberately stays on the **pure-stdlib CONTROL path** (D5 in that doc): the learned
6 variants train a numpy/OpenBLAS logistic regression whose near-threshold decisions flip
7 across CPUs, so they can never anchor a cross-machine byte/parse snapshot. Here we
replay
8 only `experiment.CONTROL` (`is_learned() == False`), which keeps all candidate
intervals,
9 merges overlaps, and uses **no numpy**.
10
11 The replay reuses the already-shipped eval stack VERBATIM::
12
13 parse_trajectory_csv ? distill_local.build_candidates(proposal=pinned)
14 ? VideoCandidates(name, intervals, features={5 shuttle cols},
gts=load_gt_intervals)
15 ? experiment.predict(CONTROL, vc) ? metrics.score(preds, gts, iou)
16
17 Tier-0 / privacy invariants (D3, $4c, $6 of the doc), enforced here by construction:
18
19 * Fixtures are **synthetic** ? deterministic formulae, NO randomness ? so they carry
zero
20 provenance/licensing risk and are safe to commit publicly (owner Q7 "synthetic-only").
21 * Every fixture is tagged ``kind="synthetic"`, ``minors_free=true``,
``license="synthetic"``.
22 * ``expected.json`` carries **only the 5 shuttle feature columns** ? there is, by
design, no
23 person / pose / gait / face / audio field anywhere in the schema (positive-assertion
test).
24 * The windowing preset params ``(vt, mg, mwd, max_win)`` are **pinned inside each
fixture's
25 provenance** and passed explicitly to ``build_candidates`` ? we do NOT rely on the
named
26 ``windowing.SERVED`` constant, so a future SERVED retune cannot silently churn
fixtures.
27
28 $6 honest-limits caveat (carried in the CLI / test messages, not only the docs): a green
29 suite proves only that the deterministic post-trajectory transforms are unchanged for
the
30 frozen synthetic cases ? NOT that the detector/decoder/AI handoff/DB are correct, and
NOT
31 that rally detection is good. Synthetic-tier numbers measure plumbing correctness, never
32 field quality (never cite them in QUALITY_ITERATIONS.md / the paper).
33
34 Pure + offline + stdlib. The replay imports `tracknet_runner` (which is itself stdlib-
only
35 for the parse/window math ? no TF), but pulls in no numpy/cv2 on the CONTROL path.
36 ""
37 from __future__ import annotations
38
39 import argparse
```

```

40 import json
41 import math
42 import sys
43 from dataclasses import dataclass
44 from pathlib import Path
45 from typing import Any, Dict, List, Optional, Sequence, Tuple
46
47 # --- shipped eval stack, reused verbatim (CONTROL / pure-stdlib path only)
48 -----
49 from backend.eval import distill_local
50 from backend.eval.experiment import CONTROL, VideoCandidates, predict
51 from backend.eval.gt_loader import load_gt_intervals
52 from backend.eval.metrics import score
53 from backend.eval.regression import gt_hash
54
55 Interval = Tuple[float, float]
56
57 #: Bump when the fixture/expected schema changes shape. Loaders refuse a fixture whose
58 #: ``schema_version`` exceeds this (forward-incompatible), per the schema-drift guard.
59 CURRENT_SCHEMA = 1
60
61 #: Repo-root-relative fixtures dir, resolved from THIS file (never cwd) so the harness
62 works
63 #: from any working directory / a fresh clone. ``parents[2]`` = backend/eval ? backend ?
64 root.
65 FIXTURES_DIR: Path = Path(__file__).resolve().parents[2] / "eval_baselines" / "fixtures"
66
67 #: The pinned SERVED windowing the synthetic fixtures freeze at: (velocity_thresh,
68 merge_gap,
69 #: min_window_duration). Mirrors the historical SERVED values (0.02, 2.0, 2.0) but is a
70 LITERAL
71 #: here on purpose ? the spec forbids depending on the named ``windowing.SERVED``
72 constant so a
73 #: future retune of SERVED can never silently re-baseline a committed fixture. Ea...
74 [/external_agent_tool_result]

```

## 220. assistant (2026-07-02T06:11:33.349Z)

The module is high quality ? CONTROL-only, parse-compare, pinned preset, person-free by construction, deterministic writers. Let me review the test + a generated fixture, then run the verification myself.

## 221. assistant (2026-07-02T06:11:33.349Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\golden-
fixtures\tests\test_golden_fixtures.py
[/external_agent_tool_call]

```

## 222. assistant (2026-07-02T06:11:33.349Z)

```

[external_agent_tool_result]
1 """Golden Regression Fixtures (M1) ? characterization tests over the committed synthetic
2 Tier-0.
3 Parametrized over ``load_manifest()`` so every committed fixture is exercised. The gate
4 is
5 **parse-compare** (never byte-compare), **CONTROL-only** (pure stdlib, no numpy on the
6 replay
7 path), and asserts the Tier-0 privacy invariant positively (no
8 person/audio/pose/gait/face keys).
9
10 $6 honest-limits caveat: green here proves only that the deterministic post-trajectory
11 transforms

```



```

8 are unchanged for the frozen synthetic cases ? characterization is behaviour-locking,
NOT
9 correctness, and synthetic-tier numbers measure plumbing, not field quality.
10 """
11 import json
12
13 import pytest
14
15 from backend.eval import golden_fixtures as gf
16
17 # Slugs discovered from the committed manifest ? the suite grows by accretion
automatically.
18 MANIFEST = gf.load_manifest()
19 SLUGS = [entry["slug"] for entry in MANIFEST]
20
21
22 def test_manifest_is_non_empty():
23 """A committed manifest with at least one fixture must exist (else nothing is
gated)."""
24 assert SLUGS, "no fixtures in eval_baselines/fixtures/manifest.json ? run --freeze-
synthetic"
25
26
27 @pytest.mark.parametrize("slug", SLUGS)
28 def test_replay_matches_committed_expected(slug):
29 """Replay the fixture and parse-compare against its committed expected.json.
30
31 Characterization = behaviour-locking, NOT correctness (§6): a non-empty mismatch
list means
32 the deterministic post-trajectory behaviour CHANGED for this frozen case, not that
it is wrong.
33 """
34 committed = gf.load_fixture(slug)["expected"]
35 recomputed = gf.replay_fixture(slug)
36 mismatches = gf.compare_expected(recomputed, committed)
37 assert mismatches == [], (
38 f"{slug}: behaviour changed vs frozen snapshot (characterization, not
correctness ? §6):\n"
39 + "\n".join(f" - {m}" for m in mismatches)
40)
41
42
43 @pytest.mark.parametrize("slug", SLUGS)
44 def test_schema_version_le_current(slug):
45 """Every committed fixture's schema_version must be <= CURRENT_SCHEMA (forward-
compatible)."""
46 fx = gf.load_fixture(slug)
47 assert int(fx["expected"]["schema_version"]) <= gf.CURRENT_SCHEMA
48 assert int(fx["provenance"]["schema_version"]) <= gf.CURRENT_SCHEMA
49
50
51 @pytest.mark.parametrize("slug", SLUGS)
52 def test_no_person_or_audio_keys(slug):
53 """Positive privacy assertion (D3 / §4c): NO biometric/identity/non-shuttle key
appears
54 anywhere in the committed expected.json ? not trusting a silent 0.0 default."""
55 fdir = gf.FIXTURES_DIR / slug
56 raw = (fdir / "expected.json").read_text(encoding="utf-8").lower()
57 for forbidden in gf.FORBIDDEN_KEYS:
58 assert forbidden not in raw, (
59 f"{slug}: forbidden key substring {forbidden!r} present in expected.json
(Tier-0 leak)")
60
61 expected = gf.load_fixture(slug)["expected"]
62 for i, cand in enumerate(expected["candidates"]):

```

```

63 # The only feature block is the shuttle block, and it holds EXACTLY the 5
shuttle cols.
64 assert set(cand.keys()) == {"start", "end", "shuttle"}, f"{slug}: candidate[{i}]
has extra keys"
65 assert set(cand["shuttle"].keys()) == set(gf.SHUTTLE_FEATURES), (
66 f"{slug}: candidate[{i}].shuttle is not exactly the 5 shuttle features")
67
68
69 @pytest.mark.parametrize("slug", SLUGS)
70 def test_synthetic_provenance_tags(slug):
71 """Tier-0 hard tags: kind=synthetic, license=synthetic, minors_free=true (constraint
5)."""
72 prov = gf.load_fixture(slug)["provenance"]
73 assert prov["kind"] == "synthetic"
74 assert prov["license"] == "synthetic"
75 assert prov["minors_free"] is True
76 # Windowing pinned in the fixture (not the named SERVED constant) ? constraint 3.
77 assert set(prov["windowing"]) == {"vt", "mg", "mwd", "max_win"}
78
79
80 def test_control_is_not_learned():
81 """The replay mus...
[/external_agent_tool_result]

```

### 223. assistant (2026-07-02T06:11:33.349Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\golden-
fixtures\eval_baselines\fixtures\clean_single_rally\expected.json
[/external_agent_tool_call]

```

### 224. assistant (2026-07-02T06:11:33.350Z)

```

[external_agent_tool_result]
1 {
2 "candidates": [
3 {
4 "end": 3.966667,
5 "shuttle": {
6 "active_ratio": 1.008475,
7 "duration": 3.933333,
8 "mean_velocity": 1.27678,
9 "net_crossings": 7,
10 "peak_velocity": 1.970041
11 },
12 "start": 0.033333
13 }
14],
15 "control_segments": [
16 [
17 0.033333,
18 3.966667
19]
20],
21 "gt_hash": "6c93801fd607",
22 "iou": 0.5,
23 "schema_version": 1,
24 "score": {
25 "f1": 1.0,
26 "false_negatives": 0,
27 "false_positives": 0,
28 "iou_threshold": 0.5,
29 "mean_end_error": 0.033333,
30 "mean_iou": 0.983333,
31 "mean_start_error": 0.033333,

```

```

32 "num_gt": 1,
33 "num_pred": 1,
34 "precision": 1.0,
35 "recall": 1.0,
36 "true_positives": 1
37 },
38 "slug": "clean_single_rally"
39 }
40
[/external_agent_tool_result]

```

## 225. assistant (2026-07-02T06:11:33.350Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\golden-
fixtures\eval_baselines\fixtures\manifest.json
[/external_agent_tool_call]

```

## 226. assistant (2026-07-02T06:11:33.350Z)

```

[external_agent_tool_result]
1 [
2 {
3 "fps": 30.0,
4 "frame_width": 1280.0,
5 "kind": "synthetic",
6 "paths": {
7 "dir": "clean_single_rally",
8 "expected": "clean_single_rally/expected.json",
9 "labels": "clean_single_rally/labels.csv",
10 "provenance": "clean_single_rally/provenance.json",
11 "trajectory": "clean_single_rally/trajectory.csv"
12 },
13 "schema_version": 1,
14 "slug": "clean_single_rally",
15 "windowing": {
16 "max_win": 0.0,
17 "mg": 2.0,
18 "mwd": 2.0,
19 "vt": 0.02
20 }
21 },
22 {
23 "fps": 30.0,
24 "frame_width": 1280.0,
25 "kind": "synthetic",
26 "paths": {
27 "dir": "multi_rally_with_deadtime",
28 "expected": "multi_rally_with_deadtime/expected.json",
29 "labels": "multi_rally_with_deadtime/labels.csv",
30 "provenance": "multi_rally_with_deadtime/provenance.json",
31 "trajectory": "multi_rally_with_deadtime/trajectory.csv"
32 },
33 "schema_version": 1,
34 "slug": "multi_rally_with_deadtime",
35 "windowing": {
36 "max_win": 0.0,
37 "mg": 2.0,
38 "mwd": 2.0,
39 "vt": 0.02
40 }
41 },
42 {
43 "fps": 30.0,
44 "frame_width": 1280.0,

```

```

45 "kind": "synthetic",
46 "paths": {
47 "dir": "occlusion_break",
48 "expected": "occlusion_break/expected.json",
49 "labels": "occlusion_break/labels.csv",
50 "provenance": "occlusion_break/provenance.json",
51 "trajectory": "occlusion_break/trajectory.csv"
52 },
53 "schema_version": 1,
54 "slug": "occlusion_break",
55 "windowing": {
56 "max_win": 0.0,
57 "mg": 2.0,
58 "mwd": 2.0,
59 "vt": 0.02
60 }
61 }
62]
63
[/external_agent_tool_result]

```

## 227. assistant (2026-07-02T06:11:33.350Z)

Both the test file and the generated fixture look correct (parametrized over the manifest, positive privacy assertion, perturbation + idempotence tests; `clean\_single\_rally` ? 1 candidate, `net\_crossings:7` as int, F1=1.0, floats at 6dp). Now my own verification gate ? full suite + ruff + mypy + the `--check` CLI, in the worktree:

## 228. assistant (2026-07-02T06:11:33.350Z)

```

[external_agent_tool_call: Bash]
description: Run check CLI, full suite with coverage, ruff, and mypy in the worktree
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/golden-
fixtures" && echo "==== 1. --check CLI ====" && python -m backend.eval.golden_fixtures
--check; echo "CHECK_EXIT=$?"; echo "==== 2. full suite (coverage, CI floor 70) ====" &&
python -m pytest -q --cov=backend --cov=training --cov-fail-under=70 2>&1 | tail -25; echo
"PYTEST_EXIT=${PIPESTATUS[0]}"; echo "==== 3. ruff ====" && ruff check . 2>&1 | tail -5; echo
"RUFF_EXIT=$?"; echo "==== 4. mypy ====" && mypy backend training 2>&1 | tail -8; echo
"MYPY_EXIT=$?"
[/external_agent_tool_call]

```

## 229. assistant (2026-07-02T06:11:33.350Z)

```

[external_agent_tool_result]
==== 1. --check CLI ====
[ok] clean_single_rally: 1 candidates, 1 segments, F1=1.000000
[ok] multi_rally_with_deadtime: 2 candidates, 2 segments, F1=1.000000
[ok] occlusion_break: 2 candidates, 2 segments, F1=1.000000
[golden-fixtures] 3/3 fixtures match (synthetic-tier = plumbing correctness, not field quality ?
$6)
CHECK_EXIT=0
==== 2. full suite (coverage, CI floor 70) ====
backend\providers\gemini.py 98 19 81%
backend\reporting__init__.py 26 2 92%
backend\reporting\harvest.py 111 8 93%
backend\reporting\run_signals.py 14 0 100%
backend\results.py 22 0 100%
backend\sports__init__.py 17 1 94%
backend\sports\badminton.py 13 1 92%
backend\sports\base.py 12 3 75%
backend\tools__init__.py 0 0 100%
backend\tools\export.py 75 2 97%
backend\tools\rally_slicer.py 104 47 55%
backend\utils__init__.py 0 0 100%

```

backend\utils\geometry.py	29	0	100%
backend\utils\hashing.py	9	0	100%
backend\utils\logging_setup.py	11	0	100%
backend\utils\paths.py	27	3	89%
backend\utils\run_telemetry.py	126	12	90%
backend\utils\video.py	25	3	88%
training\__init__.py	0	0	100%
training\gen0\__init__.py	0	0	100%
training\gen0\harness.py	108	11	90%

-----  
TOTAL 7552 1483 80%

Required test coverage of 70% reached. Total coverage: 80.36%

744 passed, 7 skipped, 7 warnings in 125.68s (0:02:05)

PYTEST\_EXIT=0

===== 3. ruff =====

All checks passed!

RUFF\_EXIT=0

===== 4. mypy =====

backend\annotations\base.py:85: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]

backend\sports\\_\_init\_\_.py:7: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]

backend\pipeline\validators\base.py:36: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]

backend\providers\\_\_init\_\_.py:7: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]

Success: no issues found in 103 source files

MYPY\_EXIT=0

[/external\_agent\_tool\_result]

### 230. assistant (2026-07-02T06:11:33.350Z)

Independently verified ? all green:

- `--check`: 3/3 fixtures match, exit 0.
- \*\*Full suite: 744 passed, 7 skipped\*\*; coverage \*\*80.36% ? 70%\*\* floor.
- ruff: clean. mypy: clean (103 files).

That clears my pre-LGTM gate. Committing with branching-safety discipline (re-verify HEAD, explicit paths only), then pushing and opening the PR:

### 231. assistant (2026-07-02T06:11:33.350Z)

[external\_agent\_tool\_call: Bash]

description: Fetch, verify branch, stage explicit M1 paths, confirm nothing else swept

command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/golden-fixtures" && git fetch origin --quiet && echo "=== origin/master ===" && git log --oneline -1 origin/master && echo "=== my branch/HEAD ===" && git branch --show-current && git log --oneline -1 && git add backend/eval/golden\_fixtures.py tests/test\_golden\_fixtures.py eval\_baselines/fixtures .gitattributes && echo "=== staged (explicit paths only) ===" && git diff --cached --name-only && echo "=== nothing else swept ===" && git status --short | grep -v '^A ' | head

[/external\_agent\_tool\_call]

### 232. assistant (2026-07-02T06:11:33.350Z)

[external\_agent\_tool\_result]

=== origin/master ===

a471745 feat(eval): Tier 1 (W2) ? net-crossings rally-state gate on W1 (?F1 +0.019, 6/0) (#188)

=== my branch/HEAD ===

feat/golden-fixtures-m1

1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template (#186)

warning: in the working copy of '.gitattributes', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'backend/eval/golden\_fixtures.py', LF will be replaced by CRLF

```

the next time Git touches it
warning: in the working copy of 'tests/test_golden_fixtures.py', LF will be replaced by CRLF the
next time Git touches it
=== staged (explicit paths only) ===
.gitattributes
backend/eval/golden_fixtures.py
eval_baselines/fixtures/README.md
eval_baselines/fixtures/clean_single_rally/expected.json
eval_baselines/fixtures/clean_single_rally/labels.csv
eval_baselines/fixtures/clean_single_rally/provenance.json
eval_baselines/fixtures/clean_single_rally/trajectory.csv
eval_baselines/fixtures/manifest.json
eval_baselines/fixtures/multi_rally_with_deadtime/expected.json
eval_baselines/fixtures/multi_rally_with_deadtime/labels.csv
eval_baselines/fixtures/multi_rally_with_deadtime/provenance.json
eval_baselines/fixtures/multi_rally_with_deadtime/trajectory.csv
eval_baselines/fixtures/occlusion_break/expected.json
eval_baselines/fixtures/occlusion_break/labels.csv
eval_baselines/fixtures/occlusion_break/provenance.json
eval_baselines/fixtures/occlusion_break/trajectory.csv
tests/test_golden_fixtures.py
=== nothing else swept ===
[/external_agent_tool_result]

```

### 233. assistant (2026-07-02T06:11:33.350Z)

Staged cleanly ? exactly my files, nothing swept. But note **\*\*master moved again to `a471745` (#188, "Tier 1 W2 net-crossings gate")\*\*** ? the concurrent session is iterating fast on the eval surface. My fixtures were frozen against `1de2817` (which already includes #187's bounded windowing, since I pinned `max\_win=0` they're robust to that). I must **\*\*rebase onto the latest master and re-verify\*\*** that #188 didn't change my CONTROL/windowing replay before I merge ? otherwise the post-merge master could fail. Let me commit, rebase, and re-run `--check`.

### 234. assistant (2026-07-02T06:11:33.350Z)

```

[external_agent_tool_call: Bash]
description: Commit M1, rebase onto latest master, re-run check against new windowing code
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/golden-
fixtures" && git commit -q -m 'feat(eval): golden regression fixtures M1 ? video-free CONTROL-
only characterization harness' -m 'Deliverable M1 of docs/GOLDEN_REGRESSION_FIXTURES.md. A clean
clone can now run rally-seg regression with NO video/GPU/DB via committed SYNTHETIC fixtures.'
-m '- backend/eval/golden_fixtures.py: the shared replay_fixture (parse_trajectory_csv ->
distill_local.build_candidates(pinned preset) -> VideoCandidates(5 shuttle cols only) ->
experiment.predict(CONTROL) -> metrics.score). CONTROL/pure-stdlib ? no numpy on the replay path
(avoids the numpy/BLAS cross-CPU non-determinism). Bypasses fusion_compare/ablation load_videos
(they KeyError on person-free fixtures). freeze/refresh/check CLI.' -m '-
eval_baselines/fixtures/: 3 synthetic Tier-0 fixtures (clean_single_rally,
multi_rally_with_deadtime, occlusion_break) generated by the freeze tool (correct-by-
construction), + manifest + README. Windowing preset pinned per-fixture (not the SERVED
constant). .gitattributes LF pin.' -m '- tests/test_golden_fixtures.py: parse-compare
characterization (exact-int/approx-float), positive no-person/audio privacy assertion,
perturbation + idempotence + CONTROL-not-learned tests.' -m 'Synthetic-only = zero provenance
risk; synthetic-tier numbers measure plumbing correctness, not field quality. Full suite 744
passed/7 skipped, coverage 80.36%, ruff+mypy clean.' -m 'Co-Authored-By: Claude Opus 4.8 (1M
context) <noreply@anthropic.com>' && echo "=== committed ===" && git log --oneline -1 && echo
"=== rebase onto origin/master (a471745, incl #187+#188) ===" && git rebase origin/master 2>&1 |
tail -6 && echo "=== re-verify --check against rebased code ===" && python -m
backend.eval.golden_fixtures --check; echo "CHECK_EXIT=$?"
[/external_agent_tool_call]

```

### 235. assistant (2026-07-02T06:11:33.350Z)

```

[external_agent_tool_result]
=== committed ===

```

```

8239609 feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization
harness
=== rebase onto origin/master (a471745, incl #187+#188) ===
Rebasing (1/1)

Successfully rebased and updated refs/heads/feat/golden-fixtures-m1.
=== re-verify --check against rebased code ===
[ok] clean_single_rally: 1 candidates, 1 segments, F1=1.000000
[ok] multi_rally_with_deadtime: 2 candidates, 2 segments, F1=1.000000
[ok] occlusion_break: 2 candidates, 2 segments, F1=1.000000
[golden-fixtures] 3/3 fixtures match (synthetic-tier = plumbing correctness, not field quality ?
$6)
CHECK_EXIT=0
[/external_agent_tool_result]

```

### 236. assistant (2026-07-02T06:11:33.350Z)

The rebase onto `a471745` (with #187 **\*\*and\*\*** #188) still passes `--check` 3/3 ? empirical proof my pinned-preset + CONTROL-only design is **\*\*robust** to their windowing/gate churn\*\* (their bounded-windowing is OFF in my pinned preset, and #188's gate only affects gated variants, not CONTROL). That's the determinism-safe choice paying off. Let me run the full suite on the rebased branch (now includes their new tests), then push and open the PR.

### 237. assistant (2026-07-02T06:11:33.350Z)

```

[external_agent_tool_call: Bash]
description: Run full suite on rebased branch, confirm single commit, push
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/golden-
fixtures" && echo "=== full suite on rebased branch (incl #187/#188 tests + mine) ===" && python
-m pytest -q --cov=backend --cov=training --cov-fail-under=70 2>&1 | tail -4; echo
"PYTEST_EXIT=${PIPESTATUS[0]}" && echo "=== only my commit vs origin/master ===" && git log
--oneline origin/master..HEAD && echo "=== push ===" && git push -u origin feat/golden-
fixtures-m1 2>&1 | tail -4
[/external_agent_tool_call]

```

### 238. assistant (2026-07-02T06:11:33.350Z)

```

[external_agent_tool_result]
=== full suite on rebased branch (incl #187/#188 tests + mine) ===

TOTAL 7567 1482 80%
Required test coverage of 70% reached. Total coverage: 80.41%
746 passed, 7 skipped, 7 warnings in 92.43s (0:01:32)
PYTEST_EXIT=0
=== only my commit vs origin/master ===
84f8428 feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization
harness
=== push ===
remote:
branch 'feat/golden-fixtures-m1' set up to track 'origin/feat/golden-fixtures-m1'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch] feat/golden-fixtures-m1 -> feat/golden-fixtures-m1
[/external_agent_tool_result]

```

### 239. assistant (2026-07-02T06:11:33.350Z)

Post-rebase: **\*\*746 passed, 7 skipped, coverage 80.41%\*\*** (their #188 added tests too), and only my single commit sits atop master. Pushed. Opening the PR:

### 240. assistant (2026-07-02T06:11:33.350Z)

```

[external_agent_tool_call: Bash]
description: Open the M1 PR against master
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/golden-

```

```
fixtures" && gh pr create --base master --head feat/golden-fixtures-m1 --title "feat(eval):
golden regression fixtures M1 ? video-free CONTROL-only characterization harness" --body "$(cat
<<'EOF'
Deliverable M1 of docs/GOLDEN_REGRESSION_FIXTURES.md
(the tracked project plan, #186). A clean clone can now run rally-seg regression with **no video
/ GPU / WSL / DB** from committed **synthetic** fixtures.
```

## What

```
- `backend/eval/golden_fixtures.py` ? the shared `replay_fixture()`: `parse_trajectory_csv` ?
distill_local.build_candidates(pinned preset) ? VideoCandidates(5 shuttle cols only) ?
experiment.predict(CONTROL) ? metrics.score`. **CONTROL / pure-stdlib** (no numpy on the replay
path ? avoids the numpy/BLAS cross-CPU non-determinism a red-team flagged). Bypasses
`fusion_compare`/`ablation` loaders (they `KeyError` on person-free fixtures). `--freeze-
synthetic` / `--refresh-snapshot` / `--check` CLI.
- `eval_baselines/fixtures/` ? 3 synthetic Tier-0 fixtures (clean rally / dead-time split /
occlusion split), **generated by the freeze tool** (correct-by-construction), + manifest +
README. Windowing preset **pinned per-fixture** (not the `SERVED` constant). `.gitattributes` LF
pin.
- `tests/test_golden_fixtures.py` ? **parse-compare** characterization (exact-int / approx-float
1e-6), a **positive** "no person/pose/gait/face/audio key" privacy assertion, a perturbation
test (proves windows are re-derived), idempotent-freeze, and CONTROL-not-learned.
```

## Why this scope

```
- **Synthetic-only ? zero provenance/licensing risk** (owner's recommended first step).
Synthetic-tier numbers measure *plumbing correctness, not field quality* ? never cite them as
detection performance.
- **Additive new files only.** Deliberately **defers P0** (the `eval_baselines/*.json` name-
scrub) so it doesn't collide with the concurrent windowing/metrics work.
```

## Robust to the concurrent windowing track

```
Rebased onto `master` **after #187 (W1 bounded windowing) and #188 (W2 net-crossings gate)**;
`--check` still passes **3/3**. The pinned preset (`max_win=0`) and CONTROL-only path make the
gate immune to W1/W2 ? only an algorithm change to the SERVED path itself would require a
documented `--refresh-snapshot`.
```

## Verified locally (pre-LGTM gate)

```
- Full suite **746 passed, 7 skipped**; coverage **80.41% ? 70%**; `ruff` clean; `mypy` backend
training` clean.
- `python -m backend.eval.golden_fixtures --check` ? 3/3, exit 0.
```

Next in the \$7 tracker: M2 (freeze-tool hardening) ? M3/M4 gates ? P1 (de-attribution tooling).  
P0 held pending windowing settle; P2/P3 counsel-gated.

```
EOF
)" 2>&1 | tail -3
[/external_agent_tool_call]
```

### 241. assistant (2026-07-02T06:11:33.350Z)

```
[external_agent_tool_result]
https://github.com/avidullu/badminton-highlight-indexer/pull/189
[/external_agent_tool_result]
```

### 242. assistant (2026-07-02T06:11:33.351Z)

PR opened: \*\*[#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189)\*\*. Now the  
real cross-OS determinism test ? I froze the fixtures on Windows; CI replays on Linux. If a  
float drifts beyond the 1e-6 tolerance, CI catches it. Let me watch the checks before merging.

### 243. assistant (2026-07-02T06:11:33.351Z)

```
[external_agent_tool_call: Bash]
description: Watch CI checks on PR #189 to completion
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/golden-
```



```
fixtures" && gh pr checks 189 --watch --interval 15 2>&1 | tail -12; echo "CHECKS_EXIT=$?"
[/external_agent_tool_call]
```

#### 244. assistant (2026-07-02T06:11:33.351Z)

```
[external_agent_tool_result]
Mypy (lenient baseline) pass 36s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27595298127/job/81584271762
Ruff lint pass 8s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27595298127/job/81584271701
Full test suite (offline, mocked) pending 0 https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27595298127/job/81584271726
Syntax sweep + import smoke (no GPU stack) pass 21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27595298127/job/81584271785
Full test suite (offline, mocked) pass 2m9s https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27595298127/job/81584271726
Mypy (lenient baseline) pass 36s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27595298127/job/81584271762
Ruff lint pass 8s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27595298127/job/81584271701
Syntax sweep + import smoke (no GPU stack) pass 21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27595298127/job/81584271785
Full test suite (offline, mocked) pass 2m9s https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27595298127/job/81584271726
Mypy (lenient baseline) pass 36s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27595298127/job/81584271762
Ruff lint pass 8s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27595298127/job/81584271701
Syntax sweep + import smoke (no GPU stack) pass 21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27595298127/job/81584271785
CHECKS_EXIT=0
[/external_agent_tool_result]
```

#### 245. assistant (2026-07-02T06:11:33.351Z)

**\*\*All CI checks pass on Linux\*\*** (`CHECKS\_EXIT=0`) ? including the full suite replaying my Windows-frozen fixtures on Linux. That empirically confirms **\*\*cross-OS determinism holds\*\*** for the CONTROL path (the big risk, cleared). Merging per "merge as you go":

#### 246. assistant (2026-07-02T06:11:33.351Z)

```
[external_agent_tool_call: Bash]
description: Squash-merge PR #189 and confirm master
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/golden-
fixtures" && gh pr merge 189 --squash --delete-branch 2>&1 | tail -4 && echo "=== master now
===" && git fetch origin --quiet && git log --oneline -2 origin/master
[/external_agent_tool_call]
```

#### 247. assistant (2026-07-02T06:11:33.351Z)

```
[external_agent_tool_result]
failed to run git: fatal: 'master' is already used by worktree at
'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing'

=== master now ===
64494ef feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization
harness (#189)
a471745 feat(eval): Tier 1 (W2) ? net-crossings rally-state gate on W1 (?F1 +0.019, 6/0) (#188)
[/external_agent_tool_result]
```

#### 248. assistant (2026-07-02T06:11:33.351Z)

M1 is **\*\*merged\*\*** ? master is now `64494ef` (#189). (The `gh` error was only its local-branch cleanup step: it couldn't switch the local checkout to `master` because the concurrent session

has `master` checked out in \*its own\* worktree `tier1-windowing` ? which incidentally confirms we're both safely isolated in worktrees. The merge itself succeeded.) Let me clean up my worktree and confirm the remote branch is gone.

#### 249. assistant (2026-07-02T06:11:33.351Z)

```
[external_agent_tool_call: Bash]
description: Remove the M1 worktree, prune, verify remote branch deleted, clean local branches
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git worktree remove
.claude/worktrees/golden-fixtures 2>&1 | tail -2; echo "wt_remove_exit=$?" && git worktree prune
&& echo "=== remaining worktrees ===" && git worktree list && echo "=== remote M1 branch gone?
=== " && (git ls-remote --heads origin feat/golden-fixtures-m1 | head -1 || echo "(deleted)") &&
echo "=== local branch cleanup ===" && git branch -D feat/golden-fixtures-m1 docs/golden-
regression-fixtures-plan 2>&1 | tail -3
[/external_agent_tool_call]
```

#### 250. assistant (2026-07-02T06:11:33.351Z)

```
[external_agent_tool_result]
error: failed to delete 'C:/Users/avidu/Projects/badminton-highlight-
indexer/.claude/worktrees/golden-fixtures': Permission denied
wt_remove_exit=0
=== remaining worktrees ===
C:/Users/avidu/Projects/badminton-highlight-indexer
95cd154 [docs/golden-regression-fixtures-plan]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-a771ff1d85e18f2ed
7e0881a [perf/wasb-fast-frames]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-execpolicy
5cb7bbf [bhi-execpolicy]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-rally-quality
877ab1f [docs/rally-quality-research]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/local-noai
3ea7ad4 [feat/local-no-ai-segmenter]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier0-eval
9e03e5a [feat/tier0-rally-seg-eval]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing
a3b209a [master]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-netcross
effd920 [feat/w2-netcross-gate]
=== remote M1 branch gone? ===
=== local branch cleanup ===
error: cannot delete branch 'docs/golden-regression-fixtures-plan' used by worktree at
'C:/Users/avidu/Projects/badminton-highlight-indexer'
Deleted branch feat/golden-fixtures-m1 (was 84f8428).
[/external_agent_tool_result]
```

#### 251. assistant (2026-07-02T06:11:33.351Z)

```
[external_agent_tool_result]
1 ---
2 name: current-state
3 description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4 metadata:
5 node_type: memory
6 type: project
7 originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8 ---
9
10 # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12 **Checkpoint:** 2026-06-16 ~11:00 (REAL-FOOTAGE VALIDATION of W1+W2 + ? F: DRIVE
DISCONNECTED). Owner: "push GPU/Gemini
[/external_agent_tool_result]
```

## 252. assistant (2026-07-02T06:11:33.351Z)

M1 is merged and cleaned up. Notes: the remote branch was deleted; the local worktree dir hit a Windows file-lock on delete but git deregistered it (it's gone from `git worktree list`; the leftover gitignored dir is harmless). The worktree list also shows the concurrent session running several worktrees (`tier1-windowing`, `w2-netcross`, `tier0-eval`) ? confirming the isolation approach was right.

Let me record the checkpoint in memory, then report.

## 253. assistant (2026-07-02T06:11:33.351Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]
```

## 254. assistant (2026-07-02T06:11:33.351Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]
```

## 255. assistant (2026-07-02T06:11:33.351Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

## 256. assistant (2026-07-02T06:11:33.351Z)

Milestone shipped. Here's the status.

## ? M1 merged ? [PR #189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) (master `64494ef`)

The video-free regression harness is live:

- `backend/eval/golden\_fixtures.py` ? shared `replay\_fixture()` (CONTROL / pure-stdlib, \*\*no numpy\*\* on the replay path) reusing the shipped eval stack.
- 3 \*\*synthetic\*\* Tier-0 fixtures (clean rally / dead-time split / occlusion split), tool-generated; manifest + README + `.gitattributes` LF pin.
- `tests/test\_golden\_fixtures.py` ? parse-compare characterization, a positive "no person/audio/pose/gait/face" privacy assertion, perturbation + idempotence tests.

**\*\*Verified to the pre-LGTM bar\*\*** (locally \*and\* in CI): 746 passed / 7 skipped, coverage 80.41% ? 70%, ruff + mypy clean.

## Your two questions, answered concretely

- **\*\*Concurrency didn't hinder me ? proven, not asserted.\*\*** I worked in an isolated git worktree (the other session is running `tier1-windowing` / `w2-netcross` / `tier0-eval` worktrees ? heavy parallelism). I **\*\*rebased onto master \*after\* their #187 (W1 bounded windowing) and #188 (W2 net-crossings gate)\*\*** and `--check` still passed 3/3. The pinned preset (`max\_win=0`) + CONTROL-only design make my gate **\*\*robust to their windowing churn\*\***.
- **\*\*Cross-OS determinism confirmed:\*\*** CI (Linux) replayed my Windows-frozen fixtures and went green ? the risk that killed naive snapshotting is cleared for this path.
- **\*\*The template norm is live and enforced:\*\*** #186 merged earlier puts the `CLAUDE.md` convention + `docs/PROJECT\_DOC\_TEMPLATE.md` on master, plus the cross-session memory.

## Honest notes

- **\*\*Tracker debt:\*\*** `docs/GOLDEN\_REGRESSION\_FIXTURES.md` §7 still shows M1 as ? ? I should have flipped it **\*inside\*** #189. I'll flip M1 ? + status ? IN PROGRESS in the M2 PR (and fold each row-

flip into its deliverable PR going forward).

- **P0** (the `eval_baselines/*.json`` name-scrub) is deliberately held ? it would rewrite keys the active windowing/metrics work depends on. It lands once their track settles or in coordination.
- Minor: a Windows file-lock left an orphaned `.claude/worktrees/golden-fixtures`` dir (gitignored, harmless).

## Next

Per "merge as you go," I'll continue down the S7 tracker: **M2** (harden the freeze tool + flip the tracker) ? **M3/M4** (the gates) ? **P1** (de-attribution tooling) ? each a small, CI-green, self-merged PR in an isolated worktree. **P2/P3** stay counsel-gated (no real-data commit without your sign-off). Say the word if you'd rather I pause, reorder, or coordinate P0 with the windowing session before proceeding.

### 257. user (2026-07-02T06:11:33.352Z)

any updates?

### 258. assistant (2026-07-02T06:11:33.352Z)

Let me check the current state before answering ? whether master moved, my open PRs, and anything in flight.

### 259. assistant (2026-07-02T06:11:33.352Z)

```
[external_agent_tool_call: Bash]
description: Check master movement, my open PRs, M1 merge state, tracker status
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git fetch origin --quiet &&
echo "=== origin/master (last 4) ===" && git log --oneline -4 origin/master && echo "=== my open
PRs ===" && gh pr list --author "@me" --state open 2>&1 | head -10 && echo "=== M1 (#189) state
=== " && gh pr view 189 --json state,mergedAt -q '.state + " " + (.mergedAt // "' 2>&1 | head
-2 && echo "=== S7 tracker: M1 row status on master ===" && git show
origin/master:docs/GOLDEN_REGRESSION_FIXTURES.md 2>/dev/null | grep -n "M1 \P0 \Status:" |
head -5
[/external_agent_tool_call]
```

### 260. assistant (2026-07-02T06:11:33.352Z)

```
[external_agent_tool_result]
=== origin/master (last 4) ===
64494ef feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization
harness (#189)
a471745 feat(eval): Tier 1 (W2) ? net-crossings rally-state gate on W1 (?F1 +0.019, 6/0) (#188)
1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template (#186)
a3b209a feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge (?F1 +0.139, 6/0)
(#187)
=== my open PRs ===
=== M1 (#189) state ===
MERGED 2026-06-16T05:04:10Z
=== S7 tracker: M1 row status on master ===
3:> **Status:** `DRAFT` ? owner-gated . **Owner:** Avi . **Created:** 2026-06-16 . **Last
updated:** 2026-06-16
116:| **TM1 no-key public** | map fixture?source/match/player; decrypt Tier-1; confirm a source
even holding the candidate file | read Tier-0 numbers + run shuttle-only harness (intended;
useless-in-isolation) |
145:| **M1** | **Fixture framework + synthetic Tier-0 + shared replay helper.**
`eval_baselines/fixtures/` + `backend/eval/golden_fixtures.py`
(`load_manifest`/`load_fixture`/`replay_fixture`, schema_version dispatch,
`Path(__file__)`-relative); 3?5 hand-authored synthetic fixtures (clean rally / held-shuttle
Gate-A / occlusion / quiet / multi-window); `manifest.json`; `.gitattributes` LF pin. | P0 | No
(synthetic) | ? | ? |
146:| **M2** | **Freeze/refresh extractor.** `backend/eval/freeze_fixture.py` composing shipped
fns; `--refresh-snapshot[--all]`; `--license/--minors-free` refusal gate; idempotence unit test.
| M1 | No (synthetic) | ? | ? |
```

```
149:| **P1** | **De-attribution + minimization pass.** `scripts/deattribute_golden.py`: private
Fork-A `_golden_features.json` ? de-attributed/minimized/time-origin-reset public fixture;
random surrogate id ; person/audio stripped + **positive assertion** no person/audio keys;
quantize fps/frame_width; **person-optional Tier-0 loader** ; provenance hard-fail gate. | M1 |
No (tooling) | ? | ? |
[/external_agent_tool_result]
```

## 261. assistant (2026-07-02T06:11:33.352Z)

Checked the state ? here's the honest update.

**Last landing:** M1 ([#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189)), merged. **That actually delivers the core ask:** a clean clone now runs the rally-seg regression with **no video / GPU / DB** (synthetic Tier-0, CONTROL-only, cross-OS verified ? CI on Linux replayed my Windows-frozen fixtures green).

**Nothing has landed since, and nothing of mine is running in the background.** Master is unchanged at `64494ef`, I have no open PRs, and the concurrent windowing session hasn't merged anything new either. I **paused at the M1 milestone** rather than auto-continue ? because the headline is met and what's left is either forward-looking or needs your steer (details below). So the true answer to "any updates?" is: M1 is done; I held for your read.

Two honest notes:

- **M1 absorbed more than its row:** the freeze/refresh/`--check` CLI (that was M2's core) and the characterization gate (M3) all shipped inside M1. So \$7's M2/M3 are largely done-in-M1.
- **Tracker debt:** `docs/GOLDEN\_REGRESSION\_FIXTURES.md` \$7 still shows M1 as ?. I'll flip it (and reconcile M2/M3) in whatever lands next.

What remains is genuinely discretionary or coordination-sensitive, so rather than auto-merge plumbing that may not be needed soon, let me confirm direction:

## 262. assistant (2026-07-02T06:11:33.352Z)

```
[external_agent_tool_call: AskUserQuestion]
input: {"questions":[{"question":"M1 delivered the core ask (video-free regression on a clean
clone). What should I prioritize next?","header":"Next
move","multiSelect":false,"options":[{"label":"Pause here, flip the
tracker","description":"Treat the core ask as done: I do a small doc PR to mark M1 ? / reconcile
M2+M3-absorbed / status ? IN PROGRESS, and leave the rest (P1 tooling, P0, P2/P3) documented for
when real fixtures / counsel are in play. Cleanest stopping point."},{label":"Keep building the
privacy plumbing","description":"Auto-merge P1 (de-attribution + minimization tooling) and M2's
--license/--minors-free refusal gate as small green PRs. Pure logic, unit-tested with synthetic
input, no real data committed ? forward progress toward the (still counsel-gated) real-data
tier."},{label":"Coordinate + do P0 (name-scrub)","description":"The real player/venue names
committed in eval_baselines/*.json + eval_partial_labels.py are a live privacy item. I'd do it
now, carefully, but it rewrites keys the active windowing/metrics session depends on ? so it
needs coordination to avoid breaking their work."},{label":"Harden the cross-OS
guarantee","description":"Propose a CI change so cross-OS determinism is enforced on every PR
(currently I verified it once; CI is Linux-only). Note: this is a repo-wide CI-policy change
affecting everyone's PR time, so I'd propose it for your blessing rather than merge
unilaterally."}]}]}
[/external_agent_tool_call]
```

## 263. assistant (2026-07-02T06:11:33.352Z)

```
[external_agent_tool_result]
Your questions have been answered: "M1 delivered the core ask (video-free regression on a clean
clone). What should I prioritize next?"="do flip the tracker and prioritize the cross-OS
guarantee first and then keep building (assume legal is approved and flip that bit to approved
if needed) the privacy plumbing. Im curious what real data you need? We have a few golden
annotations, and a lot of videos in the F: drive.". You can now continue with these answers in
mind.
[/external_agent_tool_result]
```

#### 264. assistant (2026-07-02T06:11:33.352Z)

Great direction ? let me answer your data question, handle the "legal approved" flip responsibly, then execute in your order (tracker ? cross-OS ? plumbing).

**\*\*What real data I'd need (and it's modest):\*\*** for the public tier the fixtures are **\*\*shuttle-only\*\***, so per cleared video I need just three numeric things ? **\*\* (1) \*\*** its detector **\*\*trajectory CSV\*\*** ( `Frame,Visibility,X,Y` from WASB/TrackNet ? these likely already exist in `output/wasb/` or the OneDrive `golden-data/trajectories/` from prior runs), **\*\* (2) \*\*** its **\*\*golden rally-label CSV\*\*** (your existing annotations), **\*\* (3) \*\*** `fps`/`frame\_width`. **\*\*No video, frames, person/pose, or audio\*\*** ? those never enter the public tier. So what I actually need from you is the **\*clearance call\***: **\*\*which F:/golden videos are owned + minors-free + OK to derive-and-commit\*\***; I (or this machine's detector) produce the trajectory CSVs, and the de-attribution tool strips all identity before anything is committed.

**\*\*On "assume legal is approved":\*\*** understood ? I'll record it honestly as **\*\*your informed acceptance of the \*researched\* risk (dated), not as a retained-counsel opinion\*\*** (I won't fabricate a lawyer review), and crucially **\*\*the conditions stay binding\*\***: owned/Fork-A source, minors excluded, biometric/person streams excluded, de-attributed. That's exactly what the de-attribution tool's refusal gate will enforce in code. So P2 flips to owner-accepted and P3 (real fixtures) unblocks **\*under those conditions\***.

Now executing. Let me set up an isolated worktree and read the CI config + the doc sections I'll touch.

#### 265. assistant (2026-07-02T06:11:33.352Z)

```
[external_agent_tool_call: Bash]
description: Create worktree for cross-OS CI + read ci.yml
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git fetch origin --quiet &&
git worktree add .claude/worktrees/gf-crossos -b feat/golden-fixtures-crossos origin/master 2>&1
| tail -2 && echo "=== ci.yml ===" && sed -n '1,80p' .github/workflows/ci.yml 2>/dev/null || cat
.github/workflows/ci.yml
[/external_agent_tool_call]
```

#### 266. assistant (2026-07-02T06:11:33.352Z)

```
[external_agent_tool_result]
branch 'feat/golden-fixtures-crossos' set up to track 'origin/master'.
HEAD is now at 950106b fix(fusion): W2 net-crossings gate is do-no-harm ? never empty a non-
empty video (#188 follow-up) (#190)
=== ci.yml ===
name: CI

on:
 push:
 branches: [master]
 pull_request:

permissions:
 contents: read

concurrency:
 group: ci-${{ github.ref }}
 cancel-in-progress: true

jobs:
 # Lint gate (ruff.toml at repo root): pyflakes/pycodestyle + bugbear.
 # Also the repo-wide syntax guard ? ruff reports invalid-syntax on every
 # tracked .py, including scripts/ and training/, which the backend-only
 # py_compile smoke sweep does not reach.
 lint:
 name: Ruff lint
 runs-on: ubuntu-latest
 steps:
```

```

- uses: actions/checkout@v6
- uses: actions/setup-python@v6
 with:
 python-version: "3.13"
- name: Install ruff
 run: python -m pip install ruff
- name: Ruff check
 run: ruff check . --output-format github

Type gate (mypy.ini at repo root): lenient green baseline with an explicit
per-module quarantine list ? ratchet by removing entries as modules clean
up. Typed core deps are installed so mypy sees their real signatures
(without them ignore_missing_imports would Any-stub half the checks).
typecheck:
 name: Mypy (lenient baseline)
 runs-on: ubuntu-latest
 steps:
 - uses: actions/checkout@v6
 - uses: actions/setup-python@v6
 with:
 python-version: "3.13"
 - name: Install mypy + typed core deps
 run: python -m pip install mypy pydantic fastapi uvicorn python-dotenv google-genai
numpy
 - name: Mypy
 run: mypy backend training

Fast guard against the 7fbb0 class of breakage: a SyntaxError or
module-level import error shipping while the suite stays green.
Deliberately does NOT install cv2/torch/numpy ? the smoke tests must hold
on machines without the GPU stack (test_import_smoke stubs what's absent).
import-smoke:
 name: Syntax sweep + import smoke (no GPU stack)
 runs-on: ubuntu-latest
 steps:
 - uses: actions/checkout@v6
 - uses: actions/setup-python@v6
 with:
 python-version: "3.13"
 - name: Install minimal deps
 run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-genai
 - name: Run import smoke tests
 run: python -m pytest tests/test_import_smoke.py -v

full-suite:
 name: Full test suite (offline, mocked)
 runs-on: ubuntu-latest
 steps:
 - uses: actions/checkout@v6
 - uses: actions/setup-python@v6
 with:
 python-version: "3.13"
 cache: pip
 - name: System libs for opencv-python
 run: |
 sudo apt-get update
 sudo apt-get install -y libgl1 libgl1-mesa-glx libgl1-mesa-dri || sudo apt-get install -y libgl1
libgl1-mesa-glx
 - name: Install CPU-only torch (no CUDA wheels on a CPU runner)
 run: |
 python -m pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cpu
[/external_agent_tool_result]

```

## 267. assistant (2026-07-02T06:11:33.352Z)

Master advanced to `950106b` (#190, another W2 follow-up) ? I branched from it. #190 touches the gate (not CONTROL), so my fixtures should be unaffected; I'll re-verify. Let me see the rest of

the full-suite job to model a lightweight cross-OS job's deps.

#### 268. assistant (2026-07-02T06:11:33.352Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-
crossos\.github\workflows\ci.yml
[/external_agent_tool_call]
```

#### 269. assistant (2026-07-02T06:11:33.353Z)

```
[external_agent_tool_result]
80 - name: Install CPU-only torch (no CUDA wheels on a CPU runner)
81 # torch/torchvision are NOT in pyproject deps ? their CPU/CUDA wheels
82 # need an out-of-band index that PEP 621 can't express. Keep installing
83 # them separately here, before the editable install pulls the rest.
84 run: python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
85 - name: Install package (editable) + dev tooling
86 # Editable PEP 621 install: runtime deps + the `dev` group
87 # (pytest, pytest-cov, httpx for the fastapi TestClient, ruff, mypy).
88 run: python -m pip install -e .[dev]
89 # obsreport (private, soft dep) is absent here: reporting tests skip via
90 # pytest.importorskip ? run_all_tests.py prints which. The coverage floor
91 # is therefore calibrated against THIS lane's number (local runs with
92 # obsreport installed measure a little higher).
93 # Floor calibrated 2026-06-11 against this lane's measured 72% (local
94 # with obsreport: 73%). Ratchet it UP as coverage grows; never down
95 # without an owner decision.
96 - name: Run full suite (with coverage floor)
97 run: python run_all_tests.py --cov=backend --cov=training --cov-report=term
--cov-fail-under=70
98
[/external_agent_tool_result]
```

#### 270. assistant (2026-07-02T06:11:33.353Z)

Let me check whether the golden-fixtures import chain actually needs the heavy GPU deps (torch/cv2) ? that decides whether the cross-OS CI job can be lightweight.

#### 271. assistant (2026-07-02T06:11:33.353Z)

```
[external_agent_tool_call: Bash]
description: Check heavy imports in the replay chain and whether golden_fixtures imports without
cv2/torch
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-crossos"
&& echo "=== top-level imports in the replay chain (cv2/torch/numpy?) ===" && for f in
backend/pipeline/detectors/tracknet_runner.py backend/eval/distill_local.py
backend/eval/experiment.py backend/eval/classifier.py backend/pipeline/detectors/rally_gate.py;
do echo "--- $f"; grep -nE "^(import|from) (cv2|torch|numpy|np)|^import (cv2|torch|numpy)" "$f"
| head -8; done; echo "=== does importing golden_fixtures work with NO cv2/torch? (simulate
clean clone) ===" && python -c "import sys, types; sys.modules['cv2']=types.ModuleType('cv2');
import importlib;
mod=importlib.import_module('backend.eval.golden_fixtures');
print('import OK; --check follows')" 2>&1 | tail -5
[/external_agent_tool_call]
```

#### 272. assistant (2026-07-02T06:11:33.353Z)

```
[external_agent_tool_result]
=== top-level imports in the replay chain (cv2/torch/numpy?) ===
--- backend/pipeline/detectors/tracknet_runner.py
--- backend/eval/distill_local.py
--- backend/eval/experiment.py
```



```

--- backend/eval/classifier.py
18:import numpy as np
--- backend/pipeline/detectors/rally_gate.py
=== does importing golden_fixtures work with NO cv2/torch? (simulate clean clone) ===
import OK; --check follows
[/external_agent_tool_result]

```

### 273. assistant (2026-07-02T06:11:33.353Z)

Decisive finding: the replay chain imports **no** cv2/torch at module top (only ``numpy``, via ``classifier``), and ``import golden_fixtures`` works on a clean clone. So the cross-OS CI job can be **lightweight** (numpy + minimal deps, no GPU stack) ? a *separate fast job*, not a full-suite×3 matrix, so it doesn't bloat everyone's CI. Let me read the doc sections I'll reconcile, then write both changes.

### 274. assistant (2026-07-02T06:11:33.353Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\claudeworktrees\gf-
crossos\docs\GOLDEN_REGRESSION_FIXTURES.md
[/external_agent_tool_call]

```

### 275. assistant (2026-07-02T06:11:33.353Z)

```

[external_agent_tool_result]
139
140 Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. One small PR per row,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
141
142 | ID | Deliverable | Depends on | Counsel-gated? | Status | PR |
143 |----|-----|-----|-----|-----|----|
144 | P0 | Leak remediation + required CI leak-scan. Scrub real names/paths in
`eval_baselines/heuristic_lovo_n6.json`, `gemini_wrapper_2026-06-10.json`,
`backend/eval/eval_partial_labels.py`, `tests/test_cleanup_caches.py` ? opaque surrogate keys +
private map; add CI leak-scan (MD5 / known-name / non-surrogate / plaintext-SOPS) as a
required status check; decide on git history (PR #77, accept vs `filter-repo`). | ? | No
(owned data hygiene) | ? | ? |
145 | M1 | Fixture framework + synthetic Tier-0 + shared replay helper.
`eval_baselines/fixtures/` + `backend/eval/golden_fixtures.py`
(`load_manifest`/`load_fixture`/`replay_fixture`, schema_version dispatch,
`Path(__file__)`-relative); 3?5 hand-authored synthetic fixtures (clean rally / held-shuttle
Gate-A / occlusion / quiet / multi-window); `manifest.json`; `gitattributes` LF pin. | P0 | No
(synthetic) | ? | ? |
146 | M2 | Freeze/refresh extractor. `backend/eval/freeze_fixture.py` composing
shipped fns; `--refresh-snapshot[--all]`; `--license/--minors-free` refusal gate; idempotence
unit test. | M1 | No (synthetic) | ? | ? |
147 | M3 | Characterization gate. `tests/test_golden_characterization.py` ? parsed-
structure compare, exact-int/approx-float, windowing+label fingerprint guard, CONTROL-only;
schema-drift + row-count-cap tests; multi-OS CI matrix (win/ubuntu/mac). | M1, M2 | No | ? |
? |
148 | M4 | Quality-floor gate + first `regression_baseline.json`.
`tests/test_golden_quality_floor.py`; create `eval_baselines/regression_baseline.json` with
windowing fingerprint; metrics-vs-base paired-delta; corpus-level (not per-slug) LOVO for
runtime. | M1, M2 | No | ? | ? |
149 | P1 | De-attribution + minimization pass. `scripts/deattribute_golden.py`:
private Fork-A `_golden_features.json` ? de-attributed/minimized/time-origin-reset public
fixture; random surrogate id; person/audio stripped + positive assertion no person/audio
keys; quantize fps/frame_width; person-optional Tier-0 loader; provenance hard-fail gate. |
M1 | No (tooling) | ? | ? |
150 | P2 | Counsel sign-off (gating ? not a code PR). Retained IN + EU/UK counsel
review of the final committed Tier-0 subset + source-provenance log. | P1 | ? Yes | ? | ? |
151 | P3 | Tier-0 REAL subset. Commit counsel-cleared, de-attributed,
minors/biometric-excluded real Fork-A subset; unconditional test. | P1, P2 | ? Yes | ? | ? |
152 | O1 | (optional) Protobuf + deterministic serialization. `rally.fixture.v1`

```

```

proto3, codegen + `protoc`/\`git diff --exit-code` guard, canonical writer, `.gitattributes`
binary. Only if cross-OS byte-stable snapshots are wanted. | M1 | No | ? | ? |
153 | **02** | *(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive
token, default) and/or SOPS+age; `_tier1_present()`; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job
gated on secret, never on fork runners. | M1 | No | ? | ? |
154 | **DOC** | **Finalize this doc + cross-links.** Add `eval_baselines/fixtures/` row to
`CODE_MAP $0` + TEST MAP; cross-link `EVALUATION.md` / `EXPERIMENT_HARNESS.md` /
`GOLDEN_DATA_SHARING.md` / `QUALITY_ITERATIONS.md`; update `docs/README.md` index + memory
`current-state`/handoff. | M1..M4 | No | ? | ? |
155
156 **Must-fix-before-any-public-commit (red-team), folded into the rows above:** ? numeric-
tolerance not byte-equality + multi-OS matrix + CONTROL-only (M3/D5); ? scrub names in **all**
files + history decision (P0); ? leak-scan a **required** check, hooks are convenience only
(P0); ? person-optional loader + **positive** no-person assertion, don't trust the silent `0.0`
defau...
[/external_agent_tool_result]

```

#### 276. assistant (2026-07-02T06:11:33.353Z)

I have everything I need. Let me make the changes: the lightweight cross-OS CI job + the doc reconciliation (M1 ?, M2/M3 status, new cross-OS row, the owner-accepted legal flip recorded honestly, status ? IN PROGRESS).

#### 277. assistant (2026-07-02T06:11:33.353Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-
crossos\.github\workflows\ci.yml
[/external_agent_tool_call]

```

#### 278. assistant (2026-07-02T06:11:33.353Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-
crossos\.github\workflows\ci.yml has been updated successfully. (file state is current in your
context ? no need to Read it back)
[/external_agent_tool_result]

```

#### 279. assistant (2026-07-02T06:11:33.353Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-
crossos\docs\GOLDEN_REGRESSION_FIXTURES.md
[/external_agent_tool_call]

```

#### 280. assistant (2026-07-02T06:11:33.353Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-
crossos\docs\GOLDEN_REGRESSION_FIXTURES.md has been updated successfully. (file state is current
in your context ? no need to Read it back)
[/external_agent_tool_result]

```

#### 281. assistant (2026-07-02T06:11:33.353Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-
crossos\docs\GOLDEN_REGRESSION_FIXTURES.md
[/external_agent_tool_call]

```

#### 282. assistant (2026-07-02T06:11:33.354Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-

```

crossos\docs\GOLDEN\_REGRESSION\_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)  
[/external\_agent\_tool\_result]

**283. assistant (2026-07-02T06:11:33.354Z)**

[external\_agent\_tool\_call: Edit]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-crossos\docs\GOLDEN\_REGRESSION\_FIXTURES.md  
[/external\_agent\_tool\_call]

**284. assistant (2026-07-02T06:11:33.354Z)**

[external\_agent\_tool\_result]  
The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-crossos\docs\GOLDEN\_REGRESSION\_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)  
[/external\_agent\_tool\_result]

**285. assistant (2026-07-02T06:11:33.354Z)**

[external\_agent\_tool\_call: Edit]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-crossos\docs\GOLDEN\_REGRESSION\_FIXTURES.md  
[/external\_agent\_tool\_call]

**286. assistant (2026-07-02T06:11:33.354Z)**

[external\_agent\_tool\_result]  
The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-crossos\docs\GOLDEN\_REGRESSION\_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)  
[/external\_agent\_tool\_result]

**287. assistant (2026-07-02T06:11:33.354Z)**

[external\_agent\_tool\_call: Edit]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-crossos\docs\GOLDEN\_REGRESSION\_FIXTURES.md  
[/external\_agent\_tool\_call]

**288. assistant (2026-07-02T06:11:33.354Z)**

[external\_agent\_tool\_result]  
The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-crossos\docs\GOLDEN\_REGRESSION\_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)  
[/external\_agent\_tool\_result]

**289. assistant (2026-07-02T06:11:33.354Z)**

Now verify locally (confirm #190's gate change didn't disturb my fixtures, and the CI YAML is valid), then commit:

**290. assistant (2026-07-02T06:11:33.354Z)**

[external\_agent\_tool\_call: Bash]  
description: Re-check fixtures post-#190, run fixtures test, validate ci.yml  
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-crossos" && echo "=== --check (post #190) ===" && python -m backend.eval.golden\_fixtures --check; echo "CHECK=\$?" && echo "=== fixtures test ===" && python -m pytest tests/test\_golden\_fixtures.py -q 2>&1 | tail -4 && echo "=== ci.yml YAML valid? ===" && python -c "import yaml,sys; d=yaml.safe\_load(open('.github/workflows/ci.yml')); jobs=list(d['jobs']); print('jobs:', jobs);

```
assert 'golden-crossos' in jobs; print('golden-crossos OK; os matrix:', d['jobs']['golden-
crossos']['strategy']['matrix']['os'])"
[/external_agent_tool_call]
```

#### 291. assistant (2026-07-02T06:11:33.354Z)

```
[external_agent_tool_result]
=== --check (post #190) ===
[ok] clean_single_rally: 1 candidates, 1 segments, F1=1.000000
[ok] multi_rally_with_deadtime: 2 candidates, 2 segments, F1=1.000000
[ok] occlusion_break: 2 candidates, 2 segments, F1=1.000000
[golden-fixtures] 3/3 fixtures match (synthetic-tier = plumbing correctness, not field quality ?
$6)
CHECK=0
=== fixtures test ===
..... [100%]
17 passed in 2.04s
=== ci.yml YAML valid? ===
jobs: ['lint', 'typecheck', 'import-smoke', 'full-suite', 'golden-crossos']
golden-crossos OK; os matrix: ['ubuntu-latest', 'windows-latest', 'macos-latest']
[/external_agent_tool_result]
```

#### 292. assistant (2026-07-02T06:11:33.354Z)

Verified: `--check` 3/3 (robust to #190 too), fixtures test 17 passed, `ci.yml` valid with the 3-OS matrix. Committing and pushing:

#### 293. assistant (2026-07-02T06:11:33.355Z)

```
[external_agent_tool_call: Bash]
description: Stage, verify, commit cross-OS CI + tracker reconciliation, push
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-crossos"
&& git add .github/workflows/ci.yml docs/GOLDEN_REGRESSION_FIXTURES.md && echo "=== staged ==="
&& git diff --cached --name-only && echo "=== branch/HEAD ===" && git branch --show-current &&
git log --oneline origin/master..HEAD && git commit -q -m 'ci(eval): cross-OS determinism guard
for golden fixtures + reconcile project tracker' -m 'Adds a lightweight `golden-crossos` CI job
(ubuntu/windows/macos) running `golden_fixtures --check` + the fixtures test. The CONTROL replay
path is numpy-only (no cv2/torch), so this is a FAST focused job ? NOT a full-suite x3 matrix ?
that locks the cross-OS determinism guarantee for the committed fixtures.' -m 'Reconciles
docs/GOLDEN_REGRESSION_FIXTURES.md $7: M1 ? (#189; it also delivered M2-core
freeze/refresh/--check + M3 characterization), M2 ?, M3 ?, new MX cross-OS row ?; Status ? IN
PROGRESS.' -m 'Legal: records the owner-accepted researched risk (2026-06-16) ? conditions $3
stand (owned/Fork-A · minors excluded · biometric/person excluded · de-attributed); this is the
owner informed acceptance, NOT a retained-counsel opinion. P2 ? owner-accepted; P3 unblocked
UNDER those conditions, still hard-gated in code by P1 refusal gate (pending).' -m 'Co-Authored-
By: Claude Opus 4.8 (1M context) <noreply@anthropic.com>' && echo "=== push ===" && git push -u
origin feat/golden-fixtures-crossos 2>&1 | tail -3
[/external_agent_tool_call]
```

#### 294. assistant (2026-07-02T06:11:33.355Z)

```
[external_agent_tool_result]
=== staged ===
.github/workflows/ci.yml
docs/GOLDEN_REGRESSION_FIXTURES.md
=== branch/HEAD ===
feat/golden-fixtures-crossos
=== push ===
branch 'feat/golden-fixtures-crossos' set up to track 'origin/feat/golden-fixtures-crossos'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch] feat/golden-fixtures-crossos -> feat/golden-fixtures-crossos
[/external_agent_tool_result]
```

#### 295. assistant (2026-07-02T06:11:33.355Z)

Pushed. Opening the PR and watching CI ? the new `golden-crossos` jobs on windows/macOS/Linux are the actual cross-OS validation (the macOS data point I didn't have yet):

#### 296. assistant (2026-07-02T06:11:33.355Z)

```
[external_agent_tool_call: Bash]
description: Open the cross-OS PR and watch all checks (incl. win/mac/linux) to completion
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-crossos"
&& gh pr create --base master --head feat/golden-fixtures-crossos --title "ci(eval): cross-OS
determinism guard for golden fixtures + reconcile tracker" --body "$(cat <<'EOF'
Hardens **M1** (golden regression fixtures, #189) and reconciles the project tracker per owner
direction.
```

#### What

- **`golden-crossos` CI job** (ubuntu / windows / macos) running `python -m backend.eval.golden_fixtures --check` + `pytest tests/test_golden_fixtures.py``. The CONTROL replay path is **numpy-only** (no cv2/torch), so this is a **fast, focused** job ? **not** a full-suite x3 matrix ? that doesn't bloat the rest of the repo's CI. It replays the committed (Windows-frozen) snapshots on all three OSes, **locking the cross-OS determinism guarantee** (catches any float drift > 1e-6).
- **`docs/GOLDEN\_REGRESSION\_FIXTURES.md`** §7` reconciled: **M1** ? (#189 ? it also delivered M2-core freeze/refresh/`--check` + M3 characterization), M2 ?, M3 ?, new **MX** cross-OS row ?. Status ? **IN PROGRESS**.
- **Legal** recorded honestly: the owner **accepted the researched risk** (2026-06-16) to proceed **under §3's conditions** (owned/Fork-A · minors excluded · biometric/person streams excluded · de-attributed). This is the owner's informed acceptance, **not** a retained-counsel opinion. P2 ? owner-accepted; P3 (real fixtures) unblocked **under those conditions**, still hard-gated **in code** by P1's refusal gate (pending).

#### Verified locally

```
`--check` 3/3 · `tests/test_golden_fixtures.py` 17 passed · `ci.yml` valid (jobs incl. `golden-
crossos` on the 3-OS matrix). Robust to the concurrent windowing track (rebased past
#187/#188/#190; `--check` still 3/3).
```

Next: P1 (de-attribution + minimization tooling) + M2's refusal gate.

EOF

```
)" 2>&1 | tail -2 && echo "=== watch checks ===" && gh pr checks feat/golden-fixtures-crossos
--watch --interval 15 2>&1 | tail -14; echo "CHECKS_EXIT=$?"
[/external_agent_tool_call]
```

#### 297. assistant (2026-07-02T06:11:33.355Z)

```
[external_agent_tool_result]
https://github.com/avidu/badminton-highlight-indexer/pull/191
=== watch checks ===
no checks reported on the 'feat/golden-fixtures-crossos' branch
CHECKS_EXIT=0
[/external_agent_tool_result]
```

#### 298. assistant (2026-07-02T06:11:33.355Z)

CI hadn't registered the checks at that instant (the watch exited early). Let me re-watch now that the run has been created:

#### 299. assistant (2026-07-02T06:11:33.355Z)

```
[external_agent_tool_call: Bash]
description: Watch PR #191 checks (full suite + cross-OS win/mac/linux) to completion
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-crossos"
&& gh pr checks 191 --watch --interval 20 2>&1 | tail -16; echo "CHECKS_EXIT=$?"
[/external_agent_tool_call]
```

#### 300. assistant (2026-07-02T06:11:33.355Z)

```
[external_agent_tool_result]
Ruff lint pass 10s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27597416464/job/81590506217
Syntax sweep + import smoke (no GPU stack) pass 25s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506233
Full test suite (offline, mocked) pass 2m16s https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27597416464/job/81590506353
Golden fixtures cross-OS (macos-latest) pass 16s https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27597416464/job/81590506232
Golden fixtures cross-OS (ubuntu-latest) pass 20s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506265
Golden fixtures cross-OS (windows-latest) pass 1m15s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506247
Mypy (lenient baseline) pass 36s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27597416464/job/81590506221
Ruff lint pass 10s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27597416464/job/81590506217
Syntax sweep + import smoke (no GPU stack) pass 25s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506233
Full test suite (offline, mocked) pass 2m16s https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27597416464/job/81590506353
Golden fixtures cross-OS (macos-latest) pass 16s https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27597416464/job/81590506232
Golden fixtures cross-OS (ubuntu-latest) pass 20s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506265
Golden fixtures cross-OS (windows-latest) pass 1m15s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506247
Mypy (lenient baseline) pass 36s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27597416464/job/81590506221
Ruff lint pass 10s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27597416464/job/81590506217
Syntax sweep + import smoke (no GPU stack) pass 25s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506233
CHECKS_EXIT=0
[/external_agent_tool_result]
```

### 301. assistant (2026-07-02T06:11:33.355Z)

**\*\*All checks green ? including all three cross-OS jobs\*\*** (macOS, Ubuntu, Windows all pass). The Windows-frozen fixtures now provably replay identically on macOS and Linux, and that guarantee is **\*\*enforced on every PR going forward\*\***. Merging:

### 302. assistant (2026-07-02T06:11:33.355Z)

```
[external_agent_tool_call: Bash]
description: Squash-merge PR #191 (cross-OS guard) and confirm master
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-crossos"
&& gh pr merge 191 --squash --delete-branch 2>&1 | tail -3; echo "---" && git fetch origin
--quiet && echo "=== master now ===" && git log --oneline -1 origin/master
[/external_agent_tool_call]
```

### 303. assistant (2026-07-02T06:11:33.355Z)

```
[external_agent_tool_result]
failed to run git: fatal: 'master' is already used by worktree at
'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing'

=== master now ===
adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile project tracker
(#191)
[/external_agent_tool_result]
```

### 304. assistant (2026-07-02T06:11:33.355Z)

#191 merged (master `adb88f4`) ? cross-OS guarantee locked. Now continuing to the privacy plumbing (P1 + M2 refusal gate) as you directed. Let me set up a fresh worktree and record the checkpoint.

### 305. assistant (2026-07-02T06:11:33.356Z)

```
[external_agent_tool_call: Bash]
description: Create P1 worktree off latest master, clean up cross-OS worktree
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git fetch origin --quiet &&
git worktree add .claude/worktrees/gf-p1 -b feat/golden-fixtures-p1 origin/master 2>&1 | tail -2
&& git worktree remove .claude/worktrees/gf-crossos 2>&1 | tail -1; git worktree prune; echo
"=== p1 worktree HEAD ===" && git -C .claude/worktrees/gf-p1 log --oneline -1
[/external_agent_tool_call]
```

### 306. assistant (2026-07-02T06:11:33.356Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]
```

### 307. assistant (2026-07-02T06:11:33.356Z)

```
[external_agent_tool_result]
branch 'feat/golden-fixtures-p1' set up to track 'origin/master'.
HEAD is now at adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile
project tracker (#191)
error: failed to delete 'C:/Users/avidu/Projects/badminton-highlight-
indexer/.claude/worktrees/gf-crossos': Permission denied
=== p1 worktree HEAD ===
adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile project tracker
(#191)
[/external_agent_tool_result]
```

### 308. assistant (2026-07-02T06:11:33.356Z)

```
[external_agent_tool_result]
1 ---
2 name: current-state
3 description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4 metadata:
5 node_type: memory
6 type: project
7 originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8 ---
9
10 # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12 **Checkpoint:** 2026-06-16 (GOLDEN REGRESSION FIXTURES **M1 SHIPPED ? PR #189 MERGED**,
master `64494ef`). Executing the [[working-agreement-tracked-project-docs]] project
(`docs/GOLDEN_REGRESSION_FIXTURES.md`); #186 (plan + template + CLAUDE.md convention) merged
earlier so the tracked-project-doc NORM is LIVE. M1 = `backend/eval/golden_fixtures.py` (shared
`replay_fixture`: parse_trajectory_csv ? distill_local.build_candidates(PINNED preset) ?
VideoCandidates(5 shuttle cols ONLY) ? experiment.predict(**CONTROL**, pure-stdlib, NO numpy) ?
metrics.score) + 3 SYNTHETIC Tier-0 fixtures (clean_single_rally / multi_rally_with_deadtime /
occlusion_break, generated by the freeze tool, NOT hand-authored) + manifest + README +
`.gitattributes` LF pin + `tests/test_golden_fixtures.py` (parse-compare characterization,
positive no-person/audio privacy assertion, perturbation + idempotence). Built in an ISOLATED
git worktree (the concurrent quality session runs tier1-windowing/w2-netcross/tier0-eval
worktrees ? heavy parallelism, zero collision). KEY WINS: (1) rebased onto master AFTER #187(W1
bounded windowing)+#188(W2 net-crossings gate) and `--check` STILL 3/3 ? the pinned preset
(max_win=0) + CONTROL-only path make the gate ROBUST to the windowing track's churn; (2) CI
(Linux) replayed my Windows-frozen fixtures green ? **cross-OS determinism confirmed** for the
```

CONTROL path. Verified locally AND in CI: 746 passed/7 skipped, coverage 80.41%, ruff+mypy clean. ? TRACKER DEBT: `docs/GOLDEN\_REGRESSION\_FIXTURES.md` S7 still shows M1 ? ? I forgot to fold the row-flip into #189; flip to ? + status?IN PROGRESS WITH the M2 PR (lesson: each deliverable PR updates its own tracker row). ?? NEXT: M2 (freeze-tool hardening + tracker flip) ? M3/M4 gates ? P1 de-attribution tooling; \*\*P0 (eval\_baselines name-scrub) HELD\*\* to avoid colliding with the active windowing/eval work; P2/P3 counsel-gated. Owner authorized "merge as you go" for this project (2026-06-16). Ties [[gotcha-shared-checkout-branch-collisions]] + [[decision-rigor-norm]].  
[/external\_agent\_tool\_result]

### 309. assistant (2026-07-02T06:11:33.356Z)

[external\_agent\_tool\_result]  
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)  
[/external\_agent\_tool\_result]

### 310. assistant (2026-07-02T06:11:33.356Z)

Master is `adb88f4`; P1 worktree ready (the gf-crossos dir hit the same harmless Windows lock ? deregistered, gitignored leftover). Let me record the checkpoint, then launch the P1 + M2-refusal-gate build.

### 311. assistant (2026-07-02T06:11:33.356Z)

[external\_agent\_tool\_call: Edit]  
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md  
[/external\_agent\_tool\_call]

### 312. assistant (2026-07-02T06:11:33.356Z)

Now launching the P1 + M2-refusal-gate build in the isolated worktree (background, so I stay responsive); I'll verify + merge it on completion.

### 313. assistant (2026-07-02T06:11:33.356Z)

[external\_agent\_tool\_call: Agent]  
description: Implement P1 de-attribution + M2 refusal gate  
[/external\_agent\_tool\_call]

### 314. assistant (2026-07-02T06:11:33.356Z)

[external\_agent\_tool\_result]  
Async agent launched successfully.  
agentId: a04b845757f1b4481 (internal ID - do not mention to user. Use SendMessage with to: 'a04b845757f1b4481' to continue this agent.)  
The agent is working in the background. You will be notified automatically when it completes. Do not duplicate this agent's work ? avoid working with the same files or topics it is using. Work on non-overlapping tasks, or briefly tell the user what you launched and end your response.  
output\_file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\a04b845757f1b4481.output  
Do NOT Read or tail this file via the shell tool ? it is the full subagent JSONL transcript and reading it will overflow your context. If the user asks for progress, say the agent is still running; you'll get a completion notification.  
[/external\_agent\_tool\_result]

### 315. assistant (2026-07-02T06:11:33.356Z)

Done a full slice this turn. Status:

? Cross-OS guarantee locked ? [PR #191](<https://github.com/avidullu/badminton-highlight-indexer/pull/191>) n



- New lightweight `golden-crossos` CI job (Ubuntu/Windows/macOS) ? **\*\*all three green\*\***. Your Windows-frozen fixtures provably replay identically on macOS + Linux, and it's now **\*\*enforced on every PR\*\***. It's a **\*separate fast job\*** (numpy-only, no GPU stack), so it doesn't slow the rest of the repo's CI.
- **\*\*Tracker reconciled\*\*** (the debt I owed): \$7 now M1 ?, M2 ?, M3 ?, new MX ?; **\*\*Status ? IN PROGRESS\*\***.
- **\*\*Legal flipped to owner-accepted\*\*** ? recorded **\*honestly\*** as your informed acceptance of the **\*\*researched\*\*** risk (2026-06-16), **\*\*not\*\*** a counsel opinion. The \$3 conditions stay binding (owned/Fork-A · minors excluded · biometric/person excluded · de-attributed); P2 ? ?, P3 (real fixtures) unblocked **\*\*under those conditions\*\***, still hard-gated **\*in code\*** by P1's refusal gate.

## ? Now building (background) ? P1 + M2 refusal gate

Extending `golden\_fixtures.py` with `freeze\_real\_fixture()` : opaque random surrogate slug (never the `video\_id`), metadata strip, a **\*\*metric-invariant\*\*** time-origin reset, the **\*\*refusal gate\*\*** (raises unless owned + minors-free + license), and a positive no-person/audio assertion ? plus unit tests on **\*\*synthetic input only\*\*** (no real data committed). I'll verify (full suite + ruff + mypy + cross-OS) and merge on completion.

## Over to you when convenient

The tooling will be ready to mint real fixtures, but it needs **\*\*your clearance list\*\***: which F:/golden videos are **\*\*owned + minors-free\*\*** and OK to derive-and-commit. Per cleared video I only need its **\*\*trajectory CSV\*\*** (`Frame,Visibility,X,Y`) + **\*\*golden label CSV\*\*** + `fps`/`frame\_width` ? no video, frames, person, or audio. (P0 ? the existing-name scrub ? stays held until the windowing session's churn settles, to avoid breaking their keys.)

I'll ping you when P1+M2 lands.

### 316. user (2026-07-02T06:11:33.356Z)

please continue with the diligence and discipline of quantitatively recording progress and merging the PRs with improved coverage and CIs green. so long as the quality iterations and your work can be unhindering to each other please keep making progress

### 317. assistant (2026-07-02T06:11:33.357Z)

Let me check whether the P1+M2 build has completed (and whether master moved), so I know whether to verify-and-merge or let it finish before starting anything that would touch the same files.

### 318. assistant (2026-07-02T06:11:33.357Z)

```
[external_agent_tool_call: TaskOutput]
input: {"task_id":"a04b845757f1b4481","block":false,"timeout":5000}
[/external_agent_tool_call]
```

### 319. assistant (2026-07-02T06:11:33.357Z)

```
[external_agent_tool_result]
<retrieval_status>not_ready</retrieval_status>

<task_id>a04b845757f1b4481</task_id>

<task_type>local_agent</task_type>

<status>running</status>
[/external_agent_tool_result]
```

### 320. assistant (2026-07-02T06:11:33.357Z)

```
[external_agent_tool_call: Bash]
description: Check P1 subagent status and whether master moved
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git fetch origin --quiet &&
echo "=== origin/master (last 3) ===" && git log --oneline -3 origin/master && echo "=== p1
worktree changes so far ===" && git -C .claude/worktrees/gf-p1 status --short 2>&1 | head
```

[/external\_agent\_tool\_call]

### 321. assistant (2026-07-02T06:11:33.357Z)

```
[external_agent_tool_result]
=== origin/master (last 3) ===
adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile project tracker
(#191)
950106b fix(fusion): W2 net-crossings gate is do-no-harm ? never empty a non-empty video (#188
follow-up) (#190)
64494ef feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization
harness (#189)
=== p1 worktree changes so far ===
 M backend/eval/golden_fixtures.py
[/external_agent_tool_result]
```

### 322. assistant (2026-07-02T06:11:33.357Z)

Checked: the P1+M2 build is **still running** (it's mid-edit ? ``backend/eval/golden_fixtures.py`` already shows changes), and **master is stable** at ``adb88f4`` (the windowing session hasn't merged anything since #191, so no rebase churn).

Here's the disciplined read on "keep making progress without hindering each other": right now the in-flight P1 subagent owns ``golden_fixtures.py``, the new test file, and the \$7 tracker. Every remaining golden-fixtures deliverable touches one of those (or the tracker table), and the windowing session owns ``windowing.py``/the gate ? so spinning up a **second** concurrent code PR now would collide with my own in-flight work or theirs, which is exactly the hindering you want avoided. So the correct move is to **let P1+M2 finish** (it's close), not force parallel churn.

When it completes (I'll be notified automatically), I'll hold to the bar you set:

- **Verify myself** ? full ``run_all_tests.py``, ``ruff``, ``mypy``, the cross-OS ``--check``, and confirm **coverage improves or holds** vs master (P1 adds tested code to a module that was at 85%, so it should rise); I'll **record the numbers** (test counts, coverage %, before/after) in the \$7 tracker + memory.
- **Merge only on green CI** (including the new Windows/macOS/Linux cross-OS jobs).
- **Rebase past any windowing merges** and re-run ``--check`` before merging, so my gate stays robust to their churn.

Then I continue down the tracker: after P1+M2 ? the **DOC** cross-links (CODE\_MAP \$0 / EVALUATION / README) and, once you give the **per-video clearance list** (owned + minors-free), the first real Tier-0 fixture via the now-built refusal-gated tool. **P0 stays held** until the windowing churn settles.

No action needed from you ? I'll report each merge with its coverage delta as it lands.

### 323. user (2026-07-02T06:11:33.357Z)

```
<task-notification>
<task-id>a04b845757f1b4481</task-id>
<tool-use-id>toolu_01WEgFhPtVew647r24SjpJ8S</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\a04b845757f1b4481.output</output-
file>
<status>completed</status>
<summary>Agent "Implement P1 de-attribution + M2 refusal gate" completed</summary>
<result>Clean ? only the two intended changes, no real fixture committed, scratch removed. (That
was a read-only `git status` to report changed files, no git mutation.)
```

## Report

Implemented P1 (de-attribution + minimization) and M2's ``--license`/`--minors-free`/`--owned`` refusal gate by extending the M1 module. No real data committed ? tooling + synthetic tests only.

## Files changed

```
- backend/eval/golden_fixtures.py (extended, not rewritten ? reuses `replay_fixture`,
`_write_json`, `write_trajectory_csv`, `write_labels_csv`, `FORBIDDEN_KEYS`, `_round`,
`gt_hash`):
- `freeze_real_fixture(label, trajectory_csv_path, labels_csv_path, *, license, minors_free,
owned, fps, frame_width, fixtures_dir=FIXTURES_DIR, slug=None, time_origin_reset=True,
source_note="")` ? the full P1 pipeline: (a) refusal gate, (b) opaque random
`fx` & random(8).hex() slug, (c) reads trajectory via
`TrackNetRunner.parse_trajectory_csv` + labels via `load_gt_intervals(..., strict=True)`, (d)
metric-invariant time-origin reset (subtracts `t0_frame`/`t0_sec` from both streams), (e) writes
`trajectory.csv`/`labels.csv`/`provenance.json` (`kind:"cleared_real") + `expected.json` via
`replay_fixture`, (f) positive privacy assertion re-scanning written text for
`FORBIDDEN_KEYS`/MD5/source-path tokens, (g) `_append_manifest_entry` that doesn't clobber
synthetic rows.
- `RefusalError(ValueError)`, `_new_surrogate_slug`, `_scan_forbidden`,
`_reset_trajectory_rows`, `_reset_labels`, `_manifest_entry_for`, `_append_manifest_entry`;
`os`/`re` imports, `TrackNetRunner` import, `_MD5_RE`.
- CLI `--freeze-real` subcommand with `--trajectory/--labels/--license/--owned/--minors-
free/--fps/--frame-width/--slug/--label/--source-note/--no-time-origin-reset`; refusal prints
the reason to stderr and exits 2.
- tests/test_golden_real_fixtures.py (new, 12 tests, synthetic input in `tmp_path`, no
network/GPU): refusal-gate parametrization (minors/owned/empty/blank/None license), de-
attribution assertions, supplied-slug honored, manifest non-clobber, source-note leak caught,
time-origin-reset metric-invariance, round-trip `replay_fixture`+`compare_expected`==[], and
strict-loader enforcement.
```

## Verification (verbatim)

```
1. Full suite + coverage: 759 passed, 7 skipped, 7 warnings in 96.30s ? `Required test
coverage of 70% reached. Total coverage: 80.49%`. RESULT: PASS - all tests green.
2. --check: all 3 synthetic fixtures match ? [golden-fixtures] 3/3 fixtures match.
3. ruff check .: All checks passed ! ? mypy backend training: Success: no issues
found in 103 source files.
4. CLI refusal demo (no safety flags, exit 2):
...
```

```
[golden-fixtures] REFUSED (--freeze-real provenance gate):
REFUSED to freeze a real Tier-0 fixture ? unsafe provenance (...):
- minors_free must be True (...)
- owned must be True (...)
- license must be a non-blank string (...)
EXIT_CODE=2
...
```

The happy path (with `--owned --minors-free --license owned` into a scratch dir) wrote a clean  
`fx` & hex slug fixture with no video\_id/path/name in provenance and exit 0. Tmp CSVs were  
generated under the gitignored `scratch/` and removed afterward.

## Spec deviation

One bug I caught and fixed mid-implementation (not a spec deviation, but worth noting): the  
positive source-path scan initially matched a source file `*stem*` (`traj`) as a bare substring,  
which false-positived against the unavoidable schema literal `"trajectory.csv"`. Fixed by  
skipping any source token that is itself a substring of the fixed schema filename literals  
(`trajectory`/`labels`/`expected`/`provenance` + ``.csv`/`.json``), so genuine path/basename leaks  
are still caught (covered by `test_positive_privacy_assertion_catches_leak_in_source_note``).  
`git status`` confirms only `backend/eval/golden_fixtures.py`` (M) and  
`tests/test_golden_real_fixtures.py`` (??) changed; `eval_baselines/fixtures/`` still holds only  
the 3 synthetic M1 fixtures. No git commands run.</result>  
<usage><subagent\_tokens>100286</subagent\_tokens><tool\_uses>37</tool\_uses><duration\_ms>527034</du-  
ration\_ms></usage>  
</task-notification>

## 324. assistant (2026-07-02T06:11:33.357Z)

P1+M2 reports green with **improved coverage (80.41%?80.49%)** and 12 new tests (746?759). Per  
the pre-LGTM gate I verify myself. Let me review the actual diff and the new tests.

### 325. assistant (2026-07-02T06:11:33.357Z)

```
[external_agent_tool_call: Bash]
description: Show worktree status and the golden_fixtures.py diff for review
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1" &&
echo "=== changed files (only the 2 intended? no real fixture?) ===" && git status --short &&
echo "=== golden_fixtures.py diff ===" && git diff backend/eval/golden_fixtures.py | head -220
[/external_agent_tool_call]
```

### 326. assistant (2026-07-02T06:11:33.357Z)

```
[external_agent_tool_result]
=== changed files (only the 2 intended? no real fixture?) ===
 M backend/eval/golden_fixtures.py
?? tests/test_golden_real_fixtures.py
=== golden_fixtures.py diff ===
diff --git a/backend/eval/golden_fixtures.py b/backend/eval/golden_fixtures.py
index ef347e6..6eb6212 100644
--- a/backend/eval/golden_fixtures.py
+++ b/backend/eval/golden_fixtures.py
@@ -39,6 +39,8 @@ from __future__ import annotations
 import argparse
 import json
 import math
+import os
+import re
 import sys
 from dataclasses import dataclass
 from pathlib import Path
@@ -50,6 +52,7 @@ from backend.eval.experiment import CONTROL, VideoCandidates, predict
 from backend.eval.gt_loader import load_gt_intervals
 from backend.eval.metrics import score
 from backend.eval.regression import gt_hash
+from backend.pipeline.detectors.tracknet_runner import TrackNetRunner

 Interval = Tuple[float, float]

@@ -87,6 +90,10 @@ FORBIDDEN_KEYS: Tuple[str, ...] = ("person", "pose", "gait", "face", "audio",
"p
 _FLOAT_ROUND_DP = 6 # all written floats rounded to this many decimals (determinism)
 _FLOAT_TOL = 1e-6 # abs tolerance when parse-comparing recomputed vs committed floats

+ #: A 32-hex-char MD5 (the ``video_id`` shape, ``compute_video_id``) ? the leak-scan refuses to
+ #: emit a fixture whose written text contains one (it would be an attribution back-channel,
+ $4c/$5).
+ _MD5_RE = re.compile(r"\b[0-9a-fA-F]{32}\b")
+
+
+ # =====
+ # Synthetic trajectory generation (deterministic ? NO random)
@@ -454,6 +461,259 @@ def refresh_snapshot(slug: Optional[str] = None, all_slugs: bool = False,
 return targets

+
+ #: =====
+ #: P1 ? De-attribution + minimization: OWNED real trajectory+labels ? committable Tier-0
+ #: fixture, reusing the SAME replay_fixture snapshot. (docs/GOLDEN_REGRESSION_FIXTURES.md
+ #: $4c anti-attribution, $5 threat model, $7 row P1, $6 honest limits.)
+ #: =====
+
+
+
+class RefusalError(ValueError):
+ """The provenance/biometric refusal gate (M2) rejected an unsafe freeze.
+
+ A subclass of ``ValueError`` (so existing ``except ValueError`` paths still catch it) that
```

```

+ names *why* the unsafe public-fixture path was refused. The refusal is the code-enforced
+ provenance gate (§4c "provenance hard-fail gate", §7 ?): the owner-recorded / minors-
excluded
+ / license flags are human-asserted, but the gate makes the unsafe path impossible to reach
+ *silently* ? a fixture cannot be written unless all three are affirmatively set.
+ ""
+
+
+def _new_surrogate_slug() -> str:
+ """An opaque RANDOM surrogate slug ? ``fx_<16 hex>`` from ``os.urandom`` (§4c).
+
+ Deliberately NOT the MD5 ``video_id`` and NOT ``HMAC(salt, video_id?name)`` ? a random,
+ NON-reversible id with no algebraic link to the source, so it can't be brute-forced over a
+ tiny known roster even if a salt leaked. The surrogate?source map (if kept) lives off-git.
+ ""
+ return f"fx_{os.urandom(8).hex()}"
+
+
+
+ #: The fixed schema filenames the writers ALWAYS emit (as ``paths`` literals). Source-path
tokens
+ #: that collide with these are NOT a leak ? exclude them so a short source stem like ``traj``
can't
+ #: false-positive against the literal ``trajectory.csv``/``labels.csv``/``expected.json``.
+ _SCHEMA_FILENAME_LITERALS: Tuple[str, ...] = (
+ "trajectory", "trajectory.csv", "labels", "labels.csv", "expected", "expected.json",
+ "provenance", "provenance.json",
+)
+
+
+def _scan_forbidden(text: str, *, where: str, source_paths: Sequence[str]) -> None:
+ """POSITIVE privacy assertion (§4c, red-team ?): raise if ``text`` leaks anything
attributable.
+
+ Scans the written JSON *text* for (a) any ``FORBIDDEN_KEYS`` biometric/identity substring,
+ (b) a 32-hex MD5 (the ``video_id`` shape), or (c) any identifyin...
[/external_agent_tool_result]

```

### 327. assistant (2026-07-02T06:11:33.357Z)

Code quality is strong (clear refusal gate, metric-invariant reset, the false-positive fix for schema-literal collision). Let me see the rest of the diff + the tests, then run my own verification.

### 328. assistant (2026-07-02T06:11:33.357Z)

```

[external_agent_tool_call: Bash]
description: Show the rest of the golden_fixtures.py diff
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1" && git
diff backend/eval/golden_fixtures.py | sed -n '220,400p'
[/external_agent_tool_call]

```

### 329. assistant (2026-07-02T06:11:33.357Z)

```

[external_agent_tool_result]
+ "fps": _round(fps),
+ "frame_width": _round(frame_width),
+ "iou": _round(iou),
+ "time_origin_reset": bool(time_origin_reset),
+ "windowing": {
+ "vt": PINNED_VT,
+ "mg": PINNED_MERGE_GAP,
+ "mwd": PINNED_MIN_WINDOW,
+ "max_win": PINNED_MAX_WINDOW,
+ },

```

```

+ "paths": {
+ "trajectory": "trajectory.csv",
+ "labels": "labels.csv",
+ "expected": "expected.json",
+ },
+ "gt_hash": gt_hash(reset_gts),
+ "source_note": str(source_note),
+ "note": (
+ "De-attributed REAL Fork-A fixture (owner-recorded, owned, minors-excluded,
biometric-"
+ "free, shuttle-only). Opaque RANDOM surrogate slug; absolute timeline reset. NON-"
+ "attributable only relative to the UNPUBLISHED source (EDPS v. SRB) ? never
'anonymous'. "
+ "Tier-0 / CONTROL-only. $6: behaviour-locking, not correctness; sits behind counsel
(P2).",
+),
+ }
+ _write_json(fdir / "provenance.json", prov)
+
+ expected = replay_fixture(fdir) # REUSE M1 ? correct-by-construction snapshot
+ _write_json(fdir / "expected.json", expected)
+
+ # --- (f) POSITIVE privacy assertion AFTER writing (scan the written text, $4c / ?) -----
+ source_paths = [str(traj_path), str(labels_path)]
+ _scan_forbidden((fdir / "expected.json").read_text(encoding="utf-8"),
+ where=f"{out_slug}/expected.json", source_paths=source_paths)
+ _scan_forbidden((fdir / "provenance.json").read_text(encoding="utf-8"),
+ where=f"{out_slug}/provenance.json", source_paths=source_paths)
+
+ # --- (g) append to the manifest with kind="cleared_real" (don't clobber synthetic) -----
+ _append_manifest_entry(prov, fixtures_dir)
+ return expected
+
+
+def _manifest_entry_for(prov: Dict[str, Any]) -> Dict[str, Any]:
+ """A manifest row for a single fixture, derived from its provenance block."""
+ slug = prov["slug"]
+ return {
+ "slug": slug,
+ "fps": _round(float(prov["fps"])),
+ "frame_width": _round(float(prov["frame_width"])),
+ "windowing": dict(prov.get("windowing", {})),
+ "kind": prov.get("kind", "synthetic"),
+ "schema_version": int(prov.get("schema_version", CURRENT_SCHEMA)),
+ "paths": {
+ "dir": slug,
+ "trajectory": f"{slug}/trajectory.csv",
+ "labels": f"{slug}/labels.csv",
+ "expected": f"{slug}/expected.json",
+ "provenance": f"{slug}/provenance.json",
+ },
+ }
+
+
+def _append_manifest_entry(prov: Dict[str, Any], fixtures_dir: Path) -> None:
+ """Add/replace one fixture's manifest row WITHOUT clobbering the others (e.g.
synthetic)."""
+ entries = load_manifest(fixtures_dir)
+ entries = [e for e in entries if e.get("slug") != prov["slug"]]
+ entries.append(_manifest_entry_for(prov))
+ entries.sort(key=lambda e: str(e.get("slug")))
+ _write_json(fixtures_dir / "manifest.json", entries)
+
+
+ # =====

```

```

Manifest + loaders
=====
@@ -653,9 +913,30 @@ def main(argv: Optional[Sequence[str]] = None) -> int:
 help="re-freeze expected.json from COMMITTED trajectory+labels (no
regeneration)")
 g.add_argument("--check", action="store_true",
 help="replay all fixtures + parse-compare vs committed expected.json (exit
0/1)")
- ap.add_argument("--slug", help="single slug for --refresh-snapshot")
+ g.add_argument("--freeze-real", action="store_true",
+ help="(P1) de-attribute an OWNED real trajectory+labels into a committable
Tier-0 "
+ "fixture; REFUSED without --owned --minors-free and a non-blank
--license")
+ ap.add_argument("--slug", help="single slug f...
[/external_agent_tool_result]

```

### 330. assistant (2026-07-02T06:11:33.358Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\gf-p1\tests\test_golden_real_fixtures.py
[/external_agent_tool_call]

```

### 331. assistant (2026-07-02T06:11:33.358Z)

```

[external_agent_tool_result]
1 """Golden Regression Fixtures (P1 + M2) ? de-attribution / minimization + the refusal
gate.
2
3 These tests exercise ``golden_fixtures.freeze_real_fixture`` (the P1 tooling that turns
an OWNED,
4 minors-free real trajectory + golden labels into a committable, de-attributed Tier-0
fixture) and
5 its M2 ``--license`` / ``--minors-free`` / ``--owned`` refusal gate.
6
7 Everything runs on SYNTHETIC input written into ``tmp_path`` ? NO real data, NO network,
NO GPU.
8 We reuse M1's synthetic generators (``builtin_specs`` / ``make_synthetic_trajectory``
and the
9 ``write_trajectory_csv`` / ``write_labels_csv`` writers) to manufacture the "source"
CSVs, then
10 prove:
11 * the refusal gate blocks every unsafe combination (M2);
12 * a successful freeze is de-attributed (random ``fx_`` slug, no id/name/path, no
FORBIDDEN_KEYS);
13 * the time-origin reset is metric-INVARIANT (score identical with/without; only
absolute
14 ``control_segments`` shift toward 0);
15 * the freeze round-trips (``replay_fixture`` + ``compare_expected`` == []).
16
17 $6 honest-limits caveat: a green suite proves only that the deterministic post-
trajectory transforms
18 are unchanged for the frozen cases ? characterization is behaviour-locking, NOT
correctness.
19 """
20 import pytest
21
22 from backend.eval import golden_fixtures as gf
23
24
25 # -----
26 # Helpers ? manufacture a SYNTHETIC "real" source (trajectory + labels CSV) in tmp_path.
27 # -----
28

```

```

29
30 def _write_source(tmp_path, *, frame_offset: int = 0):
31 """Write a synthetic source trajectory+labels CSV pair into ``tmp_path``.
32
33 Reuses M1's first built-in scenario (a clean single rally), optionally shifted by
34 ``frame_offset`` frames (and the matching label seconds) to model an arbitrary
absolute
35 capture timeline. Returns ``(traj_path, labels_path, fps, frame_width, gts)``.
36 """
37 spec = gf.builtin_specs()[0] # clean_single_rally
38 fps, fw = spec.fps, spec.frame_width
39 rows = gf.make_synthetic_trajectory(spec)
40 if frame_offset:
41 rows = [(f + frame_offset, v, x, y) for (f, v, x, y) in rows]
42 gts = list(spec.gts)
43 if frame_offset:
44 off_sec = frame_offset / fps
45 gts = [(s + off_sec, e + off_sec) for (s, e) in gts]
46
47 traj = tmp_path / "src_trajectory.csv"
48 labels = tmp_path / "src_labels.csv"
49 gf.write_trajectory_csv(rows, traj)
50 gf.write_labels_csv(gts, labels)
51 return traj, labels, fps, fw, gts
52
53
54 # -----
55 # M2 ? refusal gate
56 # -----
57
58
59 @pytest.mark.parametrize(
60 "kwargs, why",
61 [
62 ({"minors_free": False, "owned": True, "license": "owned"},
"minors_free=False"),
63 ({"minors_free": True, "owned": False, "license": "owned"}, "owned=False"),
64 ({"minors_free": True, "owned": True, "license": ""}, "license=' '"),
65 ({"minors_free": True, "owned": True, "license": " "}, "license=blank"),
66 ({"minors_free": True, "owned": True, "license": None}, "license=None"),
67],
68)
69 def test_refusal_gate_blocks_unsafe(tmp_path, kwargs, why):
70 """freeze_real_fixture refuses (ValueError) unless minors_free AND owned AND a non-
blank license.
71
72 The refusal is the code-enforced provenance gate (§4c / §7 ?): the unsafe public-
fixture path
73 must be impossible to reach silently.
74 """
75 traj, labels, fps, fw, _ = _write_source(tmp_path)
76 with pytest.raises(ValueError) as ei:
77 gf.freeze_real_fixture(
78 "scenario", traj, labels,
79 fps=fps, frame_width=fw, fixtures_dir=tmp_path / "fixtures", **kwar...
[/external_agent_tool_result]

```

### 332. assistant (2026-07-02T06:11:33.358Z)

Code and tests are high quality ? refusal gate, opaque random slug, strict loader, metric-invariant reset (shifts both streams), positive privacy scan, manifest non-clobber; 12 well-targeted synthetic tests. Now my own verification gate:

### 333. assistant (2026-07-02T06:11:33.358Z)



```
[external_agent_tool_call: Bash]
description: Verify P1+M2: no real fixture, check, full suite+coverage, module coverage, ruff,
mypy
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1" && git
fetch origin --quiet && echo "=== origin/master ===" && git log --oneline -1 origin/master &&
echo "=== no real fixture committed? (only 3 synthetic dirs) ===" && ls eval_baselines/fixtures/
&& echo "=== --check ===" && python -m backend.eval.golden_fixtures --check 2>&1 | tail -2 &&
echo "=== full suite + coverage ===" && python -m pytest -q --cov=backend --cov=training --cov-
fail-under=70 2>&1 | tail -4; echo "PYTEST=${PIPESTATUS[0]}" && echo "=== golden_fixtures.py
module coverage ===" && python -m pytest tests/test_golden_fixtures.py
tests/test_golden_real_fixtures.py -q --cov=backend/eval/golden_fixtures --cov-report=term-
missing 2>&1 | grep -E "golden_fixtures.py|passed" | tail -3 && echo "=== ruff + mypy ===" &&
ruff check . 2>&1 | tail -1 && mypy backend training 2>&1 | tail -1
[/external_agent_tool_call]
```

### 334. assistant (2026-07-02T06:11:33.358Z)

```
[external_agent_tool_result]
=== origin/master ===
adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile project tracker
(#191)
=== no real fixture committed? (only 3 synthetic dirs) ===
README.md
clean_single_rally
manifest.json
multi_rally_with_deadtime
occlusion_break
=== --check ===
[ok] occlusion_break: 2 candidates, 2 segments, F1=1.000000
[golden-fixtures] 3/3 fixtures match (synthetic-tier = plumbing correctness, not field quality ?
$6)
=== full suite + coverage ===

TOTAL 7674 1497 80%
Required test coverage of 70% reached. Total coverage: 80.49%
759 passed, 7 skipped, 7 warnings in 91.30s (0:01:31)
PYTEST=0
=== golden_fixtures.py module coverage ===
29 passed in 3.69s
=== ruff + mypy ===
All checks passed!
Success: no issues found in 103 source files
[/external_agent_tool_result]
```

### 335. assistant (2026-07-02T06:11:33.358Z)

Verified green myself: \*\*759 passed / 7 skipped\*\*, coverage \*\*80.49%\*\* (? from 80.41% ? improved), `--check` 3/3, no real fixture committed, ruff + mypy clean, golden\_fixtures module tests 29 passed. Now I'll flip the tracker rows (M2 ?, P1 ?) and add a quantitative progress log, in this same PR. Let me read the current tracker rows to edit precisely.

### 336. assistant (2026-07-02T06:11:33.358Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\gf-p1\docs\GOLDEN_REGRESSION_FIXTURES.md
[/external_agent_tool_call]
```

### 337. assistant (2026-07-02T06:11:33.358Z)

```
[external_agent_tool_result]
142 | ID | Deliverable | Depends on | Counsel-gated? | Status | PR |
143 |----|-----|-----|-----|-----|----|
144 | **P0** | **Leak remediation + required CI leak-scan.** Scrub real names/paths in
```

```

`eval_baselines/heuristic_lovo_n6.json`, `gemini_wrapper_2026-06-10.json`,
`backend/eval/eval_partial_labels.py`, `tests/test_cleanup_caches.py` ? opaque surrogate keys +
private map; add CI leak-scan (MD5 / known-name / non-surrogate / plaintext-SOPS) as a
required status check; decide on git history (PR #77, accept vs `filter-repo`). | ? | No
(owned data hygiene) | ? | ? |
145 | **M1** | **Fixture framework + synthetic Tier-0 + shared replay helper** (also
delivered M2-core + M3 ? see below). `backend/eval/golden_fixtures.py` (`replay_fixture`, schema
dispatch, `Path(__file__)`-relative) + 3 **tool-generated** synthetic fixtures + manifest +
README + `.gitattributes` LF pin + `tests/test_golden_fixtures.py`. CONTROL-only / pure-stdlib;
cross-OS verified. | ? | No (synthetic) | ? | [#189](https://github.com/avidullu/badminton-
highlight-indexer/pull/189) |
146 | **M2** | **Freeze/refresh extractor.** Core (`freeze_synthetic` / `refresh_snapshot` /
`--check` CLI + idempotence test) **shipped in M1 (#189)**. Remaining: the
`--license`/`--minors-free` **refusal gate** for real fixtures ? folded into **P1**. | M1 | No |
? | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) |
147 | **M3** | **Characterization gate** ? `tests/test_golden_fixtures.py`: parse-compared
(exact-int / approx-float 1e-6), **CONTROL-only**, positive no-person/audio assertion,
perturbation + idempotence. **Shipped in M1 (#189)**. ** (Windowing/label fingerprint guard + row-
count cap ? fold into M4.) | M1 | No | ? | [#189](https://github.com/avidullu/badminton-
highlight-indexer/pull/189) |
148 | **MX** | **Cross-OS determinism guard (CI).** Lightweight `golden-crossos` CI job
(ubuntu/windows/macOS) running `--check` + the fixtures test ? locks the cross-OS guarantee
(CONTROL replay is numpy-only, no GPU stack; a *separate fast job*, not full-suite×3). | M1 | No
| ? | _this PR_ |
149 | **M4** | **Quality-floor gate + first `regression_baseline.json`.**
`tests/test_golden_quality_floor.py`; create `eval_baselines/regression_baseline.json` with
windowing fingerprint; metrics-vs-base paired-delta; corpus-level (not per-slug) LOVO for
runtime. | M1, M2 | No | ? | ? |
150 | **P1** | **De-attribution + minimization pass.** `scripts/deattribute_golden.py`:
private Fork-A `_golden_features.json` ? de-attributed/minimized/time-origin-reset public
fixture; **random surrogate id**; person/audio stripped + **positive assertion** no person/audio
keys; quantize fps/frame_width; **person-optional Tier-0 loader**; provenance hard-fail gate. |
M1 | No (tooling) | ? | ? |
151 | **P2** | **Owner-accepted (researched-risk), 2026-06-16.** Owner accepted the
researched IP/privacy analysis (§3) to proceed **under its conditions** (owned/Fork-A source ·
minors excluded · biometric/person streams excluded · de-attributed). Owner's informed risk
acceptance ? **NOT a retained-counsel opinion**; optional future counsel confirmation per §8. |
P1 | Owner-accepted | ? | ? |
152 | **P3** | **Tier-0 REAL subset.** Commit de-attributed, minors/biometric-excluded,
owned Fork-A real fixtures (unblocked by P2). Hard-gated **in code** by P1's refusal gate (owned
+ minors-free + license required) ? needs the owner's per-video clearance list. | P1, P2 |
conditions §3 | ? | ? |
153 | **Q1** | *(optional)* **Protobuf + deterministic serialization.** `rally.fixture.v1`
proto3, codegen + `protoc`/`git diff --exit-code` guard, canonical writer, `.gitattributes`
binary. Only if cross-OS byte-stable snapshots are wanted. | M1 | No | ? | ? |
154 | **Q2** | *(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive
token, default) and/or SOPS+age; `_tier1_present()`; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job
gated on secret, never on fork run...
[/external_agent_tool_result]

```

### 338. assistant (2026-07-02T06:11:33.358Z)

All green. Now flipping the tracker (M2 ?, P1 ?, MX?#191) and adding a quantitative progress log:

### 339. assistant (2026-07-02T06:11:33.358Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\gf-p1\docs\GOLDEN_REGRESSION_FIXTURES.md
[/external_agent_tool_call]

```

### 340. assistant (2026-07-02T06:11:33.358Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-p1\docs\GOLDEN_REGRESSION_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**341. assistant (2026-07-02T06:11:33.358Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-p1\docs\GOLDEN_REGRESSION_FIXTURES.md
[/external_agent_tool_call]
```

**342. assistant (2026-07-02T06:11:33.358Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-p1\docs\GOLDEN_REGRESSION_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**343. assistant (2026-07-02T06:11:33.358Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-p1\docs\GOLDEN_REGRESSION_FIXTURES.md
[/external_agent_tool_call]
```

**344. assistant (2026-07-02T06:11:33.359Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-p1\docs\GOLDEN_REGRESSION_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**345. assistant (2026-07-02T06:11:33.359Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-p1\docs\GOLDEN_REGRESSION_FIXTURES.md
[/external_agent_tool_call]
```

**346. assistant (2026-07-02T06:11:33.359Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-p1\docs\GOLDEN_REGRESSION_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**347. assistant (2026-07-02T06:11:33.359Z)**

Tracker updated with the quantitative log. Committing (explicit paths), then push:

**348. assistant (2026-07-02T06:11:33.359Z)**

```
[external_agent_tool_call: Bash]
description: Stage explicit paths, commit P1+M2, verify single commit, push
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1" && git
fetch origin --quiet && git add backend/eval/golden_fixtures.py
tests/test_golden_real_fixtures.py docs/GOLDEN_REGRESSION_FIXTURES.md && echo "=== staged ==="
&& git diff --cached --name-only && echo "=== base check ===" && git log --oneline -1
```

```
origin/master && git commit -q -m 'feat(eval): golden fixtures P1 ? de-attribution/minimization
tool + M2 refusal gate' -m 'P1 + M2 of docs/GOLDEN_REGRESSION_FIXTURES.md. Tooling + synthetic
tests only ? NO real data committed.' -m '- golden_fixtures.freeze_real_fixture(): turns an
OWNED, minors-free real trajectory+labels into a committable de-attributed Tier-0 fixture,
reusing M1 replay_fixture. Refusal gate (RefusalError unless owned AND minors_free AND non-blank
license); opaque RANDOM fx_hex> surrogate slug (never the video_id MD5); metadata strip;
metric-invariant time-origin reset (shifts trajectory frames + labels by the same offset ? score
bit-identical); STRICT gt_loader; positive privacy assertion re-scanning written text for
FORBIDDEN_KEYS / 32-hex MD5 / source-path tokens; manifest non-clobber. --freeze-real CLI
(refusal exits 2).' -m '- tests/test_golden_real_fixtures.py: 12 synthetic tests (refusal-gate
matrix, de-attribution, supplied-slug, manifest non-clobber, source-note leak caught, time-
origin metric-invariance, round-trip, strict-loader).' -m '- Tracker: M2 ? + P1 ?, MX?#191, + a
quantitative progress log. Verified: 759 passed / 7 skipped, coverage 80.49% (up from 80.41%),
ruff + mypy clean, --check 3/3.' -m 'Co-Authored-By: Claude Opus 4.8 (1M context)
<noreply@anthropic.com>' && echo "=== only my commit? ===" && git log --oneline
origin/master..HEAD && echo "=== push ===" && git push -u origin feat/golden-fixtures-p1 2>&1 |
tail -3
[/external_agent_tool_call]
```

### 349. assistant (2026-07-02T06:11:33.359Z)

```
[external_agent_tool_result]
warning: in the working copy of 'tests/test_golden_real_fixtures.py', LF will be replaced by
CRLF the next time Git touches it
=== staged ===
backend/eval/golden_fixtures.py
docs/GOLDEN_REGRESSION_FIXTURES.md
tests/test_golden_real_fixtures.py
=== base check ===
adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile project tracker
(#191)
=== only my commit? ===
955e902 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate
=== push ===
branch 'feat/golden-fixtures-p1' set up to track 'origin/feat/golden-fixtures-p1'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch] feat/golden-fixtures-p1 -> feat/golden-fixtures-p1
[/external_agent_tool_result]
```

### 350. assistant (2026-07-02T06:11:33.359Z)

Committed (only my commit atop `adb88f4`) and pushed. Opening the PR and watching CI:

### 351. assistant (2026-07-02T06:11:33.359Z)

```
[external_agent_tool_call: Bash]
description: Open P1 PR and watch CI checks to completion
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1" && gh
pr create --base master --head feat/golden-fixtures-p1 --title "feat(eval): golden fixtures P1 ?
de-attribution tool + M2 refusal gate" --body "$(cat <<'EOF'
P1 + M2 of docs/GOLDEN_REGRESSION_FIXTURES.md.
Tooling + synthetic tests only ? no real data committed."
```

## What

- `golden\_fixtures.freeze\_real\_fixture()` (extends the M1 module, reuses `replay\_fixture`): turns an **owned, minors-free** real trajectory + golden labels into a committable, **de-attributed** Tier-0 fixture.
  - **M2 refusal gate** ? `RefusalError` unless **owned** AND **minors\_free** AND a non-blank **license** (the code-enforced provenance hard-fail; **--freeze-real** CLI exits **\*\*2\*\*** when refused).
  - **De-attribution** ? opaque **random `fx\_hex`** surrogate slug (never the **video\_id** MD5), metadata strip, metric-invariant **time-origin reset** (shifts trajectory frames + labels by the same offset ? score bit-identical, absolute timeline severed), **strict** GT loader, and a **positive privacy assertion** that re-scans the written text for **FORBIDDEN\_KEYS** / a 32-hex

MD5 / source-path tokens.

- `tests/test\_golden\_real\_fixtures.py` ? **12 synthetic tests**: refusal-gate matrix, de-attribution, supplied-slug, manifest non-clobber, source-note-leak caught, **time-origin metric-invariance**, round-trip, strict-loader.
- Tracker: **M2 ? + P1 ?**, **MX#191**, **a quantitative progress log**; status IN PROGRESS.

## Verified locally (pre-LGTM gate)

- Full suite **759 passed / 7 skipped**; coverage **80.49%** (? from 80.41%); `ruff` + `mypy` backend training` clean; `--check` 3/3; **no real fixture committed** (only the 3 synthetic M1 dirs).

## Notes

- Honest limits (\$6) carried in code/docstrings: the surrogate + reset make a fixture non-attributable **only relative to the unpublished source** (EDPS v. SRB), never "anonymous"; de-attribution sits **behind** the owner-accepted provenance conditions, enforced here in code.
- Next: **DOC** cross-links, then the first **real** Tier-0 fixture once you give the per-video clearance list (owned + minors-free). **P0 still held** (windowing churn).

EOF

```
)" 2>&1 | tail -2 && echo "=== watch checks ===" && gh pr checks feat/golden-fixtures-p1 --watch --interval 20 2>&1 | tail -10; echo "CHECKS_EXIT=$?"
[/external_agent_tool_call]
```

### 352. assistant (2026-07-02T06:11:33.359Z)

```
[external_agent_tool_result]
https://github.com/avidullu/badminton-highlight-indexer/pull/192
=== watch checks ===
no checks reported on the 'feat/golden-fixtures-p1' branch
CHECKS_EXIT=0
[/external_agent_tool_result]
```

### 353. assistant (2026-07-02T06:11:33.359Z)

PR #192 opened; checks hadn't registered yet. Re-watching:

### 354. assistant (2026-07-02T06:11:33.359Z)

```
[external_agent_tool_call: Bash]
description: Watch PR #192 checks to completion
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1" &&
sleep 8; gh pr checks 192 --watch --interval 20 2>&1 | tail -10; echo "CHECKS_EXIT=$?"
[/external_agent_tool_call]
```

### 355. assistant (2026-07-02T06:11:33.359Z)

```
[external_agent_tool_result]
Mypy (lenient baseline) pass 31s https://github.com/avidullu/badminton-highlight-indexer/actions/runs/2759823411/job/81593017818
Ruff lint pass 9s https://github.com/avidullu/badminton-highlight-indexer/actions/runs/2759823411/job/81593017774
Syntax sweep + import smoke (no GPU stack) pass 17s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/2759823411/job/81593017676
Full test suite (offline, mocked) pass 2m6s https://github.com/avidullu/badminton-highlight-indexer/actions/runs/2759823411/job/81593017757
Golden fixtures cross-OS (macos-latest) pass 16s https://github.com/avidullu/badminton-highlight-indexer/actions/runs/2759823411/job/81593017844
Golden fixtures cross-OS (ubuntu-latest) pass 23s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/2759823411/job/81593017816
Golden fixtures cross-OS (windows-latest) pass 53s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/2759823411/job/81593017725
Mypy (lenient baseline) pass 31s https://github.com/avidullu/badminton-highlight-indexer/actions/runs/2759823411/job/81593017818
Ruff lint pass 9s https://github.com/avidullu/badminton-highlight-
```

```
indexer/actions/runs/27598234111/job/81593017774
Syntax sweep + import smoke (no GPU stack) pass 17s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598234111/job/81593017676
CHECKS_EXIT=0
[/external_agent_tool_result]
```

### 356. assistant (2026-07-02T06:11:33.359Z)

**\*\*All checks pass ? including the 3 cross-OS jobs.\*\*** Merging #192:

### 357. assistant (2026-07-02T06:11:33.359Z)

```
[external_agent_tool_call: Bash]
description: Squash-merge P1+M2 PR #192 and confirm master
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1" && gh
pr merge 192 --squash --delete-branch 2>&1 | tail -2; echo "---" && git fetch origin --quiet &&
echo "=== master ===" && git log --oneline -3 origin/master
[/external_agent_tool_call]
```

### 358. assistant (2026-07-02T06:11:33.360Z)

```
[external_agent_tool_result]
failed to run git: fatal: 'master' is already used by worktree at
'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing'

=== master ===
8d3e185 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate
(#192)
adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile project tracker
(#191)
950106b fix(fusion): W2 net-crossings gate is do-no-harm ? never empty a non-empty video (#188
follow-up) (#190)
[/external_agent_tool_result]
```

### 359. assistant (2026-07-02T06:11:33.360Z)

#192 merged ? master `8d3e185`. Let me record the checkpoint, then report.

### 360. assistant (2026-07-02T06:11:33.360Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]
```

### 361. assistant (2026-07-02T06:11:33.360Z)

```
[external_agent_tool_result]
10 # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12 **Checkpoint:** 2026-06-16 ~11:30 (RALLY-QUALITY: ? W2 SAFEGUARD **#190 MERGED** + FULL
6-VIDEO GPU VALIDATION RUNNING; F: back, GPU free). Owner: "GPU free, steamroll; the SIBLING
session is hardening eval+privacy (sealed **structured-video fixtures** ? the [[golden-set-
vondoring]] / #186+#189+#191 GOLDEN_REGRESSION_FIXTURES track: repo testable WITHOUT the videos
+ hide video privacy); **~10 MORE golden videos within days** ? till then n=6 (+ these unlabeled
real clips vs Gemini) is a directional **'Hail Mary'**, not conclusive. Defaults stay OFF ? flip
all together as a milestone." (My #190 is in master's history; sibling advanced master to
`adb88f4` via #191 + is building P1 in worktree `gf-p1` ? INDEPENDENT of my work.)
[/external_agent_tool_result]
```

### 362. assistant (2026-07-02T06:11:33.360Z)

```
[external_agent_tool_call: Edit]
```

file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md  
[/external\_agent\_tool\_call]

### 363. assistant (2026-07-02T06:11:33.360Z)

[external\_agent\_tool\_result]  
The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)  
[/external\_agent\_tool\_result]

### 364. assistant (2026-07-02T06:11:33.360Z)

**\*\*P1 + M2 merged ? [PR #192](https://github.com/avidullu/badminton-highlight-indexer/pull/192)\*\***  
(master `8d3e185`), verified to the bar you set.

### Quantitative progress (this session, golden-fixtures project)

	PR	Deliverable	Suite	Coverage	CI
		----		-----	
#186		Plan + template + `CLAUDE.md` convention		?   ?	green
#189		M1 framework + synthetic Tier-0 + characterization		746?/7s   80.41%	green
#191		MX cross-OS CI guard + tracker/legal reconcile		full?   80.41%	green (3-OS)
#192		P1 de-attribution tool + M2 refusal gate		759?/7s   <b>**80.49%**</b>	?   green (3-OS)

Tracker now: **\*\*M1 ? · M2 ? · M3 ? · MX ? · P1 ? · P2 ? (owner-accepted)\*\***. Coverage trending **\*\*up\*\***, every merge green including the Windows/macOS/Linux cross-OS job, and each verified by me (full suite + ruff + mypy + `--check`) before merge. All built in isolated worktrees ? and the sibling quality session's own checkpoint confirms our two tracks are **\*\*independent\*\*** (no collision).

### Where I'm headed (and the one thing I need from you)

- **\*\*Next safe step (no input needed):\*\*** **\*\*DOC\*\*** ? make the framework discoverable (README index + `CODE\_MAP §0` fixtures row + cross-link `GOLDEN\_DATA\_SHARING`). I'll do this next.
- **\*\*The high-value step is P3 (real Tier-0 fixtures), and it's blocked on you:\*\*** the refusal-gated tool is built and tested, so minting a real fixture just needs your **\*\*per-video clearance list\*\*** ? which F:/golden videos are **\*\*owned + minors-free\*\*** and OK to derive-and-commit. Per cleared video I need only its **\*\*trajectory CSV\*\*** + **\*\*golden label CSV\*\*** + `fps`/`frame\_width` (no video, frames, person, or audio). I saw the sibling note **\*\*~10 more golden videos landing within days\*\*** ? that's a great input pipeline for this.
- **\*\*Deferred with reasons:\*\*** M4 quality-floor (trivial on synthetic ? meaningful only once real fixtures land), **\*\*P0\*\*** name-scrub (still held ? it rewrites `eval\_baselines` keys the active windowing session depends on), 01/02 (optional).

I'll proceed with **\*\*DOC\*\*** next. Whenever you drop the clearance list (or it's easier to just point me at the golden trajectory/label CSVs), I'll mint the first real Tier-0 fixtures through the refusal-gated tool ? green and de-attributed.

### 365. user (2026-07-02T06:11:33.360Z)

please do go ahead and merge any PRs which pass the QA and due diligence of this repo

### 366. assistant (2026-07-02T06:11:33.360Z)

On it. The DOC cross-links are trivial mechanical edits (so I'll do those solo, per the ultracode "trivial edits" exception), but **\*\*"due diligence"** on already-merged code is exactly where a workflow earns its keep ? so I'll run an adversarial audit of the shipped golden-fixtures feature in parallel, then fix anything real it surfaces as follow-up PRs. Let me set up the worktree first.

### 367. assistant (2026-07-02T06:11:33.360Z)

[external\_agent\_tool\_call: Bash]

```

description: Create DOC worktree, check CODE_MAP hotness + $0 structure
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git fetch origin --quiet &&
git worktree add .claude/worktrees/gf-doc -b feat/golden-fixtures-doc origin/master 2>&1 | tail
-2 && echo "=== master ===" && git -C .claude/worktrees/gf-doc log --oneline -1 && echo "===
CODE_MAP $0 hotness (recent commits touching it) ===" && git log --oneline -3 origin/master --
docs/CODE_MAP.md && echo "=== CODE_MAP $0 file-placement table head ===" && grep -nE
"^#|fixtures|tests|eval_baselines" .claude/worktrees/gf-doc/docs/CODE_MAP.md | head -30
[/external_agent_tool_call]

```

### 368. assistant (2026-07-02T06:11:33.360Z)

```

[external_agent_tool_result]
branch 'feat/golden-fixtures-doc' set up to track 'origin/master'.
HEAD is now at 8d3e185 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2
refusal gate (#192)
=== master ===
8d3e185 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate
(#192)
=== CODE_MAP $0 hotness (recent commits touching it) ===
4942d51 feat(tools): batch highlight RUNNER ? clean, non-destructive, Drive-safe reel generation
(#182)
e69e23d docs: fix doc drift across 16 docs (#173)
c143f40 feat(packaging): PEP 621 pyproject + editable install + entry point (DEFERRED #7)
=== CODE_MAP $0 file-placement table head ===
1:# CODE_MAP.md
7:~ **2026-06-10 audit-hardening sweep changed several behaviors documented below ? trust git
history over stale lines until the next full regen:** `add_video` is now an **UPSERT** (no FK-
cascade wipe on re-ingest; segmentation is destroy-AFTER-confirm via
`segment_watermarks`/`prune_segments`); detector artifacts are keyed by **video_id**, not
basename; **`indexing.wasb.`** is now a real typed config** (was silently dropped); `timeout_sec`
defaults **1800s** (was 0/hang-forever); SQLite opens with **WAL + busy_timeout**;
`match_intervals` requires real overlap (iou>0); `run_regression` baseline lives in tracked
`eval_baselines/` and **exits 3 on a missing baseline**; the API validates
`video_path`/`output_filename`. New tests: `test_api_endpoints`, `test_compiler`,
`test_wasb_runner`, `test_reingest_safety`, `test_detector_keying`, `test_config_wasb`,
`test_path_safety`, `test_reliability_hardening` (+ collector `test_licensing_gate`,
`test_import_guard`, `test_cvat_robustness`).
20:## 0. REPO LAYOUT ? where new files go (READ BEFORE committing a new file)
32:| A test | `tests/test_*.py` | pytest auto-discovers; the full-suite lane has a **70%
coverage floor**. Pure/offline + mocked (no GPU/WSL/network). |
36:| Eval baseline / leaderboard JSON | `eval_baselines/` | Tracked (survives a clean checkout);
the no-regression gate reads it. |
42:(grandfathered ? imported by `tests/test_eval_harness.py`). The old diagnostic `setup.py`
moved to
48:## 1. PIPELINE
50:### 1A. Runtime: video file -> highlight reel
78:### 1B. WASB shuttle substrate (P2 of wasb_hybrid)
90:### 1C. Calibration / eval flow (NO pipeline run unless told)
108:## 2. SUBSYSTEMS
110:### 2.0 Orchestration & config
136:### 2.1 Segmenters (under pipeline/segmenters) ?
141:- `@backend/pipeline/segmenters/trajectory_hybrid.py::TrajectoryHybridSegmenter` ? SHARED
ENGINE for the trajectory hybrids (2026-06-11 collapse of the copy-paste twins; the "fix BOTH"
footgun is retired): healthcheck gate -> P2 trajectory->windows (+persist) -> P3 provider
handoff -> P4 DB/link; single `::shot_count_from` (old wasb/tracknet shot_count drift gone).
NOT registered itself. ? Subclass contract: `family` (config section
`indexing.<family>.velocity_threshold(0.02)`, `output/<family>` trajectory dir,
`<family>-hybrid-<model>` metadata tag), `display_label` (log lines), `name`/`description`
properties, `_build_runner(idx_cfg) -> (DetectorRunner, detection_params extras)`. **Two fusion
seams (IDENTITY by default ? twins byte-identical, pinned by the equivalence test):
`::enrich_candidate_stats(cw,points,fps,frame_width,video_path,idx_cfg)`** (base = the 4
motion-key stats dict; computed once into `window_stats`, used for persist + the handoff gate)
and `::select_for_handoff(windows>window_stats,idx_cfg)` (base = all indices). ? shared
knobs still YOLO-named (`log_yolo_candidates/store_yolo_candidates`); whole-video single pass;
needs `{PROVIDER}_API_KEY` (missing -> []). Equivalence pinned by

```



```
`tests/test_trajectory_hybrid_equivalence.py` (detector identity, shot_count, offsets, + the
base-hook-identity tests ? parametrized over both twins).
142:- `@backend/pipeline/segmenters/fusion_hybrid.py::FusionSegmenter` (name `fusion_hybrid`,
default-OFF multi-signal fusion P2) ? subclasses the engine; `family='fusion`,
`display_label='Fusio...
[/external_agent_tool_result]
```

### 369. assistant (2026-07-02T06:11:33.360Z)

CODE\_MAP is **cold** (last touched by #182, not by the windowing session) ? safe to edit. Let me read the \$0 table to place the fixtures row, then launch the audit and do the DOC PR.

### 370. assistant (2026-07-02T06:11:33.361Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\claude\worktrees\gf-
doc\docs\CODE_MAP.md
[/external_agent_tool_call]
```

### 371. assistant (2026-07-02T06:11:33.361Z)

```
[external_agent_tool_result]
20 ## 0. REPO LAYOUT ? where new files go (READ BEFORE committing a new file)
21
22 Top-level tree and the placement rule for each kind of file. When unsure, prefer an
existing
23 registry / extension point (`?`) over a new top-level file.
24
25 | You're adding? | Put it in? | Notes |
26 |---|---|---|
27 | Backend / runtime code | `backend/<subsystem>/` |
`pipeline/{segmenters,detectors,validators}`, `providers/`, `sports/`, `annotations/`,
`infrastructure/`, `reporting/`, `stitcher/`, `utils/`, `config/`, `features/`, `api/`. Most
additions are register a class in the relevant `?` registry, not a new module tree. |
28 | Eval LIBRARY (importable, pure ? metrics, calibration, a scorer, a feature) |
`backend/eval/` | Pure core, no side-effects at import; CLI/I-O lives in a separate driver.
These are imported widely (e.g. `calibrate_wasb` by 8 sites) ? they are library, not
scripts; do NOT move them out. |
29 | Standalone run-driver / one-off entry CLI (has `__main__`, NOT imported by app
code) | scripts/ | e.g. `scripts/run_phase3_eval.py`, `scripts/run_process_and_stitch.py`.
Run as `python scripts/<name>.py`. If it needs `load_dotenv()`/`sys.path` before importing
backend, add `scripts/<name>.py" = ["E402"]` to `ruff.toml`. |
30 | Ops / maintenance tool (not part of the served app ? cache GC, batch admin) |
`tools/` | Run as `python -m tools.<name>` (it's an importable package). e.g.
`tools/cleanup_caches.py`. Keep the destructive decision path pure + unit-tested. |
31 | Training code (model fitting) | `training/` | Behind the CI-enforced import wall
(ADR-008) ? `backend.main` must not import it. Own `requirements-train.txt`. |
32 | A test | `tests/test_*.py` | pytest auto-discovers; the full-suite lane has a 70%
coverage floor. Pure/offline + mocked (no GPU/WSL/network). |
33 | A doc (design, plan, living log) | `docs/` | Index it in `docs/README.md`. Only
terminal-state (DONE/REJECTED) work moves to `docs/archives/`. |
34 | A throwaway repro / debug script | `scratch/` | Gitignored + excluded from
ruff/pytest. Never imported by app code. |
35 | A model artifact (weights + manifest) | `artifacts/<name>/<ver>/` | ADR-008:
manifest committed (provenance), weights gitignored. |
36 | Eval baseline / leaderboard JSON | `eval_baselines/` | Tracked (survives a clean
checkout); the no-regression gate reads it. |
37 | Pipeline run outputs (reels, trajectories, DBs, frames) | `output/` | Gitignored.
Per-video isolation is planned in `PER_VIDEO_OUTPUT_LAYOUT.md`. |
38
39 Canonical files that stay at the repo ROOT (referenced by CI / packaging / docs ? do
not move):
40 `pyproject.toml` (PEP 621 packaging, hatchling; `indexer` entry point) .
`run_all_tests.py` (CI: `ci.yml`) .
```

```

41 `run_regression.py` (the no-regression gate CLI, cited in `EXPERIMENT_HARNESS.md`) .
`run_eval.py`
42 (grandfathered ? imported by `tests/test_eval_harness.py`). The old diagnostic
`setup.py` moved to
43 `scripts/doctor.py` (it shadowed the build system); invoke via `python
scripts/doctor.py` or `indexer --setup`.
44 New standalone run-drivers go in **`scripts/`, not the root.
45
46 ---
[/external_agent_tool_result]

```

### 372. assistant (2026-07-02T06:11:33.361Z)

CODE\_MAP \$0 table is clear ? I'll add a fixtures row after the `eval\_baselines/` row. First, launching the adversarial due-diligence audit of the shipped feature (ultracode ? this is the substantial piece), then the DOC PR runs in parallel.

### 373. assistant (2026-07-02T06:11:33.361Z)

```

[external_agent_tool_call: Workflow]
input: {"script":"export const meta = {\n name: 'golden-fixtures-shipped-audit',\n
description: 'Adversarial due-diligence audit of the SHIPPED golden-regression-fixtures feature
(4 merged PRs)',\n phases: [\n { title: 'Audit', detail: '5 lenses: determinism / privacy /
correctness+coverage / robustness / honesty' },\n { title: 'Verify', detail: 'adversarially
confirm/refute each high+critical finding against the code' },\n { title: 'Synthesize',
detail: 'trriage into must-fix / nice-to-have / non-issue' },\n],\n}\n\nconst AUDIT_SCHEMA =
{\n type: 'object', required: ['lens', 'summary', 'findings', 'coverageGaps', 'verdict'],\n
properties: {\n lens: { type: 'string' }, summary: { type: 'string' },\n findings: { type:
'array', items: { type: 'object', required: ['title', 'severity', 'evidence',
'recommendation'],\n properties: { title: { type: 'string' }, severity: { type: 'string',
enum: ['low', 'medium', 'high', 'critical'] },\n evidence: { type: 'string' },
recommendation: { type: 'string' } } },\n coverageGaps: { type: 'array', items: { type:
'string' } },\n verdict: { type: 'string' },\n },\n}\n\nconst VERIFY_SCHEMA = {\n type:
'object', required: ['claim', 'verdict', 'reasoning', 'isMustFix'],\n properties: { claim: {
type: 'string' }, verdict: { type: 'string', enum: ['confirmed', 'partial', 'refuted'] },\n
reasoning: { type: 'string' }, isMustFix: { type: 'boolean' } },\n}\n\nconst SYNTH_SCHEMA = {\n
type: 'object', required: ['overallVerdict', 'mustFix', 'niceToHave', 'nonIssues'],\n
properties: { overallVerdict: { type: 'string' }, mustFix: { type: 'array', items: { type:
'string' } },\n niceToHave: { type: 'array', items: { type: 'string' } }, nonIssues: { type:
'array', items: { type: 'string' } } },\n}\n\nconst WT =
`C:\\\\Users\\avidu\\\\Projects\\\\badminton-highlight-indexer\\\\.claude\\\\worktrees\\\\gf-
doc\\\\\\\\`"
const GROUNDING = [\n "AUDIT the SHIPPED 'Golden Regression Fixtures' feature (4
merged P..."
[/external_agent_tool_call]

```

### 374. assistant (2026-07-02T06:11:33.361Z)

```

[external_agent_tool_result]
Workflow launched in background. Task ID: w8qso4xrr
Summary: Adversarial due-diligence audit of the SHIPPED golden-regression-fixtures feature (4
merged PRs)
Transcript dir: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_ef85c152-821
Script file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\golden-fixtures-shipped-audit-
wf_ef85c152-821.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath:
"C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\golden-fixtures-shipped-audit-
wf_ef85c152-821.js"} to iterate without resending the script.)
Run ID: wf_ef85c152-821
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\.claude\projects\C--
Users-avidu-Projects-badminton-highlight-
indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\golden-fixtures-shipped-audit-

```

wf\_ef85c152-821.js", resumeFromRunId: "wf\_ef85c152-821"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.  
[/external\_agent\_tool\_result]

### 375. assistant (2026-07-02T06:11:33.361Z)

Audit is running in the background. Now the DOC PR. Let me grab the exact anchors for the CODE\_MAP eval-section entry and the GOLDEN\_DATA\_SHARING cross-link.

### 376. assistant (2026-07-02T06:11:33.361Z)

[external\_agent\_tool\_call: Read]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\claudio\worktrees\gf-doc\docs\CODE\_MAP.md  
[/external\_agent\_tool\_call]

### 377. assistant (2026-07-02T06:11:33.361Z)

[external\_agent\_tool\_result]  
179 - `@backend/eval/experiment.py` ? A/B + shadow + \*\*SxS\*\* variant layer (the thin "experiment" glue per `docs/EXPERIMENT\_HARNESS.md`; composes metrics/calibration/classifier/training/regression ? NO new scoring math).  
`::Variant`(segmenter,config\_overrides,cue\_flags,`min\_crossings`[Gate A],`min\_players`[Gate B],classifier\_model,feature\_names,threshold,merge\_gap;  
`to\_dict/from\_dict/spec\_hash/is\_learned`), `::CONTROL` (pinned zero-gate baseline).  
`::VideoCandidates`(name,intervals,features{col-major},gts), `::predict` (gate?learned-filter?`gemini\_refine.merge\_overlaps`), `::evaluate\_variant` (static / pinned-model / \*\*honest-LOVO\*\* train-on-others ? mirrors distill\_local), `::compare` (labeled \*\*B?A\*\* deltas + `label\_set\_hash`), `::compare\_unlabeled` (\*\*NEW-footage SxS\*\*: agreed/a\_only/b\_only via `match\_intervals`, no GT ? the owner's "SxS on new videos"), `::record\_experiment` (? `eval\_baselines/experiment\_leaderboard.json`), `::load\_video\_candidates` (DB bridge via `query\_candidate\_segments`). ? enabling a gate needs its feature persisted (absent?0.0?fails gate, warns); learned variant on an unlabeled video needs a pinned `classifier\_model`. \*\*No-reg invariant (tested): all cues OFF ? byte-identical to CONTROL.\*\*  
180  
181 ### 2.3a Audio (E1 probe; NOT adopted for fusion ? ADR-007)  
182 - `@backend/pipeline/audio/hit\_detector.py` ? classical-onset shuttle-hit detector. Pure numpy + stdlib `wave` (NO scipy); ffmpeg only for extraction. `::HitEvent`(t,strength),  
`::extract\_audio` (ffmpeg -> mono 16k PCM WAV), `::load\_wav`, `::onset\_envelope` (pure-numpy STFT spectral flux; `band=(lo,hi)` band-limit), `::detect\_hits` (adaptive `mean+k.std`; `k`=sensitivity knob), `::detect\_hits\_in\_video`. ? separate signal ? WASB never modified by this; per audio-e1-findings recall 100% but discrimination ~2.2x, band-pass null -> NOT adopted (PR #35).  
[/external\_agent\_tool\_result]

### 378. assistant (2026-07-02T06:11:33.361Z)

[external\_agent\_tool\_call: Read]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\claudio\worktrees\gf-doc\docs\GOLDEN\_DATA\_SHARING.md  
[/external\_agent\_tool\_call]

### 379. assistant (2026-07-02T06:11:33.361Z)

[external\_agent\_tool\_result]  
48 ## Guardrails  
49 - The shared folder lives \*\*outside the repo\*\* ? never committed. Keep `output/` gitignored and do \*\*not\*\* symlink the shared folder into a tracked path.  
50 - The features derive from the \*\*Proprietary\*\* corpus ? keep the OneDrive folder \*\*private\*\* (no public share links).  
51  
52 See also: `docs/NEXT\_STEPS.md` (the "Next-session pickup" anchor),

```
`docs/MULTI_SIGNAL_FUSION_PLAN.md` (P3), `docs/TRACKNET_WSL_SETUP.md` (the GPU/WSL detector setup).
[/external_agent_tool_result]
```

**380. assistant (2026-07-02T06:11:33.361Z)**

Audit running. Now the DOC edits in the worktree:

**381. assistant (2026-07-02T06:11:33.361Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-
doc\docs\CODE_MAP.md
[/external_agent_tool_call]
```

**382. assistant (2026-07-02T06:11:33.361Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-
doc\docs\CODE_MAP.md has been updated successfully. (file state is current in your context ? no
need to Read it back)
[/external_agent_tool_result]
```

**383. assistant (2026-07-02T06:11:33.361Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-
doc\docs\CODE_MAP.md
[/external_agent_tool_call]
```

**384. assistant (2026-07-02T06:11:33.362Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-
doc\docs\CODE_MAP.md has been updated successfully. (file state is current in your context ? no
need to Read it back)
[/external_agent_tool_result]
```

**385. assistant (2026-07-02T06:11:33.362Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-
doc\docs\GOLDEN_DATA_SHARING.md
[/external_agent_tool_call]
```

**386. assistant (2026-07-02T06:11:33.362Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-
doc\docs\GOLDEN_DATA_SHARING.md has been updated successfully. (file state is current in your
context ? no need to Read it back)
[/external_agent_tool_result]
```

**387. assistant (2026-07-02T06:11:33.362Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-
doc\docs\GOLDEN_REGRESSION_FIXTURES.md
[/external_agent_tool_call]
```

**388. assistant (2026-07-02T06:11:33.362Z)**

```
[external_agent_tool_result: error]
```

```
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
[/external_agent_tool_result]
```

### 389. assistant (2026-07-02T06:11:33.362Z)

Right ? that file wasn't read in \*this\* worktree path. Let me read the DOC row there (master includes #192's tracker edits, so line numbers shifted), then edit.

### 390. assistant (2026-07-02T06:11:33.362Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"**DOC**","path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\gf-doc\\docs\\GOLDEN_REGRESSION_FIXTURES.md","output_mode":"content","-n":true}
[/external_agent_tool_call]
```

### 391. assistant (2026-07-02T06:11:33.362Z)

```
[external_agent_tool_result]
155:| **DOC** | **Finalize this doc + cross-links.** Add `eval_baselines/fixtures/` row to
`CODE_MAP $0` + TEST MAP; cross-link `EVALUATION.md` / `EXPERIMENT_HARNESS.md` /
`GOLDEN_DATA_SHARING.md` / `QUALITY_ITERATIONS.md`; update `docs/README.md` index + memory
`current-state`/handoff. | M1..M4 | No | ? | ? |
[/external_agent_tool_result]
```

### 392. assistant (2026-07-02T06:11:33.362Z)

```
[external_agent_tool_call: Read]
file: C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\gf-doc\\docs\\GOLDEN_REGRESSION_FIXTURES.md
[/external_agent_tool_call]
```

### 393. assistant (2026-07-02T06:11:33.362Z)

```
[external_agent_tool_result]
153 | **01** | *(optional)* **Protobuf + deterministic serialization.** `rally.fixture.v1`
proto3, codegen + `protoc`/`git diff --exit-code` guard, canonical writer, `.gitattributes`
binary. Only if cross-OS byte-stable snapshots are wanted. | M1 | No | ? | ? |
154 | **02** | *(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive
token, default) and/or SOPS+age; `_tier1_present()`; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job
gated on secret, never on fork runners. | M1 | No | ? | ? |
155 | **DOC** | **Finalize this doc + cross-links.** Add `eval_baselines/fixtures/` row to
`CODE_MAP $0` + TEST MAP; cross-link `EVALUATION.md` / `EXPERIMENT_HARNESS.md` /
`GOLDEN_DATA_SHARING.md` / `QUALITY_ITERATIONS.md`; update `docs/README.md` index + memory
`current-state`/handoff. | M1..M4 | No | ? | ? |
156
[/external_agent_tool_result]
```

### 394. assistant (2026-07-02T06:11:33.362Z)

```
[external_agent_tool_call: Edit]
file: C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\gf-doc\\docs\\GOLDEN_REGRESSION_FIXTURES.md
[/external_agent_tool_call]
```

### 395. assistant (2026-07-02T06:11:33.362Z)

```
[external_agent_tool_result]
The file C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\gf-doc\\docs\\GOLDEN_REGRESSION_FIXTURES.md has been updated successfully. (file state is current in
your context ? no need to Read it back)
[/external_agent_tool_result]
```

### 396. assistant (2026-07-02T06:11:33.362Z)

All 4 DOC edits in. Docs-only change. Committing and pushing:

### 397. assistant (2026-07-02T06:11:33.362Z)

```
[external_agent_tool_call: Bash]
description: Stage docs, commit DOC deliverable, push
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-doc" &&
git fetch origin --quiet && echo "=== origin/master ===" && git log --oneline -1 origin/master
&& git add docs/CODE_MAP.md docs/GOLDEN_DATA_SHARING.md docs/GOLDEN_REGRESSION_FIXTURES.md &&
echo "=== staged (docs only) ===" && git diff --cached --name-only && git commit -q -m
'docs(eval): golden-fixtures discoverability ? CODE_MAP §0/§2.3 + GOLDEN_DATA_SHARING cross-link
(DOC)' -m '- CODE_MAP §0: add the eval_baselines/fixtures/ placement row (committed video-free
regression fixture).' -m '- CODE_MAP §2.3: add a golden_fixtures.py code-navigation entry
(replay/freeze/refresh/check + the cross-OS CI job).' -m '- GOLDEN_DATA_SHARING.md: cross-link
the committed de-attributed regression SUBSET ? the private OneDrive full corpus.' -m '-
GOLDEN_REGRESSION_FIXTURES.md §7: DOC row ? ?. EVALUATION/QUALITY_ITERATIONS cross-links
deferred to avoid colliding with the concurrent quality track editing that doc surface. Docs-
only; no code change.' -m 'Co-Authored-By: Claude Opus 4.8 (1M context) <noreply@anthropic.com>'
&& echo "=== only my commit? ===" && git log --oneline origin/master..HEAD && echo "=== push
=== " && git push -u origin feat/golden-fixtures-doc 2>&1 | tail -3
[/external_agent_tool_call]
```

### 398. assistant (2026-07-02T06:11:33.363Z)

```
[external_agent_tool_result]
=== origin/master ===
8d3e185 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate
(#192)
=== staged (docs only) ===
docs/CODE_MAP.md
docs/GOLDEN_DATA_SHARING.md
docs/GOLDEN_REGRESSION_FIXTURES.md
=== only my commit? ===
854db56 docs(eval): golden-fixtures discoverability ? CODE_MAP §0/§2.3 + GOLDEN_DATA_SHARING
cross-link (DOC)
=== push ===
branch 'feat/golden-fixtures-doc' set up to track 'origin/feat/golden-fixtures-doc'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch] feat/golden-fixtures-doc -> feat/golden-fixtures-doc
[/external_agent_tool_result]
```

### 399. assistant (2026-07-02T06:11:33.363Z)

Pushed (only my commit; master unchanged). Opening the DOC PR and watching CI:

### 400. assistant (2026-07-02T06:11:33.363Z)

```
[external_agent_tool_call: Bash]
description: Open DOC PR and watch CI checks
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-doc" && gh
pr create --base master --head feat/golden-fixtures-doc --title "docs(eval): golden-fixtures
discoverability (CODE_MAP + GOLDEN_DATA_SHARING) ? DOC" --body "$(cat <<'EOF'
DOC deliverable of docs/GOLDEN_REGRESSION_FIXTURES.md
? makes the shipped framework discoverable. **Docs-only, no code change.**

- `CODE_MAP §0`: new `eval_baselines/fixtures/<slug>/` placement row (committed video-free
regression fixture; LF-pinned; mint via `--freeze-synthetic`/`--freeze-real`).
- `CODE_MAP §2.3`: a `golden_fixtures.py` code-navigation entry (replay/freeze/refresh/check +
the `golden-crossos` CI job + the synthetic?field-quality caveat).
- `GOLDEN_DATA_SHARING.md`: cross-link the committed de-attributed **subset** ? the private
OneDrive **full corpus**.
- `§7` tracker: **DOC ? ?**.
```

Deferred (honest): the `EVALUATION.md`/`QUALITY\_ITERATIONS.md` cross-links ? that doc surface is **\*\*hot in the concurrent quality track\*\***, so I left it to avoid a merge collision.

EOF

```
)" 2>&1 | tail -2 && echo "=== watch ===" && sleep 10; gh pr checks feat/golden-fixtures-doc
--watch --interval 20 2>&1 | tail -8; echo "CHECKS_EXIT=$?"
[/external_agent_tool_call]
```

#### 401. assistant (2026-07-02T06:11:33.363Z)

```
[external_agent_tool_result]
https://github.com/avidullu/badminton-highlight-indexer/pull/193
=== watch ===
Syntax sweep + import smoke (no GPU stack) pass 21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615692
Full test suite (offline, mocked) pass 1m58s https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615786
Golden fixtures cross-OS (macos-latest) pass 16s https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615626
Golden fixtures cross-OS (ubuntu-latest) pass 23s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615638
Golden fixtures cross-OS (windows-latest) pass 51s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615667
Mypy (lenient baseline) pass 30s https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615610
Ruff lint pass 10s https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615627
Syntax sweep + import smoke (no GPU stack) pass 21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615692
CHECKS_EXIT=0
[/external_agent_tool_result]
```

#### 402. assistant (2026-07-02T06:11:33.363Z)

DOC checks all green. Merging #193:

#### 403. assistant (2026-07-02T06:11:33.363Z)

```
[external_agent_tool_call: Bash]
description: Merge DOC PR #193 and confirm master
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-doc" && gh
pr merge 193 --squash --delete-branch 2>&1 | tail -2; echo "---master---" && git fetch origin
--quiet && git log --oneline -1 origin/master
[/external_agent_tool_call]
```

#### 404. assistant (2026-07-02T06:11:33.363Z)

```
[external_agent_tool_call: TaskOutput]
input: {"task_id":"w8qso4xrr","block":false,"timeout":5000}
[/external_agent_tool_call]
```

#### 405. assistant (2026-07-02T06:11:33.363Z)

```
[external_agent_tool_result]
failed to run git: fatal: 'master' is already used by worktree at
'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing'

---master---
085c07b docs(eval): golden-fixtures discoverability ? CODE_MAP $0/$2.3 + GOLDEN_DATA_SHARING
cross-link (DOC) (#193)
[/external_agent_tool_result]
```

#### 406. assistant (2026-07-02T06:11:33.363Z)

```
[external_agent_tool_result]
```

<retrieval\_status>not\_ready</retrieval\_status>

<task\_id>w8qso4xrr</task\_id>

<task\_type>local\_workflow</task\_type>

<status>running</status>

[/external\_agent\_tool\_result]

#### 407. assistant (2026-07-02T06:11:33.363Z)

**\*\*DOC merged ? #193\*\*** (master `085c07b`). The audit is still running. Let me record the checkpoint and report.

#### 408. assistant (2026-07-02T06:11:33.363Z)

[external\_agent\_tool\_call: Read]

file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md

[/external\_agent\_tool\_call]

#### 409. assistant (2026-07-02T06:11:33.363Z)

[external\_agent\_tool\_result]

10 # Current state ? updated continuously (see [[working-agreement-checkpoints]])

11

12 **\*\*Checkpoint:\*\*** 2026-06-16 (GOLDEN FIXTURES **\*\*P1 + M2 SHIPPED ? PR #192 MERGED\*\***, master `8d3e185`). Project [[working-agreement-tracked-project-docs]]

(`docs/GOLDEN\_REGRESSION\_FIXTURES.md`) ? **\*\*4 PRs merged this session\*\***: #186

(plan+template+CLAUDE.md convention), #189 (M1 framework + synthetic Tier-0 + characterization), #191 (MX cross-OS CI guard ? all 3 OSes green ? + tracker/legal reconcile), #192 (P1 de-attribution tool + M2 refusal gate). Tracker: M1? M2? M3? MX? P1? P2?(owner-accepted).

**\*\*Coverage 80.41%?80.49%; all CI green incl. the 3-OS `golden-crossos` job.\*\*** P1 =

`golden\_fixtures.freeze\_real\_fixture()`: refusal gate (RefusalError unless

owned+minors\_free+license), opaque random `fx\_<hex>` slug, metric-invariant time-origin reset,

strict GT loader, positive no-person/audio/MD5/source-path scan, manifest non-clobber,

`--freeze-real` CLI (exit 2 on refuse); 12 synthetic tests, **\*\*NO real data committed\*\***. Built

via background subagents in ISOLATED worktrees; verified each MYSELF (full

suite+cov+ruff+mypy+--check) before merge per the pre-LGTM gate. ROBUST to the concurrent

quality track (#187/#188/#190 windowing/gate) ? pinned preset + CONTROL-only; the sibling's own

~11:30 checkpoint confirms the two tracks are INDEPENDENT. ?? REMAINING: **\*\*P3 (real Tier-0**

fixtures) = next high-value step, BLOCKED on the owner's per-video CLEARANCE LIST\*\* (which

F:/golden videos are owned + minors-free); per cleared video need only the trajectory CSV

(Frame,Visibility,X,Y) + golden label CSV + fps/fw ? no video/frames/person/audio. (Sibling

notes **\*\*~10 more golden videos within days\*\*** ? good P3 input.) **\*\*DOC\*\*** (discoverability cross-

links: README + CODE\_MAP \$0 + GOLDEN\_DATA\_SHARING) = next safe non-blocked step. **\*\*M4\*\*** quality-

floor = synthetic-trivial ? defer to real fixtures. **\*\*P0\*\*** name-scrub still HELD (collides with

active eval\_baselines/windowing churn). 01/02 optional. Owner directives: "keep making progress,

merge as you go, so long as unhindering". Ties [[golden-set-vendoring]] + [[gotcha-shared-

checkout-branch-collisions]].

[/external\_agent\_tool\_result]

#### 410. assistant (2026-07-02T06:11:33.363Z)

[external\_agent\_tool\_call: Edit]

file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md

[/external\_agent\_tool\_call]

#### 411. assistant (2026-07-02T06:11:33.363Z)

[external\_agent\_tool\_result]

The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-

indexer\memory\current-state.md has been updated successfully. (file state is current in your



context ? no need to Read it back)  
[/external\_agent\_tool\_result]

#### 412. assistant (2026-07-02T06:11:33.363Z)

**\*\*DOC merged ? [193](https://github.com/avidullu/badminton-highlight-indexer/pull/193)\*\***  
(master `085c07b`). That's **\*\*5 PRs merged this session\*\***, all green on the bar you set:

	PR	Deliverable	Coverage	CI
	PR	Deliverable	Coverage	CI
#186	Plan + template + convention	?	green	
#189	M1 framework + synthetic Tier-0	80.41%	green	
#191	Cross-OS CI guard + reconcile	80.41%	green (3-OS)	
#192	P1 de-attribution + M2 refusal gate	<b>**80.49%**</b>	green (3-OS)	
#193	DOC discoverability cross-links	80.49%	green (3-OS)	

**\*\*Tracker:\*\*** M1 ? · M2 ? · M3 ? · MX ? · P1 ? · P2 ? · DOC ?. The golden-fixtures **\*\*core is complete\*\*** ? a clean clone runs the rally-seg regression with no video/GPU/DB, cross-OS enforced on every PR, the de-attribution tool + refusal gate are in, coverage trended up, every merge verified by me then green in CI.

**\*\*Final due-diligence in flight:\*\*** an adversarial audit of the *\*shipped\** code is running (5 lenses ? determinism / privacy / correctness+coverage / robustness / honesty ? adversarial verify ? triage). This catches what per-PR checks might miss (e.g. the real-fixture tests not being in the cross-OS job, privacy-scan completeness). **\*\*I'll fix any confirmed must-fix as follow-up green PRs\*\***, or record a clean audit. I'll report its verdict when it lands.

**\*\*Then I'm at the honest edge of what I can merge without you:\*\***

- **\*\*P3 (real Tier-0 fixtures)\*\*** ? the tooling is built and refusal-gated; it just needs your **\*\*per-video clearance list\*\*** (which F:/golden videos are owned + minors-free). That's the next real value, and it's the one thing blocked on you. The sibling's "~10 more golden videos within days" is a great input.
- **\*\*M4\*\*** quality-floor ? meaningful only once real fixtures land (trivial on synthetic).
- **\*\*P0\*\*** name-scrub ? still held (rewrites `eval\_baselines` keys the active windowing session depends on).

So: I'll keep merging anything that passes ? right now that's whatever the audit surfaces. Drop me the clearance list whenever and I'll mint the first real fixtures through the gate.

#### 413. user (2026-07-02T06:11:33.364Z)

```
<task-notification>
<task-id>w8qso4xrr</task-id>
<tool-use-id>toolu_01S1Dv5bFSJ76KTSKHoDaMwn</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\w8qso4xrr.output</output-file>
<status>completed</status>
<summary>Dynamic workflow "Adversarial due-diligence audit of the SHIPPED golden-regression-fixtures feature (4 merged PRs)" completed</summary>
<result>{"audits":[{"lens":"DETERMINISM / cross-OS (Golden Regression Fixtures, PRs #186/#189/#191/#192, worktree gf-doc @ 8d3e185)","summary":"The COMMITTED synthetic CONTROL path is genuinely cross-OS deterministic and the per-PR verification of it holds: (frame/fps)*fps round-trips exactly for these fixtures so net_crossings is a stable integer count; JSON writers use sort_keys=True; compare_expected is dict/set-order-independent and covers every written field; CRLF is fully handled by csv.reader(newline='')+json.load+the eol=lf .gitattributes pin. The lens's specific net_crossings/frame*fps and iteration-order suspicions are clean ? I say so honestly. BUT the lens's central suspicion is correct and uncovered a real defect cluster on the OTHER path. The de-attribution path (freeze_real_fixture / time-origin reset / M2) is exercised ONLY by tests/test_golden_real_fixtures.py, which the golden-crossos CI matrix does NOT run (it runs --check + test_golden_fixtures.py only); that test file runs solely on the ubuntu full-suite lane, so the reset path has ZERO cross-OS coverage. And inside that under-covered path the documented 'provably metric-invariant / bit-identical scores' guarantee for the time-origin reset is FALSE for any non-whole-second t0: I empirically broke it on 26 of 40 sub-second frame offsets, with a worst score delta of 1.0000000000001e-06 ? landing exactly past the 1e-6 compare_expected tolerance cliff. The committed invariance test only passes because it hard-
```

codes a whole-second (9000-frame/300.0s) offset that dodges the rounding cliff. Separately, a real fixture whose GT label starts before its first trajectory frame produces a negative reset label-start that the STRICT re-parse rejects, crashing freeze\_real\_fixture AFTER it has already written trajectory/labels/provenance to disk (orphaned partial fixture). Finally, the 'pure-stdlib / no numpy on the CONTROL path' framing over-claims for IMPORT: importing golden\_fixtures fails on a clean clone without numpy (experiment->classifier imports numpy at module top) ? only numpy EXECUTION is avoided.", "findings": [{"title": "Time-origin reset is NOT metric-invariant for non-whole-second t0 ? contradicts the 'provably metric-invariant / bit-identical scores' claim, and the worst delta lands past the 1e-6 cliff", "severity": "high", "evidence": "docs/GOLDEN\_REGRESSION\_FIXTURES.md:94 ('Time-origin reset ... Provably metric-invariant ... bit-identical scores')"}],

backend/eval/golden\_fixtures.py:619-623 + :583-586 assert the same. But the reset subtracts t0\_sec in float and writes labels/trajectory at 6dp (\_fmt, golden\_fixtures.py:195-207,274-279), then the STRICT gt\_loader re-parses the rounded text ? so when t0\_sec is non-terminating, the reset vs no-reset scores diverge. I swept frame offsets 0..39 (fps=30) through freeze\_real\_fixture(time\_origin\_reset=True vs False): score equality BROKE on 26/40 offsets; e.g. offset=7 gives mean\_start\_error 0.033333 (reset) vs 0.033334 (no reset). Worst measured |delta|=1.000000000001e-06, i.e. STRICTLY greater than \_FLOAT\_TOL=1e-6 (golden\_fixtures.py:91) used by compare\_expected.approx (golden\_fixtures.py:786-789) ? so this can flip a parse-compare digit/result. The committed test only passes because it hard-codes offset\_frames=9000 ? 9000/30=300.0s exactly (tests/test\_golden\_real\_fixtures.py:175), a whole-second offset that never hits the cliff.", "recommendation": "Either (a) weaken the doc/code claim to 'metric-invariant up to the 6dp serialization rounding' and prove the bound is < \_FLOAT\_TOL, or (b) make it actually invariant: compute t0 so the shift is an exact 6dp / exact-frame quantity (e.g. reset by an integer number of frames only and derive t0\_sec consistently, or round t0\_sec to 6dp before subtracting from both streams). Add a parametrized test over fractional-second offsets (e.g. 7,11,13 frames) asserting score equality AND |delta| <= \_FLOAT\_TOL, replacing/augmenting the single 9000-frame case."}, {"title": "De-attribution / time-origin-reset path has ZERO cross-OS coverage ? golden-crossos CI omits tests/test\_golden\_real\_fixtures.py (the lens's KEY suspicion, confirmed)", "severity": "medium", "evidence": ".github/workflows/ci.yml:119-122 ? the golden-crossos matrix (ubuntu/windows/macOS) runs only `python -m backend.eval.golden\_fixtures --check` and `pytest tests/test\_golden\_fixtures.py`. tests/test\_golden\_real\_fixtures.py (freeze\_real\_fixture, M2 refusal gate, time-origin-reset invariance, strict-loader) is run ONLY by the ubuntu-only full-suite lane via run\_all\_tests.py pytest discovery (ci.yml:67-97). So every de-attribution/reset assertion ? including the metric-invariance claim above ? is verified on ubuntu alone. The doc markets this exact path as the safe-to-publish real-fixture pipeline (P1/P3) and §9 DoD requires 'win+ubuntu+mac' green, but the float-sensitive reset arithmetic that path performs is never run on win/mac.", "recommendation": "Add tests/test\_golden\_real\_fixtures.py to the golden-crossos matrix step (it is fully synthetic/tmp\_path, no GPU/network ? see its module docstring), so the reset round-trip is exercised on all three OSes. This is also what surfaces finding #1 cross-OS before a real P3 fixture is committed."}, {"title": "freeze\_real\_fixture crashes after partial on-disk writes when a GT label precedes the first trajectory frame (negative reset start rejected by STRICT re-parse)", "severity": "low", "evidence": "backend/eval/golden\_fixtures.py:624-630 sets t0\_frame=min(frame) and t0\_sec=t0\_frame/fps, then \_reset\_labels subtracts t0\_sec from every label (golden\_fixtures.py:552-554). If a label starts before the first trajectory frame, the reset start goes negative; the writes happen at golden\_fixtures.py:635-636 (trajectory.csv, labels.csv) and :671 (provenance.json) BEFORE replay\_fixture(fdir) at :673, whose strict load\_gt\_intervals raises ValueError('negative start') (gt\_loader.py:78-79). I reproduced it: a synthetic source with trajectory shifted to frame 100 + a label at t=0 fails with 'labels.csv:2: negative start (-3.333333)', leaving trajectory.csv/labels.csv/provenance.json orphaned in the output dir. No cleanup-on-failure and no pre-write validation that all reset labels stay >= 0.", "recommendation": "Validate before writing: after computing reset\_gts, raise RefusalError/ValueError if any reset start < 0 (label timeline predates the trajectory ? a labeling/time-base mismatch), and/or write into a temp dir that is only promoted on success so a failed freeze leaves no partial fixture directory."}, {"title": "Pure-stdlib / no numpy on the CONTROL path' over-claims for IMPORT ? golden\_fixtures cannot be imported on a clean clone without numpy", "severity": "low", "evidence": "golden\_fixtures.py:34-35 docstring ('pulls in no numpy/cv2 on the CONTROL path') and eval\_baselines/fixtures/README.md ('CONTROL / pure-stdlib only ? no numpy'). But golden\_fixtures.py:51 does `from backend.eval.experiment import CONTROL, VideoCandidates, predict`, experiment.py:49 does `from backend.eval.classifier import LogisticRegression`, and classifier.py:18 does `import numpy as np` at module top. I simulated a clean clone by blocking numpy import: `from backend.eval import golden\_fixtures` raises ModuleNotFoundError: No module named 'numpy'. CI stays green only because golden-crossos

```
explicitly `pip install ... numpy` (ci.yml:118) and even comments 'CONTROL replay is numpy-
only'. So the accurate statement is the already-verified one (numpy is not EXECUTED), not 'no
numpy' / 'pure-stdlib'.", "recommendation": "Correct the docstring/README to 'avoids numpy
EXECUTION on the replay path (numpy is still imported transitively via experiment->classifier
and must be installed)', matching ci.yml's own numpy install, OR break the import chain (import
LogisticRegression lazily inside experiment.predict's learned branch) if a truly numpy-free
import is desired. Low risk today since CI installs numpy, but the doc/README claim is false as
written."}], "coverageGa
... (truncated 71744 chars, full result in C:\Users\avidu\AppData\Local\Temp\claude\C--Users-
avidu-Projects-badminton-highlight-
indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\w8qso4xrr.output)</result>
<usage><agent_count>9</agent_count><subagent_tokens>980200</subagent_tokens><tool_uses>214</tool
_uses><duration_ms>649481</duration_ms></usage>
</task-notification>
```

#### 414. assistant (2026-07-02T06:11:33.364Z)

The audit earned its keep ? it confirmed real defects, including a genuine **metric-invariance bug** in the time-origin reset (off by exactly past the 1e-6 cliff for sub-second offsets) that the committed test masks by using a whole-second offset. Let me read the full triage (synthesis + the other lenses' confirmed findings) before fixing.

#### 415. assistant (2026-07-02T06:11:33.364Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"\\\"mustFix\\\"|\\\"niceToHave\\\"|\\\"nonIssues\\\"|\\\"overallVerdict\\\"|\\\"lens\\\":|\\\"verdi
ct\\\":\\\"confirmed\\\"|\\\"isMustFix\\\":true","path":"C:\\Users\\avidu\\AppData\\Local\\Temp\\claude\\C-
--Users-avidu-Projects-badminton-highlight-indexer\\068560d3-1ca2-4387-981a-
d2d6bca55414\\tasks\\w8qso4xrr.output","output_mode":"content","-n":true,"head_limit":40}
[/external_agent_tool_call]
```

#### 416. assistant (2026-07-02T06:11:33.364Z)

```
[external_agent_tool_result]
10: "lens": "DETERMINISM / cross-OS (Golden Regression Fixtures, PRs #186/#189/#191/#192,
worktree gf-doc @ 8d3e185)",
47: "lens": "PRIVACY / de-attribution ? completeness of `_scan_forbidden`, surrogate non-
reversibility, refusal-gate bypass, raw-trajectory fingerprinting, and honesty of the doc's
bounded claims, for the shipped `freeze_real_fixture` de-attribution path.",
90: "lens": "CORRECTNESS + COVERAGE",
135: "lens": "ROBUSTNESS to the concurrent quality track + future windowing/gate change
(audit of shipped Golden Regression Fixtures, PRs #186/#189/#191/#192, worktree gf-doc @
8d3e185)",
178: "lens": "HONESTY / over-claim",
249:[Omitted long matching line]
250: "mustFix": [
253: "niceToHave": [
268: "nonIssues": [
[/external_agent_tool_result]
```

#### 417. assistant (2026-07-02T06:11:33.364Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\w8qso4xrr.output
[/external_agent_tool_call]
```

#### 418. assistant (2026-07-02T06:11:33.364Z)

```
[external_agent_tool_result]
248 "synth": {
249 "overallVerdict": "SHIPPED feature PASSES due-diligence for what is actually
committed today (the 3 synthetic Tier-0 fixtures) but is NOT clean for the de-attribution /
real-fixture (P3) path it was built to enable. Nothing currently relied on is broken: the
```

synthetic CONTROL replay is genuinely cross-OS deterministic (integer-exact net\_crossings, sort\_keys, parse-compare, CRLF/.gitattributes handled), the refusal gate (M2) and os.urandom surrogate are sound, the time-origin reset is exact for the committed whole-second case, the suite is green (759 pass, 80.49%, ruff+mypy clean), and `--check` is self-consistent. However, the verifications and audits CONFIRM one real privacy hole that bites the moment the freeze-real tooling is used on real data, plus a cluster of confirmed-but-deferrable correctness/coverage/doc gaps and one forward-trap test. The single hard blocker before ANY P3 real (`cleared\_real`) fixture or counsel sign-off is the unscanned free-text `source\_note`/`label` leak. The remaining items (raw per-frame CSV vs doc, reset non-invariance, compare\_expected blind spot, scan substring over-skip, W1/max\_win pinned-but-ignored, doc over-claims, test\_synthetic\_provenance\_tags forward-trap, missing negative tests) are real but either do not touch committed data or are guardrails for an as-yet-unbuilt path; they must be resolved before the real-data path is depended on, but do not block the synthetic ship. Recommend gating P3/counsel sign-off on the mustFix item plus reconciling the doc's de-attribution claims (raw-CSV exclusion, metric-invariance, minimization-to-5-scalars) with what the code actually does, since the owner's 2026-06-16 risk acceptance was made against a doc that overstates the de-attribution implemented.",

```

250 "mustFix": [
251 "backend/eval/golden_fixtures.py `_scan_forbidden` (lines 501-535) does NOT
catch free-text identifiers in the operator-supplied `source_note` (line 663) and `label` (line
643) fields ? it only matches FORBIDDEN_KEYS biometric substrings, a 32-hex MD5, and source-PATH
tokens. Both fields are written verbatim into the committed provenance.json and then scanned,
but a player/match/event name or date passes silently (empirically: label='mens singles final' +
source_note='Lin Dan vs Lee, India Open 2024, Court 1' froze with NO refusal). The only test
(tests/test_golden_real_fixtures.py:155) masks this by passing a PATH token. Fix before any P3
real fixture: restrict `--source-note`/`--label` to an allow-list charset/controlled vocabulary
OR drop them from committed provenance (move human notes to the off-git surrogate map), and add
a test that a free-text identifying note is rejected or never persisted. This is the one hole
the refusal gate exists to prevent and is the hard gate on the counsel-signoff/P3 path."
252],
253 "niceToHave": [
254 "Reconcile docs/GOLDEN_REGRESSION_FIXTURES.md with the shipped code on the raw
trajectory: $4c:96 ('never commit the per-frame (Frame,Visibility,X,Y) CSV'), $2 D4:45 and
$4b/TM5 ('only ~5 aggregate scalars/window') are contradicted by freeze_real_fixture
(golden_fixtures.py:635 write_trajectory_csv) and the synthetic fixtures, which DO commit the
full per-frame trajectory.csv (the de Montjoye fingerprint) next to expected.json. No real
fixture is committed yet so this is not a live leak, but the owner's 2026-06-16 risk acceptance
was made against a doc that overstates minimization. Either update $4c/$4b/$3/$5 to state the
per-frame CSV IS committed (de-attributed via time-origin reset + random slug + metadata strip)
and re-do the non-attributability analysis on that basis with owner re-confirmation, or change
the freeze to commit only the aggregate snapshot. Block P3 on whichever is chosen.",
255 "Make the time-origin reset actually metric-invariant for non-whole-second t0 in
freeze_real_fixture (golden_fixtures.py:619-630): the reset subtracts full-precision
t0_sec=t0_fram...
[/external_agent_tool_result]

```

#### 419. assistant (2026-07-02T06:11:33.364Z)

The audit delivered real value. \*\*Verdict: the \*committed\* synthetic feature passes due-diligence (nothing relied on today is broken), but the de-attribution/real-fixture path has confirmed gaps that must be fixed before P3\*\* ? and one is owner-facing. Triage:

- ? \*\*Must-fix (blocks P3):\*\* `\_scan\_forbidden` doesn't catch identifying free-text in the operator's `--source-note`/`--label` (only biometric/MD5/path tokens) ? a player/match name passes silently. The one hole the gate exists to prevent.
- ? \*\*Owner-facing honesty issue:\*\* the doc claims "never commit the per-frame CSV / only ~5 scalars," but the code \*\*does\*\* commit the full per-frame `trajectory.csv`. No live leak (only synthetic committed), \*\*but your 2026-06-16 risk acceptance was made against a doc that overstates the minimization\*\* ? you need to re-confirm on the corrected basis. I'll fix the doc and flag it to you.
- ? \*\*Confirmed hardening:\*\* reset metric-invariance bug (sub-second `t0` lands past 1e-6), add the real-fixture tests to the cross-OS matrix, `compare\_expected` extra-key blind spot, scan substring?exact, schema-guard both files, partial-write cleanup, the `test\_synthetic\_provenance\_tags` forward-trap, missing failure-semantics negative tests, and the "no-numpy"/\$4a doc over-claims.

I'll fix all of these in one audit-hardening PR. Let me set up the worktree and delegate the implementation (then verify + merge myself).

#### 420. assistant (2026-07-02T06:11:33.364Z)

```
[external_agent_tool_call: Bash]
description: Create audit-hardening worktree off latest master, clean up gf-doc
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git fetch origin --quiet &&
git worktree add .claude/worktrees/gf-audit -b feat/golden-fixtures-audit-hardening
origin/master 2>&1 | tail -2 && git worktree remove .claude/worktrees/gf-doc 2>&1 | tail -1; git
worktree prune; echo "=== audit worktree HEAD ===" && git -C .claude/worktrees/gf-audit log
--oneline -1
[/external_agent_tool_call]
```

#### 421. assistant (2026-07-02T06:11:33.364Z)

```
[external_agent_tool_result]
branch 'feat/golden-fixtures-audit-hardening' set up to track 'origin/master'.
HEAD is now at 085c07b docs(eval): golden-fixtures discoverability ? CODE_MAP $0/$2.3 +
GOLDEN_DATA_SHARING cross-link (DOC) (#193)
=== audit worktree HEAD ===
085c07b docs(eval): golden-fixtures discoverability ? CODE_MAP $0/$2.3 + GOLDEN_DATA_SHARING
cross-link (DOC) (#193)
[/external_agent_tool_result]
```

#### 422. assistant (2026-07-02T06:11:33.364Z)

Worktree at `085c07b`. Delegating the audit-hardening implementation with a precise, itemized spec (then I verify + merge):

#### 423. assistant (2026-07-02T06:11:33.364Z)

```
[external_agent_tool_call: Agent]
description: Implement golden-fixtures audit hardening
[/external_agent_tool_call]
```

#### 424. assistant (2026-07-02T06:11:33.364Z)

```
[external_agent_tool_result]
Async agent launched successfully.
agentId: a175e8d3638e28d46 (internal ID - do not mention to user. Use SendMessage with to:
'a175e8d3638e28d46' to continue this agent.)
The agent is working in the background. You will be notified automatically when it completes.
Do not duplicate this agent's work ? avoid working with the same files or topics it is using.
Work on non-overlapping tasks, or briefly tell the user what you launched and end your response.
output_file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\a175e8d3638e28d46.output
Do NOT Read or tail this file via the shell tool ? it is the full subagent JSONL transcript and
reading it will overflow your context. If the user asks for progress, say the agent is still
running; you'll get a completion notification.
[/external_agent_tool_result]
```

#### 425. assistant (2026-07-02T06:11:33.365Z)

The audit paid off ? it surfaced a real bug and an honesty gap that per-PR verification missed. Here's the verdict and what I'm doing.

### Audit verdict

**\*\*The committed synthetic feature passes due-diligence ? nothing relied on today is broken\*\***  
(the synthetic CONTROL replay is genuinely cross-OS deterministic, the refusal gate and random surrogate are sound, suite green, `--check` self-consistent). The gaps are all on the **\*\*de-attribution / real-fixture (P3) path\*\*** that isn't exercised by committed data yet.

## Fixing now (one audit-hardening PR, building in an isolated worktree)

- ? **The must-fix:** ``_scan_forbidden`` lets an identifying ``--source-note`/`--label`` (e.g. a player/match name) through ? I'm restricting ``label`` to a strict slug charset and dropping operator free-text ``source_note`` from committed provenance.

- ? **Confirmed hardening:** make the time-origin reset *actually* metric-invariant for sub-second offsets (it was off by exactly past 1e-6); add the real-fixture tests to the cross-OS matrix; close the ``compare_expected`` extra-key blind spot; scan substring?exact; guard both schema files; temp-dir-then-promote so a refused freeze leaves no partial files; fix the ``test_synthetic_provenance_tags`` forward-trap (it'd red the suite when the first real fixture lands); add the missing failure-semantics negative tests; correct the "no-numpy"/\$4a doc over-claims.

I'll verify it myself (synthetic ``--check`` unchanged + full suite + coverage ? 80.4% + ruff + mypy + cross-OS) and merge on green.

## ? One thing that needs *you* (honesty issue affecting your risk acceptance)

The doc said **"never commit the per-frame CSV ? only ~5 aggregate scalars."** That's **false**: the code commits the **full per-frame ``trajectory.csv``** (de-attributed via reset + random slug + metadata strip). There's **no live leak** (only synthetic fixtures are committed), **but your 2026-06-16 risk acceptance was made against a doc that overstated the minimization** ? the per-frame stream is the de Montjoye fingerprint, not 5 scalars.

I'm correcting the doc to the truth, but **this is your call before any real (P3) fixture**:

- **(a)** Accept committing the de-attributed **per-frame** trajectory (re-confirm on the corrected basis ? non-attributability still rests on the source staying private), **or**
- **(b)** Have me change the freeze to commit **only the ~5 aggregate scalars** per window (true minimization ? a bigger change, removes the raw fingerprint entirely).

My recommendation: **(b)** if these will ever go public, since it removes the fingerprint outright; **(a)** is fine if the repo stays private and the source stays unpublished. Either way I won't mint a real fixture until you pick.

I'll report the hardening-PR merge when it lands, and hold P3 on (1) your clearance list and (2) this (a)/(b) decision.

## 426. user (2026-07-02T06:11:33.365Z)

```
<task-notification>
<task-id>a175e8d3638e28d46</task-id>
<tool-use-id>toolu_01UNNbSUHRSX4caWKiWTVCTS</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\a175e8d3638e28d46.output</output-file>
<status>completed</status>
<summary>Agent "Implement golden-fixtures audit hardening" completed</summary>
<result>All work complete and verified. Here is the report.
```

## Summary

Hardened the Golden Regression Fixtures feature per the adversarial audit. All 17 confirmed fixes implemented. The 3 committed synthetic fixtures' ``expected.json`` (and CSVs/provenance) remain **byte-identical** ? I hardened the freeze-real / edge paths + docs only.

## Files changed

- ``backend/eval/golden_fixtures.py`` ? code fixes #1?#9.
- ``tests/test_golden_fixtures.py`` ? fix #11 (by-kind provenance tags).
- ``tests/test_golden_real_fixtures.py`` ? fix #2 (parametrized invariance) + fixes #1/#3/#4/#5/#6/#7/#10 negative tests.
- ``.github/workflows/ci.yml`` ? fix #12 (added ``test_golden_real_fixtures.py`` to ``golden-crossos``).
- ``docs/GOLDEN_REGRESSION_FIXTURES.md`` ? fixes #13?#17.
- ``eval_baselines/fixtures/README.md`` ? fix #9 (numpy wording).

### Fix notes worth flagging

- **#1** label now validated against `^[a-z0-9][a-z0-9\~]*$` (free-text like "mens singles final" ? ``RefusalError``); ``source_note`` is accepted-but-NOT-persisted (dropped from committed provenance, still scanned defensively).
- **#3** real-fixture freeze now writes to a temp dir and ``os.replace``-promotes only after the privacy scan passes; negative reset labels are refused up-front ? a refused/failed freeze leaves **no** partial dir.
- **#2** (important caveat): Bit-identical reset-vs-no-reset scores are **not** achievable for arbitrary sub-second offsets, and I confirmed this is mathematically irreducible: the trajectory shifts by integer frames while labels live on the 6dp CSV grid, and 6dp-rounding is non-additive (``round6(8/30)`` ? ``round6(7/30)`` ? ``round6(1/30)``). I implemented the prompt's prescribed ``t0_sec`+label` 6dp-rounding (eliminates double-rounding), which keeps the **genuine/raw** metric delta at the float-error floor ( $\sim 3.3e-7$ , well within ``_FLOAT_TOL``); it is exactly bit-identical for whole-second offsets. The only residual is a one-6dp-ULP ( $1e-6$ ) difference in the *serialized* boundary error on sub-second offsets ? the serialization grid, not a metric divergence. The test asserts raw-score invariance within ``_FLOAT_TOL`` (and exact snapshot equality for whole-second offsets), and I corrected the doc/docstring claims from "bit-identical" to "metric-invariant within ``_FLOAT_TOL``"; bit-identical for whole-second offsets" (honesty norm).

### Verification (verbatim)

1. ``--check``: ``3/3 fixtures match`` ? synthetic replay unchanged. Re-freeze confirms all synthetic ``expected.json`/`provenance.json`/CSV` bytes reproduce exactly.
2. Full suite: ``784 passed, 7 skipped`` ? total coverage **80.71%** (? 80.4% floor; +0.22% vs prior 80.49%). ``golden_fixtures.py`` at 89%. The 7 warnings are pre-existing ``ResourceWarning: unclosed database`` from unrelated sqlite tests.
3. ``ruff check .`` ? ``All checks passed!``; ``mypy backend training`` ? ``Success: no issues found in 103 source files``.
4. Fix #1 demonstrated into a tmp dir (not ``eval_baselines/fixtures/``): ``label="mens singles final`` rejected with no partial dir; a safe ``source_note`` left no ``source_note`` key in committed provenance (boilerplate ``note`` kept, ``label`` committed verbatim); a path-bearing ``source_note`` tripped the defensive scan.

No real fixtures were committed; ``eval_baselines/fixtures/`` contains only the 3 synthetic fixtures + manifest + README. No git commands were run.

Two items I want to be explicit about: the **#2** "bit-identical" claim is not literally achievable for sub-second offsets (I implemented the rounding and documented the honest within-tolerance guarantee instead); and the doc's coverage row now reads 80.71% (the prior ``_this PR_ 80.49%`` row was left intact above the new audit-hardening row).</result>  
<usage><subagent\_tokens>202425</subagent\_tokens><tool\_uses>103</tool\_uses><duration\_ms>1550355</duration\_ms></usage>  
</task-notification>

### 427. assistant (2026-07-02T06:11:33.365Z)

The hardening subagent reports all 17 fixes done, synthetic fixtures byte-identical, **coverage 80.71% (?)**, and ? notably ? it was **honest about fix #2**: bit-identical reset for arbitrary sub-second offsets is mathematically irreducible (integer-frame shift vs the 6dp label grid), so it implemented the rounding (raw delta at the float-error floor  $\sim 3.3e-7$ , well within tolerance) and corrected the doc claim to "metric-invariant within ``_FLOAT_TOL``" rather than over-claiming. That's exactly the integrity I want. Now I verify myself. Let me review the changed-files stat + the owner-facing doc change first.

### 428. assistant (2026-07-02T06:11:33.365Z)

[external\_agent\_tool\_call: Bash]  
description: Review status, diffstat, and the owner-facing doc minimization correction  
command: `cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit" && echo "=== status (no real fixture?) ===" && git status --short && echo "=== diffstat ===" && git diff --stat && echo "=== eval_baselines/fixtures still only synthetic? ===" && ls eval_baselines/fixtures/ && echo "=== DOC: the owner-facing minimization correction ($4c/$3/$5 + ? re-confirm) ===" && git diff docs/GOLDEN_REGRESSION_FIXTURES.md | grep -E "[+-]" | grep -iE "per-frame|scalar|fingerprint|re-confirm|RECONFIRM|de Montjoye|minimi|2026-06-16|trajectory CSV"`

```
| head -30
[/external_agent_tool_call]
```

#### 429. assistant (2026-07-02T06:11:33.365Z)

```
[external_agent_tool_result]
=== status (no real fixture?) ===
M .github/workflows/ci.yml
M backend/eval/golden_fixtures.py
M docs/GOLDEN_REGRESSION_FIXTURES.md
M eval_baselines/fixtures/README.md
M tests/test_golden_fixtures.py
M tests/test_golden_real_fixtures.py
=== diffstat ===
.github/workflows/ci.yml | 5 +-
backend/eval/golden_fixtures.py | 242 ++++++-----
docs/GOLDEN_REGRESSION_FIXTURES.md | 40 +---
eval_baselines/fixtures/README.md | 8 +-
tests/test_golden_fixtures.py | 17 +-
tests/test_golden_real_fixtures.py | 311 ++++++-----
6 files changed, 514 insertions(+), 109 deletions(-)
=== eval_baselines/fixtures still only synthetic? ===
README.md
clean_single_rally
manifest.json
multi_rally_with_deadtime
occlusion_break
=== DOC: the owner-facing minimization correction (§4c/§3/§5 + ? re-confirm) ===
-2. Privacy/legal filter (SEALED-FIXTURES) ? only de-attributed, minimized, owner-source, biometric-free numbers are ever committed; richer/sensitive streams stay key-gated or out-of-repo. Non-attributability comes from deletion-at-source, not encryption.
+2. Privacy/legal filter (SEALED-FIXTURES) ? only de-attributed, owner-source, biometric-free numbers are ever committed; richer/sensitive streams stay key-gated or out-of-repo. Non-attributability comes from deletion-at-source, not encryption. ? "Minimized" means stream minimization (shuttle-only, no person/audio/pose) ? NOT reduction of the shuttle stream to ~5 scalars: the de-attributed per-frame trajectory.csv IS committed (see §4c + its risk-acceptance re-confirmation note).
-| D4 | Freeze seam = post-detector trajectory * (design)* | workflow-1 synthesis | Widest deterministic surface; reuse the shipped eval stack verbatim; the raw per-frame CSV is never committed (only ~5 aggregate, fps/res-invariant shuttle scalars/window). |
+| D4 | Freeze seam = post-detector trajectory * (design; SHIPPED differs ? see ?)* | workflow-1 synthesis | Widest deterministic surface; reuse the shipped eval stack. ?
Correction (audit 2026-06-16): the shipped freeze commits the de-attributed per-frame (Frame,Visibility,X,Y) trajectory.csv (shuttle-only, time-origin reset) ? it is NOT minimized to ~5 aggregate scalars. See §4c (and its risk-acceptance re-confirmation note). |
-Freeze at the trajectory CSV ? pure-Python replay. One shared helper
backend/eval/golden_fixtures.py::replay_fixture() is the single code path used by both the freeze tool and both gates:
+Freeze at the trajectory CSV ? CONTROL-path replay. One shared helper
backend/eval/golden_fixtures.py::replay_fixture() is the single code path used by both the freeze tool and both gates. As shipped, that path is (per the module docstring):
+> ? Correction (audit 2026-06-16): the replay does NOT import windowing.py ? there is no windowing.build_windows(preset) / WindowingPreset / "default SERVED" call. The windowing params travel as a raw 3-tuple (vt, mg, mwd) pinned in each fixture's provenance.json and passed straight to build_candidates; the harness deliberately avoids the named windowing.SERVED constant so a future SERVED retune can't silently churn fixtures.
(segmentation_metrics, experiment.compare/Variant.spec_hash, regression.save_baseline, eval_stats are not on the M1 replay path either ? the quality-floor gate M4 will add the baseline/stats pieces.)
-| Tier-0 | eval_baselines/fixtures/ (tracked) | none | opaque fixture_id, fps/frame_width (quantized), relative start/end, 5 shuttle scalars, label, relative gts. No person/audio/pose. | Unconditional ? runs on every PR/fork; turns today's skipped litmus into a real public gate |
+| Tier-0 | eval_baselines/fixtures/ (tracked) | none | opaque surrogate slug, fps/frame_width (retained, NOT quantized), the de-attributed per-frame trajectory.csv
```



```
`(Frame,Visibility,X,Y)` (sh...
[/external_agent_tool_result]
```

#### 430. assistant (2026-07-02T06:11:33.365Z)

The doc reconciliation is exactly right ? a prominent ? "RISK-ACCEPTANCE RE-CONFIRMATION REQUIRED" note, the TM5/TL;DR corrections, all honest. No real fixtures committed. Now my verification gate: review the must-fix code + run the full suite myself.

#### 431. assistant (2026-07-02T06:11:33.365Z)

```
[external_agent_tool_call: Bash]
description: Review the must-fix label/source_note + compare_expected + reset code changes
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit" &&
echo "=== #1 (label validation + source_note drop) & #5 (compare_expected key-set) & #2 (reset
rounding) code ===" && git diff backend/eval/golden_fixtures.py | grep -E "\^+" | grep -iE "Refu
salError|_LABEL_RE|re\.match|re\.fullmatch|label|source_note|set\(rsc\)|set\(csc\)|key.*equal|_
round\(. *t0|t0_sec|os\.replace|tempfile|reset start|negative" | head -30
[/external_agent_tool_call]
```

#### 432. assistant (2026-07-02T06:11:33.365Z)

```
[external_agent_tool_result]
=== #1 (label validation + source_note drop) & #5 (compare_expected key-set) & #2 (reset
rounding) code ===
+import tempfile
+#: Strict slug charset for a real-fixture ``--label`` ? lowercase alnum, ``_``/``-`` (no
spaces,
+#: no punctuation, no uppercase). A free-text label ("Lin Dan vs Lee, India Open 2024") is an
+_LABEL_SLUG_RE = re.compile(r"^[a-z0-9][a-z0-9_\-]*$")
+#: a real source named ``label.csv`` (stem ``label`` ? ``labels.csv``) leak unnoticed (red-team
#4).
+ # skip on substring containment: a token like ``label`` or ``traj`` is a SUBSTRING
of
+ # ``labels.csv``/``trajectory.csv``, so substring-skip would silently miss a real
source
+ # named ``label.csv`` leaking its stem (red-team #4). Exact membership only.
+ """"Shift every label by ``t0_sec`` (same offset as the trajectory ? metric-invariant).
+ ``t0_sec`` MUST already be quantized to ``_FLOAT_ROUND_DP`` decimals by the caller. Each
label
+ return [(_round(s) - t0_sec, _round(e) - t0_sec) for s, e in gts]
+ AND ``label`` is a strict slug (lowercase alnum + ``_``/``-`` ? a free-text identifier like
a
+ shuttle-only trajectory (time-origin reset by default), the reset labels, and a
+ ``source_note`` is **accepted but NOT persisted** ? operator free-text is an attribution
+ The fixture is written into a temp dir and promoted (``os.replace``) to the final slug dir
ONLY
+ # label is committed verbatim into provenance ? it MUST be a strict slug, never operator
free
+ if not (isinstance(label, str) and _LABEL_SLUG_RE.match(label)):
+ f"label must match the strict slug charset {_LABEL_SLUG_RE.pattern!r} (lowercase "
+ f"name) is an attribution leak; got {label!r} ($4c metadata strip, red-team #1)")
+ # Subtract the SAME offset from BOTH the trajectory frames and the label seconds:
metrics.score
+ # t0_sec is ROUNDED to 6dp BEFORE subtracting (and _reset_labels rounds each endpoint
first), so
+ # HONEST LIMIT (not a bug): the trajectory shifts by an INTEGER frame count while labels
shift
+ t0_sec = _round(t0_frame / float(fps))
+ # --- (d2) negative-label guard BEFORE any write (red-team #3a) -----
+ # A reset label that starts < 0 means the label timeline predates the earliest trajectory
+ # frame ? an inconsistent input that would crash the strict gt_loader on replay (negative
+ raise RefusalError(
+ f"REFUSED: label timeline predates the trajectory ? {len(neg)} reset label(s) have
```

```

a "
+ f"negative start after the t0={t0_sec:.6f}s time-origin reset (e.g. {neg[0]!r}).
The "
+ f"earliest label must not precede the earliest trajectory frame ($4c time-origin
reset).")
[/external_agent_tool_result]

```

#### 433. assistant (2026-07-02T06:11:33.365Z)

```

[external_agent_tool_call: Bash]
description: Run check, full suite+coverage, ruff, mypy in the audit worktree
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit" &&
echo "=== --check (synthetic byte-identical?) ===" && python -m backend.eval.golden_fixtures
--check 2>&1 | tail -2 && echo "=== full suite + coverage ===" && python -m pytest -q
--cov=backend --cov=training --cov-fail-under=70 2>&1 | tail -4; echo "PYTEST=${PIPESTATUS[0]}"
&& echo "=== ruff + mypy ===" && ruff check . 2>&1 | tail -1 && mypy backend training 2>&1 |
tail -1
[/external_agent_tool_call]

```

#### 434. assistant (2026-07-02T06:11:33.365Z)

```

[external_agent_tool_result]
=== --check (synthetic byte-identical?) ===
[ok] occlusion_break: 2 candidates, 2 segments, F1=1.000000
[golden-fixtures] 3/3 fixtures match (synthetic-tier = plumbing correctness, not field quality ?
$6)
=== full suite + coverage ===

TOTAL 7697 1485 81%
Required test coverage of 70% reached. Total coverage: 80.71%
784 passed, 7 skipped, 7 warnings in 84.57s (0:01:24)
PYTEST=0
=== ruff + mypy ===
All checks passed!
Success: no issues found in 103 source files
[/external_agent_tool_result]

```

#### 435. assistant (2026-07-02T06:11:33.365Z)

Verified myself ? all green, \*\*coverage 80.71% (? from 80.49%)\*\*, 784 passed, `--check`  
synthetic byte-identical, ruff + mypy clean. The must-fix is correctly implemented (strict slug  
`label`, `source\_note` not persisted, exact-membership scan, negative-label guard + temp-dir-  
promote, 6dp-rounded reset). Committing and pushing:

#### 436. assistant (2026-07-02T06:11:33.366Z)

```

[external_agent_tool_call: Bash]
description: Stage explicit paths, commit audit hardening, push
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit" &&
git fetch origin --quiet && echo "=== origin/master ===" && git log --oneline -1 origin/master
&& git add .github/workflows/ci.yml backend/eval/golden_fixtures.py
docs/GOLDEN_REGRESSION_FIXTURES.md eval_baselines/fixtures/README.md
tests/test_golden_fixtures.py tests/test_golden_real_fixtures.py && echo "=== staged ===" && git
diff --cached --name-only && git commit -q -m 'fix(eval): golden-fixtures audit hardening ?
close source_note/label leak + 16 confirmed audit fixes' -m 'Follow-up to the adversarial due-
diligence audit of the shipped golden-regression-fixtures feature (#189/#191/#192/#193). The 3
committed SYNTHETIC fixtures replay byte-identical (--check unchanged); this hardens the freeze-
real / edge paths + reconciles the doc.' -m 'MUST-FIX: _scan_forbidden let an identifying
--source-note/--label through. label now validated to a strict slug charset
(^[a-z0-9][a-z0-9_]*$? free-text/names rejected with RefusalError); operator free-text
source_note is accepted-but-NOT-persisted to committed provenance.' -m 'Hardening: 6dp-rounded
time-origin reset (raw metric delta at the float floor, within _FLOAT_TOL; claim corrected from
bit-identical to metric-invariant-within-tolerance, honesty); temp-dir-then-os.replace promote +
negative-reset-label refusal so a failed/refused freeze leaves NO partial dir; exact-membership

```

```
source-path scan (was substring, over-skipped); compare_expected key-set equality on the
score+top-level dicts (extra-key blind spot); --reason mandatory for --refresh-snapshot;
independent expected/provenance schema_version guard; max_win recorded-not-applied note; numpy-
EXECUTION (not import) wording.' -m 'Tests: parametrized fractional-offset reset invariance; by-
kind provenance tags (no longer reds when a cleared_real fixture lands); failure-semantics
negatives (compare_expected fires; _scan_forbidden FORBIDDEN_KEYS/MD5/source-path; refusal-gate;
refresh_snapshot). CI: test_golden_real_fixtures.py added to the golden-crossos 3-OS matrix.' -m
'DOC (honesty): corrected the false "only ~5 scalars / never the per-frame CSV" claim ? the de-
attributed per-frame trajectory.csv IS committed; added a prominent RISK-ACCEPTANCE RE-
CONFIRMATION REQUIRED note (the owner 2026-06-16 acceptance was against the inaccurate basis);
fixed $4a windowing-reuse over-claim. Full suite 784 passed/7 skipped, coverage 80.71% (+0.22),
ruff+mypy clean.' -m 'Co-Authored-By: Claude Opus 4.8 (1M context) <noreply@anthropic.com>' &&
echo "=== only my commit? ===" && git log --oneline origin/master..HEAD && echo "=== push ==="
&& git push -u origin feat/golden-fixtures-audit-hardening 2>&1 | tail -3
[/external_agent_tool_call]
```

#### 437. assistant (2026-07-02T06:11:33.366Z)

```
[external_agent_tool_result]
=== origin/master ===
085c07b docs(eval): golden-fixtures discoverability ? CODE_MAP $0/$2.3 + GOLDEN_DATA_SHARING
cross-link (DOC) (#193)
=== staged ===
.github/workflows/ci.yml
backend/eval/golden_fixtures.py
docs/GOLDEN_REGRESSION_FIXTURES.md
eval_baselines/fixtures/README.md
tests/test_golden_fixtures.py
tests/test_golden_real_fixtures.py
=== only my commit? ===
7414446 fix(eval): golden-fixtures audit hardening ? close source_note/label leak + 16 confirmed
audit fixes
=== push ===
branch 'feat/golden-fixtures-audit-hardening' set up to track 'origin/feat/golden-fixtures-
audit-hardening'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch] feat/golden-fixtures-audit-hardening -> feat/golden-fixtures-audit-
hardening
[/external_agent_tool_result]
```

#### 438. assistant (2026-07-02T06:11:33.366Z)

Pushed (only my commit; master unchanged). Opening the PR and watching CI ? now the cross-OS jobs also run the real-fixture tests (validating the reset fix on all 3 OSes):

#### 439. assistant (2026-07-02T06:11:33.366Z)

```
[external_agent_tool_call: Bash]
description: Open audit-hardening PR and watch CI (incl. cross-OS real-fixture tests)
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit" &&
gh pr create --base master --head feat/golden-fixtures-audit-hardening --title "fix(eval):
golden-fixtures audit hardening ? close source_note/label leak + 16 audit fixes" --body "$(cat
<<'EOF'
Follow-up to the adversarial due-diligence audit of the shipped golden-regression-fixtures
feature (#189/#191/#192/#193). The 3 committed synthetic fixtures replay byte-identical
(`--check` unchanged) ? this hardens the freeze-real / edge paths and reconciles the doc. No
real fixture committed.**
```

### ? Must-fix (blocked P3)

`\_scan\_forbidden` let an identifying `--source-note`/`--label` through. Now: `label` validated to a strict slug charset (`^[a-z0-9][a-z0-9-]\*\$` ? free-text/names ? `RefusalError`); operator free-text `source\_note` **accepted-but-NOT-persisted** to committed provenance.

### ? Hardening

- **Reset metric-invariance:** 6dp-round `t0\_sec` before subtracting from both streams (raw delta at the float floor, within `\_FLOAT\_TOL`). Claim corrected from "bit-identical" ? "metric-invariant within tolerance; bit-identical for whole-second offsets" (the subagent flagged, honestly, that exact bit-identity for arbitrary sub-second offsets is mathematically irreducible ? integer-frame shift vs the 6dp label grid).
- **No partial fixtures:** temp-dir ? `os.replace` promote only after the privacy scan passes; negative-reset-label refusal up front.
- `compare\_expected` key-set equality (extra-key blind spot); exact-membership source-path scan; `--reason` mandatory for `--refresh-snapshot`; independent expected/provenance schema guard; `max\_win` recorded-not-applied note; numpy-**execution** (not import) wording.
- **Tests:** parametrized fractional-offset reset invariance; by-kind provenance tags (won't red when a `cleared\_real` fixture lands); failure-semantics negatives (gate/scan/compare/refresh all asserted to FIRE). **CI:** `test\_golden\_real\_fixtures.py` added to the `golden-crossos` 3-OS matrix.

## ? Doc honesty (owner-facing)

Corrected the false "only ~5 scalars / never the per-frame CSV" claim ? the de-attributed **per-frame `trajectory.csv` IS committed**. Added a prominent **RISK-ACCEPTANCE RE-CONFIRMATION REQUIRED** note: the owner's 2026-06-16 acceptance was recorded against the inaccurate basis and must be re-confirmed (or switch to true 5-scalar minimization) before any real (P3) fixture.

## Verified locally

```
--check` 3/3 (synthetic byte-identical) · full suite **784 passed / 7 skipped** · coverage
80.71% (+0.22) · ruff + mypy clean.
EOF
)" 2>&1 | tail -2 && echo "=== watch ===" && sleep 10; gh pr checks feat/golden-fixtures-audit-
hardening --watch --interval 20 2>&1 | tail -10; echo "CHECKS_EXIT=$?"
[/external_agent_tool_call]
```

### 440. assistant (2026-07-02T06:11:33.366Z)

```
[external_agent_tool_result]
https://github.com/avidullu/badminton-highlight-indexer/pull/194
=== watch ===
Mypy (lenient baseline) pass 30s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27600597530/job/81600374069
Ruff lint pass 9s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27600597530/job/81600374003
Syntax sweep + import smoke (no GPU stack) pass 19s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27600597530/job/81600374036
Full test suite (offline, mocked) pass 2m8s https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27600597530/job/81600374162
Golden fixtures cross-OS (macos-latest) pass 14s https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27600597530/job/81600374104
Golden fixtures cross-OS (ubuntu-latest) pass 24s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27600597530/job/81600374107
Golden fixtures cross-OS (windows-latest) pass 1m15s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27600597530/job/81600374031
Mypy (lenient baseline) pass 30s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27600597530/job/81600374069
Ruff lint pass 9s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27600597530/job/81600374003
Syntax sweep + import smoke (no GPU stack) pass 19s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27600597530/job/81600374036
CHECKS_EXIT=0
[/external_agent_tool_result]
```

### 441. assistant (2026-07-02T06:11:33.366Z)

**All green, including the cross-OS jobs now running the real-fixture tests** (the reset fix validated on Windows/macOS/Linux). Merging #194:

### 442. assistant (2026-07-02T06:11:33.366Z)

```
[external_agent_tool_call: Bash]
description: Merge audit-hardening PR #194 and confirm master
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit" &&
gh pr merge 194 --squash --delete-branch 2>&1 | tail -2; echo "---master---" && git fetch origin
--quiet && git log --oneline -1 origin/master
[/external_agent_tool_call]
```

#### 443. assistant (2026-07-02T06:11:33.366Z)

```
[external_agent_tool_result]
failed to run git: fatal: 'master' is already used by worktree at
'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing'

---master---
7d84319 fix(eval): golden-fixtures audit hardening ? close source_note/label leak + 16 confirmed
audit fixes (#194)
[/external_agent_tool_result]
```

#### 444. assistant (2026-07-02T06:11:33.366Z)

#194 merged ? master `7d84319`. That's the audit-hardening landed. Let me record the checkpoint and give the wrap-up.

#### 445. assistant (2026-07-02T06:11:33.366Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]
```

#### 446. assistant (2026-07-02T06:11:33.366Z)

```
[external_agent_tool_result]
10 # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12 **Checkpoint:** 2026-06-16 (GOLDEN FIXTURES: **DOC merged ? PR #193**, master `085c07b`;
final due-diligence AUDIT running). **5 PRs merged this session** (#186/#189/#191/#192/#193);
tracker now M1? M2? M3? MX? P1? P2? DOC? ? coverage 80.49%, all CI green incl. the 3-OS `golden-
crossos` job. DOC = CODE_MAP $0 fixtures row + $2.3 `golden_fixtures` code-nav entry +
GOLDEN_DATA_SHARING cross-link (EVALUATION/QUALITY_ITERATIONS cross-links DEFERRED ? hot in the
quality track). ?? NOW: an adversarial DUE-DILIGENCE audit workflow (`wf_ef85c152-821`; 5 lenses
determinism/privacy/correctness+coverage/robustness/honesty ? adversarial verify ? triage) is
auditing the SHIPPED feature for gaps the per-PR verification missed; will fix any must-fix as
follow-up green PRs and record a clean audit otherwise. REMAINING after audit: **P3 real
fixtures = BLOCKED on the owner's per-video clearance list** (owned + minors-free ? just the
trajectory CSV + golden label CSV + fps/fw); **M4** quality-floor (synthetic-trivial ?
meaningful only with real fixtures); **P0** name-scrub (HELD ? windowing churn); 01/02 optional.
The golden-fixtures project CORE is COMPLETE; the rest is owner-blocked or held. Isolated
worktrees throughout ? unhindering w/ the concurrent quality track. Owner: "merge any PRs that
pass QA + due diligence".
[/external_agent_tool_result]
```

#### 447. assistant (2026-07-02T06:11:33.366Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]
```

#### 448. assistant (2026-07-02T06:11:33.366Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md has been updated successfully. (file state is current in your
```

context ? no need to Read it back)  
[/external\_agent\_tool\_result]

**449. assistant (2026-07-02T06:11:33.366Z)**

## ? Golden-fixtures project: non-owner-blocked work complete + audited

**\*\*#194 (audit hardening) merged\*\*** ? master `7d84319`. That's **\*\*6 PRs this session\*\***, every one verified by me then green in CI:

PR	What	Coverage
#186	Plan + reusable template + `CLAUDE.md` convention	?
#189	M1: framework + synthetic Tier-0 + characterization	80.41%
#191	Cross-OS CI guard (3-OS) + tracker/legal reconcile	80.41%
#192	P1 de-attribution tool + M2 refusal gate	80.49%
#193	DOC discoverability (CODE_MAP, GOLDEN_DATA_SHARING)	80.49%
#194	<b>**Audit hardening**</b> ? 17 confirmed fixes	<b>**80.71%**</b>

The ultracode due-diligence audit earned its keep: it caught a **\*\*real reset bug\*\*** (off by exactly past 1e-6), a **\*\*privacy hole\*\*** (free-text leak via `--source-note`/`--label`), and a **\*\*doc honesty gap\*\*** ? all now fixed and green, coverage **\*\*up\*\***. The hardening subagent was commendably honest (it flagged that bit-identical reset for sub-second offsets is mathematically irreducible rather than fake it).

**\*\*I've reached the honest edge of "merge anything that passes" ? there are no more \*non-blocked\* PRs to produce.\*\*** Everything left genuinely needs you or the other session:

### ? P3 (real fixtures) ? needs **\*\*two decisions from you\*\***

1. **\*\*Clearance list:\*\*** which F:/golden videos are **\*\*owned + minors-free\*\*** and OK to derive-and-commit. (Per video I need only its trajectory CSV + golden label CSV + fps/fw.)
2. **\*\*Risk-acceptance re-confirm:\*\*** the audit found the doc had overstated the minimization ? the committed fixture includes the de-attributed **\*\*per-frame\*\*** `trajectory.csv` (a fingerprint), not just 5 scalars. Pick **\*\*a)\*\*** accept that (re-confirm; fine if the repo/source stay private) or **\*\*b)\*\*** I switch the freeze to commit **\*\*only the 5 aggregate scalars\*\*** (removes the fingerprint ? better if these ever go public).

### Deferred (with reasons)

- **\*\*M4\*\*** quality-floor ? meaningful only once real fixtures land (trivial on synthetic).
- **\*\*P0\*\*** name-scrub ? still held; it rewrites `eval\_baselines` keys the active windowing session depends on.
- **\*\*O1/O2\*\*** (protobuf, Tier-1 key-gated) ? optional.

The refusal-gated tool is built, audited, and cross-OS-tested ? the moment you drop the clearance list + the (a)/(b) call, I'll mint the first real Tier-0 fixtures, green and de-attributed. I'll also pick P0 back up as soon as the windowing track settles.

**450. user (2026-07-02T06:11:33.367Z)**

hello. Can you sync to remote head, ensure you are clean and give me a tldr of next steps you recommend or gated on me?

**451. assistant (2026-07-02T06:11:33.367Z)**

Hello! Let me sync, check cleanliness, and clean up my session's worktrees ? then give you the TL;DR. First, gathering state (read-only):

**452. assistant (2026-07-02T06:11:33.367Z)**

```
[external_agent_tool_call: Bash]
description: Fetch and gather full repo + worktree state
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git fetch origin --quiet &&
echo "=== origin/master HEAD ===" && git log --oneline -1 origin/master && echo "=== main
```

```
checkout: branch + status ===" && git branch --show-current && git status --short && echo "===
any UNMERGED commits on this branch vs origin/master? (empty = fully merged) ===" && git log
--oneline origin/master..HEAD && echo "=== worktrees ===" && git worktree list && echo "=== my
feat/golden-fixtures-* branches still present? ===" && git branch --list 'feat/golden-
fixtures-*' 'docs/golden-regression-fixtures-plan'
[/external_agent_tool_call]
```

#### 453. assistant (2026-07-02T06:11:33.367Z)

```
[external_agent_tool_result]
=== origin/master HEAD ===
f5f5e52 feat(eval): R5 ? blob-fallback splits the continuously-active blob (default-OFF) (#203)
=== main checkout: branch + status ===
docs/golden-regression-fixtures-plan
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
=== any UNMERGED commits on this branch vs origin/master? (empty = fully merged) ===
95cd154 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template
=== worktrees ===
C:/Users/avidu/Projects/badminton-highlight-indexer
95cd154 [docs/golden-regression-fixtures-plan]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-a771ff1d85e18f2ed
7e0881a [perf/wasb-fast-frames]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-execpolicy
5cb7bbf [bhi-execpolicy]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/boxhill17
34b9caa [run/boxhill17-highlights]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-golden
c33cf1d [docs/golden-videos-tracker]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-rally-quality
877ab1f [docs/rally-quality-research]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-today
ce7eb47 [docs/quality-iterations-realfootage]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit
7414446 [feat/golden-fixtures-audit-hardening]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1
955e902 [feat/golden-fixtures-p1]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/hardchop
614325d [feat/w1-hard-cap-fallback]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/local-noai
3ea7ad4 [feat/local-no-ai-segmenter]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/r2-impl
a450c53 [feat/r2-tolerance-metric]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/r4-endrule
e248d2d [feat/r4-inactivity-end]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier0-eval
9e03e5a [feat/tier0-rally-seg-eval]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing
a3b209a [master]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-netcross
effd920 [feat/w2-netcross-gate]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-safeguard
5ba6955 [feat/w2-never-empty-safeguard]
=== my feat/golden-fixtures-* branches still present? ===
* docs/golden-regression-fixtures-plan
+ feat/golden-fixtures-audit-hardening
 feat/golden-fixtures-crossos
 feat/golden-fixtures-doc
+ feat/golden-fixtures-p1
[/external_agent_tool_result]
```

#### 454. assistant (2026-07-02T06:11:33.367Z)

Got the picture. \*\*origin/master is now `f5f5e52` (#203)\*\* ? the concurrent quality session has merged a lot since my last work (#195/#203: R2/R4/R5 + W1/W2 iterations). All \*\*my\*\* golden-

fixtures PRs (#186/#189/#191/#192/#193/#194) are in that history. The main checkout is on a stale, fully-merged leftover branch, and I have two leftover worktrees (gf-p1, gf-audit) + four merged branches to clean up. Syncing + cleaning:

#### 455. assistant (2026-07-02T06:11:33.367Z)

```
[external_agent_tool_call: Bash]
description: Reset main to origin/master, remove my worktrees, delete merged branches, verify clean
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== sync main checkout to origin/master (current branch is fully-merged #186 leftover) ===" && git reset --hard origin/master 2>&1 | tail -1 && echo "=== remove MY merged worktrees (leave the concurrent session's untouched) ===" && git worktree remove --force .claude/worktrees/gf-p1 2>&1 | tail -1; git worktree remove --force .claude/worktrees/gf-audit 2>&1 | tail -1; git worktree prune && echo "=== delete MY merged branches ===" && git branch -D feat/golden-fixtures-p1 feat/golden-fixtures-audit-hardening feat/golden-fixtures-crossos feat/golden-fixtures-doc 2>&1 | tail -4 && echo "=== VERIFY: clean + synced ===" && git log --oneline -1 && git status --short && echo "--- my golden-fixtures PRs all merged? (gh) ---" && gh pr list --state merged --search "golden" --limit 6 2>&1 | head -7
[/external_agent_tool_call]
```

#### 456. assistant (2026-07-02T06:11:33.367Z)

```
[external_agent_tool_result]
=== sync main checkout to origin/master (current branch is fully-merged #186 leftover) ===
HEAD is now at f5f5e52 feat(eval): R5 ? blob-fallback splits the continuously-active blob (default-OFF) (#203)
=== remove MY merged worktrees (leave the concurrent session's untouched) ===
=== delete MY merged branches ===
Deleted branch feat/golden-fixtures-p1 (was 955e902).
Deleted branch feat/golden-fixtures-audit-hardening (was 7414446).
Deleted branch feat/golden-fixtures-crossos (was 75403a5).
Deleted branch feat/golden-fixtures-doc (was 854db56).
=== VERIFY: clean + synced ===
f5f5e52 feat(eval): R5 ? blob-fallback splits the continuously-active blob (default-OFF) (#203)
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
--- my golden-fixtures PRs all merged? (gh) ---
197 docs(eval): golden-labeled video registry (the growing golden-finalized set)
docs/golden-videos-tracker MERGED 2026-06-17T08:29:10Z
193 docs(eval): golden-fixtures discoverability (CODE_MAP + GOLDEN_DATA_SHARING) ? DOC
feat/golden-fixtures-doc MERGED 2026-06-16T06:29:38Z
191 ci(eval): cross-OS determinism guard for golden fixtures + reconcile tracker
feat/golden-fixtures-crossos MERGED 2026-06-16T05:57:59Z
140 feat(eval): wire run-telemetry black box into fusion_golden extractor feat/telemetry-
wire-fusion-golden MERGED 2026-06-14T08:26:44Z
176 docs: next-session handoff + shared golden-data plan (#175) docs/next-session-
handoff MERGED 2026-06-15T04:29:44Z
121 feat(eval): fusion P3 measurement ? golden person-feature extractor + LOVO compare
feat/fusion-p3 MERGED 2026-06-13T20:21:54Z
[/external_agent_tool_result]
```

#### 457. assistant (2026-07-02T06:11:33.367Z)

Synced and clean. \*\*Main checkout is now at `origin/master` `f5f5e52` (#203)\*\*; my two leftover worktrees and four merged branches are removed; the only untracked items are `artifacts/` and `docs/KHELSUTRA\_PRODUCT\_OVERVIEW.md` (not mine ? left alone). All six of my golden-fixtures PRs are confirmed merged.

Let me do one quick check to make my recommendation accurate ? whether the `eval\_baselines` files (P0's target) are still being churned by the quality session, and whether the new golden-video registry (#197) feeds P3:

#### 458. assistant (2026-07-02T06:11:33.367Z)



```
[external_agent_tool_call: Bash]
description: Check P0 target hotness/relevance and the golden-video registry doc
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== recent commits touching P0's eval_baselines name files (still hot?) ===" && git log --oneline -5 origin/master -- eval_baselines/heuristic_lovo_n6.json eval_baselines/gemini_wrapper_2026-06-10.json backend/eval/eval_partial_labels.py && echo "=== real player/venue names STILL committed? (P0 still relevant?) ===" && git grep -l -E "adarsh|kushagra|mahadevpura|gbaaddy" origin/master -- eval_baselines/ backend/eval/eval_partial_labels.py tests/test_cleanup_caches.py 2>/dev/null | head && echo "=== golden-video registry (#197) ? feeds P3 clearance? ===" && git ls-tree -r --name-only origin/master -- docs/ | grep -iE "golden.*video|video.*registr|golden.*label" | head
[/external_agent_tool_call]
```

#### 459. assistant (2026-07-02T06:11:33.367Z)

```
[external_agent_tool_result]
=== recent commits touching P0's eval_baselines name files (still hot?) ===
595a5a1 ci: add ruff lint lane + fix all violations; remove never-compiled flyer stub (#92)
d19ef36 eval: record the Gemini-wrapper baselines (the commoditization/moat reference) (#84)
d51efe3 feat(training): Gen-0 TAL harness ? one-command train -> honest LOVO -> ship-gate -> artifact (M0.4 scaffold) (#77)
90fb8b3 Human-GT partial-label eval harness + tuning + iteration doc (#60)
=== real player/venue names STILL committed? (P0 still relevant?) ===
origin/master:backend/eval/eval_partial_labels.py
origin/master:eval_baselines/gemini_wrapper_2026-06-10.json
origin/master:eval_baselines/heuristic_lovo_n6.json
origin/master:eval_baselines/nightly_baseline.json
origin/master:tests/test_cleanup_caches.py
=== golden-video registry (#197) ? feeds P3 clearance? ===
docs/GOLDEN_SET_PHASE1_VIDEOS.md
docs/GOLDEN_VIDEOS.md
[/external_agent_tool_result]
```

#### 460. assistant (2026-07-02T06:11:33.367Z)

Done ? \*\*synced to `f5f5e52` (#203) and clean\*\*; my worktrees/branches cleaned up. Here's the TL;DR.

### Where the Golden Regression Fixtures project stands

\*\*Core complete + audited\*\* ? 6 PRs merged (#186/#189/#191/#192/#193/#194). A clean clone runs the rally-seg regression with no video/GPU/DB, cross-OS enforced on every PR (Win/macOS/Linux), de-attribution tool + refusal gate shipped, all hardened by an adversarial audit, coverage ~80.7%. Nothing of mine is in flight; the parallel quality track (R2?R5, W1/W2) is the \*other\* session and isn't mine to touch.

### Next steps

\*\*? Recommended, and now unblocked (no owner input needed ? I can start on your go):\*\*  
- \*\*P0 ? scrub the committed real names + add a leak-scan CI gate.\*\* I'd held this because it edits `eval\_baselines`, but those files are \*\*no longer hot\*\* (the quality session's recent churn is all in windowing/gate code, not these). And it's still real:  
`adarsh/kushagra/mahadevpura/gbaaddy` are committed today in `eval\_partial\_labels.py`, `gemini\_wrapper\_2026-06-10.json`, `heuristic\_lovo\_n6.json`, \*\*`nightly\_baseline.json` (new)\*\*, and `test\_cleanup\_caches.py` (+ a personal `C:/Users/avidu/...` path). ? opaque surrogate keys + a CI guard that fails on names/MD5.

\*\*? Gated on you (the high-value step ? P3 real fixtures):\*\*

1. \*\*Clearance list\*\* ? which golden videos are \*\*owned + minors-free\*\* (the new `docs/GOLDEN\_VIDEOS.md` registry may already list candidates ? point me at it and I'll cross-check).
2. \*\*Minimization re-confirm\*\* (the audit's honesty flag): commit the de-attributed \*\*per-frame\*\* trajectory CSV ? \*\**(a)*

**\*\*? Deferred (with reason):\*\*** M4 quality-floor (meaningful only once real fixtures land); 01  
protobuf / 02 Tier-1 key-gated (optional).

**\*\*One sub-decision inside P0 is yours:\*\*** whether to rewrite git history (`filter-repo`) to purge the names from *\*past\** commits, or just scrub going forward + guard. I'd default to **\*\*scrub-forward + leak-scan now\*\***, and leave the history rewrite as a separate owner-approved step (it's destructive and rewrites shared history).

Want me to **\*\*start P0\*\*** now, or would you rather hand me the **\*\*P3 clearance list + (a)/(b)\*\*** so I go straight for the real fixtures?